Block practice · Nerves, senses, hormones and drugs

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Identify brain regions and match them to their functions

Identify parts of the eye and state their functions

Match hormones to glands and effects, and contrast them with nerves

Draw, label and compare the structure of a neurone

Define a reflex and name the neurones on the receptor-to-effector path

Describe how an impulse crosses a synapse

Explain how the eye focuses on near and distant objects

Explain eye defects and how they are corrected or treated

Explain the effects and dangers of alcohol and other drugs

Show how an image forms on the retina and reaches the brain

Plan an investigation measuring a response or treatment in people

Explain validity problems and fixes in investigations on people

Predict how a drug, toxin or disease changes synaptic transmission

Practice workbook · Identify brain regions and match them to their functions

Labelling or locating the cerebral hemispheres, cerebellum, medulla, hypothalamus and pituitary, and pairing each with functions such as balance, memory, breathing or temperature control.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATCH-BRAIN-AREA-TO-FUNCTION

1. Question 1 · part b · 4 marks

June 2021 Paper 2

Worked-example retry · Attempt part b only.

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1 The diagram shows a section through a human brain.



OLevels.com SCHOOL

(a) Using lines, label these structures on the diagram.

- cerebral hemisphere
- cerebellum
- pituitary gland
- hypothalamus

(4)

(b) State which structure in part (a) is responsible for each of these functions.

(4)

Function	Structure
control of muscle tone and posture	
intelligence and thought	
monitoring of body temperature	
release of ADH	

(Total for Question 1 = 8 marks)

1. Question 1 · part b · 4 marks · Working and answer

Working and answer

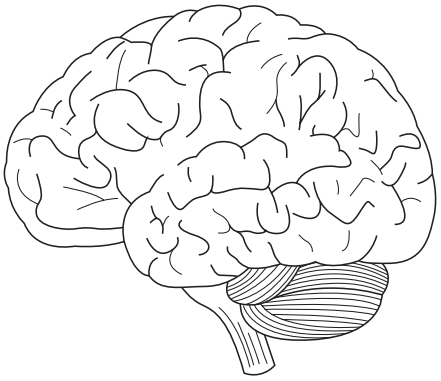
OLevels.com SCHOOL

2. Question 7 · part a · 3 marks

June 2019 Paper 1

Further practice · Attempt part a only.

7 (a) The diagram shows a human brain.



Add lines labelled X, Y and Z to the diagram to show the areas of the brain that control these functions.

(3)

X voluntary actions

Y balance

Z breathing rate

(b) Parkinson's disease affects the cells in the brain that help to control body movement.

The affected cells are unable to communicate effectively with neurones that cause muscles to contract.

(i) Name the type of neurone that causes muscles to contract.

(1)

.....

(ii) Give one difference between neurones that cause muscles to contract and neurones that transmit nerve impulses from receptor organs.

(1)

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20

2. Question 7 · part a · 3 marks · Working and answer

Working and answer

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3. Question 6 · part b · 1 mark

June 2019 Paper 2R

Further practice · Attempt part b only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Alzheimer's disease is a degenerative disease that usually starts slowly and gets worse over time. Scientists want to understand the causes of the disease and discover treatments for the memory loss that is a major feature of Alzheimer's. Scientists use the brains of people who have died to try to find out what happens in the brains of people who have Alzheimer's.

The most common early symptom is short-term memory loss, which is difficulty in remembering recent events. People with this condition are no longer able to transfer information about their daily life from their short-term to long-term memory. For example, they are unable to remember if they went shopping.

The condition causes plaques of an insoluble protein called B-amyloid to build up in the gap between neurones. Over time, these plaques appear in the neurones themselves in the regions of the brain associated with memory.

The B-amyloid is formed when an essential protein called amyloid precursor protein (APP) is not broken down correctly.

(a) Describe two differences between short-term memory and long-term memory (lines 8 to 9). **(2)**

1 .....

.....

2 .....

.....

(b) State the area of the brain responsible for memory. **(1)**

.....

(c) State two symptoms of Alzheimer's disease, other than memory loss. **(2)**

1 .....

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2 .....

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3. Question 6 · part b · 1 mark · Working and answer

Working and answer

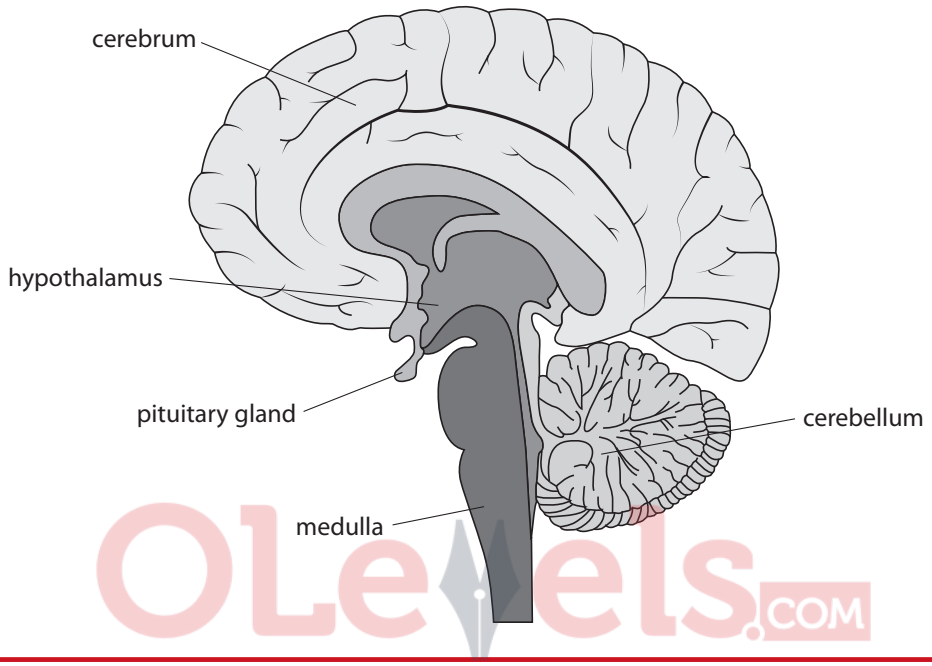
OLevels.com SCHOOL

4. Question 7 · part a.i · 3 marks

January 2020 Paper 2

Further practice · Attempt part a.i only.

7 The diagram shows parts of the human brain.



(a) (i) The table lists parts of the brain and some functions of the brain.

Complete the table by putting a tick (✓) to show if the function is correct for each part.

(3)

Part	Function	Tick
medulla	controls heart rate and breathing	
pituitary gland	involved in language	
hypothalamus	responsible for osmoregulation	
cerebellum	controls voluntary movements	

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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 5 9 9 1 0 A 0 1 8 2 4

4. Question 7 · part a.i · 3 marks

January 2020 Paper 2

Further practice · Attempt part a.i only.

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(ii) The cerebrum is divided into the left and right cerebral hemispheres.

Each hemisphere contains different areas. One of these is the motor area.

State the function of the motor area.

(1)

.....

.....

(b) Nerve cells deep inside the brain produce the neurotransmitter dopamine.

Parkinson's disease is a brain disorder that destroys these nerve cells.

Explain why the destruction of these nerve cells causes the symptoms associated with Parkinson's disease.

(3)

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(Total for Question 7 = 7 marks)

4. Question 7 · part a.i · 3 marks · Working and answer

Working and answer

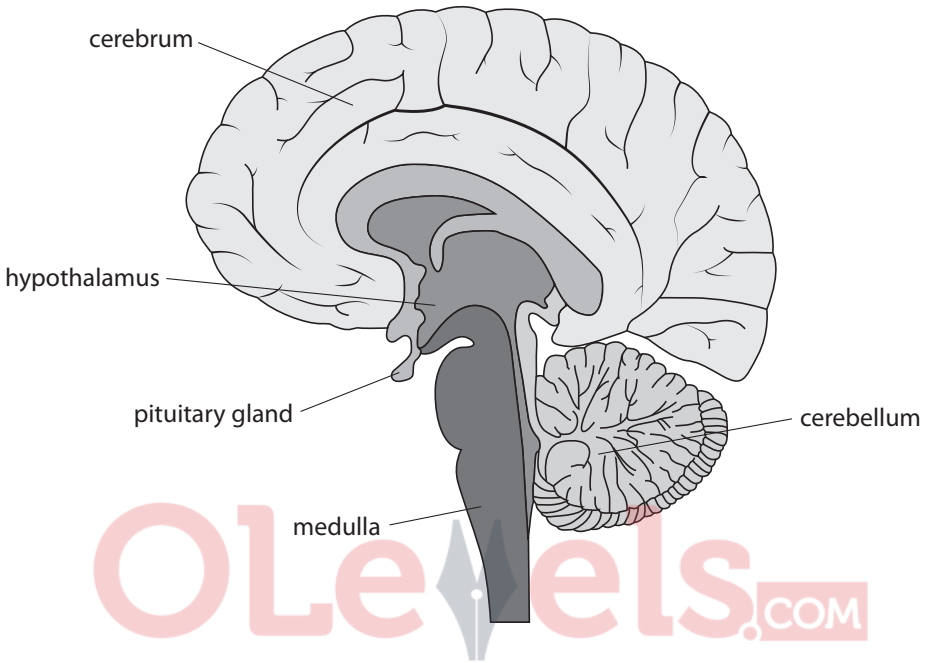
OLevels.com SCHOOL

5. Question 7 · part a.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part a.ii only.

7 The diagram shows parts of the human brain.



(a) (i) The table lists parts of the brain and some functions of the brain.

Complete the table by putting a tick (✓) to show if the function is correct for each part.

(3)

Part	Function	Tick
medulla	controls heart rate and breathing	
pituitary gland	involved in language	
hypothalamus	responsible for osmoregulation	
cerebellum	controls voluntary movements	

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P 5 9 9 1 0 A 0 1 8 2 4

5. Question 7 · part a.ii · 1 mark

January 2020 Paper 2

Further practice · Attempt part a.ii only.

DO NOT WRITE IN THIS AREA

(ii) The cerebrum is divided into the left and right cerebral hemispheres.

Each hemisphere contains different areas. One of these is the motor area.

State the function of the motor area.

(1)

.....

.....

(b) Nerve cells deep inside the brain produce the neurotransmitter dopamine.

Parkinson's disease is a brain disorder that destroys these nerve cells.

Explain why the destruction of these nerve cells causes the symptoms associated with Parkinson's disease.

(3)

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(Total for Question 7 = 7 marks)

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OLevels.com SCHOOL

Working and answer

6. Question 6 · part a.i · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.i only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The effects of alcohol

Humans have been drinking alcohol for thousands of years. Drinking too much alcohol over many years can cause diabetes, malnutrition and diseases of the central nervous system and the liver. One immediate side effect of drinking too much alcohol is increased urination.

Beer is about 95% water and only 5% alcohol. The liver converts that 5% of alcohol into the same volume of water and some carbon dioxide. So if a person drinks 200 cm^3 of beer (one unit), the end result is 200 cm^3 of water. However, this person will not just urinate 200 cm^3 of urine, in fact the person urinates a total of 320 cm^3 of urine. So in general each unit of alcohol makes the person urinate an extra 120 cm^3 of urine more than the normal urine output.

A person with a mass of 60 kg produces 60 cm^3 of urine an hour and a person with a mass of 80 kg produces 80 cm^3 an hour and so on. Therefore if a 60 kg person drinks 200 cm^3 of beer in an hour they will produce 60 cm^3 of urine plus 320 cm^3, making a total of 380 cm^3.

Alcohol interferes with the mechanism that regulates water levels in our body because alcohol affects a small gland called the pituitary gland. One hormone released by the pituitary gland is anti-diuretic hormone (ADH). Alcohol reduces how much ADH is released by the pituitary. Each unit of alcohol that a person drinks forces the kidneys to produce an extra 120 cm^3 of urine more than the normal output.

Logic suggests that drinking lots of water would replace the 120 cm^3 of urine produced, but it is not that simple. The body only retains half the water that a person drinks. The other half leaves the body in urine, so the person will become dehydrated even if they drink lots of water.

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6. Question 6 · part a.i · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.i only.

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(a) (i) Where in the body is the pituitary gland (line 16)?

(1)

- ☒ **A** brain
- ☒ **B** kidney
- ☒ **C** liver
- ☒ **D** pancreas

(ii) Which system includes the pituitary gland (line 16)?

(1)

- ☒ **A** digestive
- ☒ **B** endocrine
- ☒ **C** nervous
- ☒ **D** skeletal

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(b) A person weighs 75 kg and drinks 400 cm^3 of beer in an hour (lines 11–14).

(i) Calculate the volume of urine produced by the person in this hour.

(3)

volume of urine = ... cm^3

(ii) Calculate the percentage increase in the volume of urine produced by this person in this hour.

(3)

percentage increase = ...%

6. Question 6 · part a.i · 1 mark · Working and answer

Working and answer

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7. Question 1 · part a · 4 marks

June 2021 Paper 2

Further practice · Attempt part a only.

DO NOT WRITE IN THIS AREA

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1 The diagram shows a section through a human brain.



OLevels.com SCHOOL

(a) Using lines, label these structures on the diagram.

- cerebral hemisphere
- cerebellum
- pituitary gland
- hypothalamus

(4)

(b) State which structure in part (a) is responsible for each of these functions.

(4)

Function	Structure
control of muscle tone and posture	
intelligence and thought	
monitoring of body temperature	
release of ADH	

(Total for Question 1 = 8 marks)

7. Question 1 · part a · 4 marks · Working and answer

Working and answer

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8. Question 3 · part a · 1 mark

November 2021 Paper 2

Further practice · Attempt part a only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Sweating is normal when we are hot. It is part of our natural cooling mechanism. If you sweat visibly when you do not need to cool down, it can be a problem.

The sweating process is controlled by the brain, which sends signals down the nerves to the sweat glands.

Sweating is crucial to survival. We sweat to control body temperature and the sweat glands in our armpits can produce several litres of sweat in 24 hours. How much we sweat varies hugely from individual to individual and even from day to day.

Hyperhidrosis occurs when some areas of the body start sweating excessively. This affects about 1% of the population. There are two main types. The most common type is primary focal hyperhidrosis, which affects many parts of the body. There is no known cause, but it usually begins in childhood and often runs in families. Another type is secondary hyperhidrosis, which often doesn't begin until after the teenage years, and usually has an underlying medical cause. These causes can include diabetes, menopause or chronic infection. It can also be a result of eating certain foods. Stress can play a major role too.

(a) Which part of the brain controls sweating (line 4)?

(1)

- ☒ **A** cerebellum
- ☒ **B** cerebral hemispheres
- ☒ **C** hypothalamus
- ☒ **D** pituitary gland

(b) Which type of nerve sends signals to the sweat glands (lines 4 and 5)?

(1)

- ☒ **A** connector
- ☒ **B** motor
- ☒ **C** relay
- ☒ **D** sensory

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8. Question 3 · part a · 1 mark · Working and answer

Working and answer

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9. Question 3 · part d · 1 mark

June 2024 Paper 2

Further practice · Attempt part d only.

3 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

People make lifestyle choices that can sometimes have a negative effect on their health. People who drink too much alcohol are at risk of brain and liver damage, certain types of cancer and mental illness. Liver damage can affect the digestion of fat. Alcohol can also affect the eyes, increasing the risk of developing cataracts that cause the lenses in the eyes to become cloudy. Alcohol is also a depressant. It increases the activity of a chemical in parts of the brain that control balance, speech, motor skills and memory. In addition, alcohol suppresses an area of the brain called the amygdala. The amygdala is responsible for the fight or flight function which prepares the body for action, initiating the release of stress hormones. These hormones increase glucose usage by the brain and other parts of the body.

When a person stops drinking alcohol after years of drinking a lot of alcohol, the amygdala in the brain becomes fully active again, which is the cause of all forms of alcohol withdrawal. Over time, if a person regularly drinks heavily, the central nervous system gets used to the suppressing effect of the alcohol, which means the brain and body are affected if the alcohol level suddenly drops. An individual can go straight into fight or flight mode as the alcohol leaves the system and levels of stress hormones begin to return to normal. Symptoms of withdrawal can be as severe as seizures and hallucinations or less severe symptoms such as loss of appetite and insomnia (an inability to sleep).

(a) Drinking alcohol and diet are two lifestyle choices.

Describe how a lifestyle choice, other than drinking alcohol and diet, can have a harmful effect on health (Lines 1 and 2).

(2)

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P 7 4 6 3 0 A 0 8 2 4

9. Question 3 · part d · 1 mark

June 2024 Paper 2

Further practice · Attempt part d only.

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(b) Explain how drinking alcohol affects the digestion of fat (Lines 3 to 4).

(2)

(c) State how cataracts affect vision (Lines 4 to 5).

(1)

(d) Name a part of the brain, other than the amygdala, that is affected by alcohol (Lines 5 to 7).

(1)

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(e) Explain how alcohol acts as a depressant on the nervous system (Line 5).

(3)

9. Question 3 · part d · 1 mark · Working and answer

Working and answer

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Practice workbook · Identify parts of the eye and state their functions

Naming labelled eye structures, choosing structures from descriptions, matching parts to jobs, and describing roles of cornea, lens, retina, fovea, iris and the blind spot.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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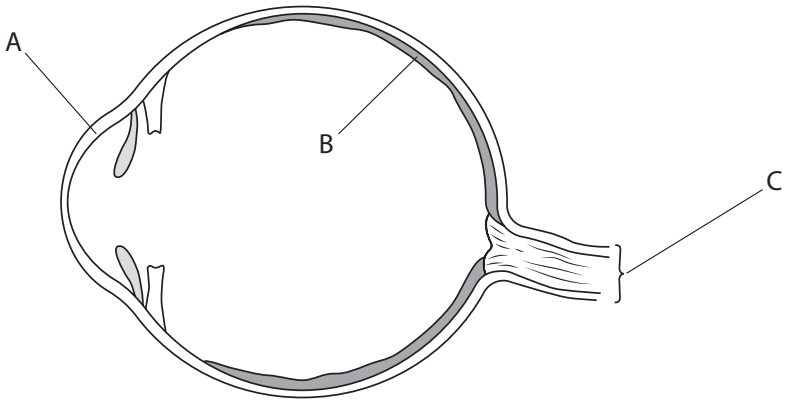
Type ID: HBI-IDENTIFY-EYE-PARTS-AND-ROLES

1. Question 2 · part a.ii · 3 marks

June 2025 Paper 1R

Worked-example retry · Attempt part a.ii only.

2 The diagram shows a section through the eye with the lens missing.



(a) (i) Complete the diagram by drawing in the lens and suspensory ligaments as they would appear when the eye is looking at a near object.

(2)

(ii) Name the parts labelled A, B and C.

(3)

A .....

B .....

C .....

(b) A person is looking at a near object.

Describe the changes that occur in the eye if the person then looks at a distant object.

(4)

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(Total for Question 2 = 9 marks)

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1. Question 2 · part a.ii · 3 marks · Working and answer

Working and answer

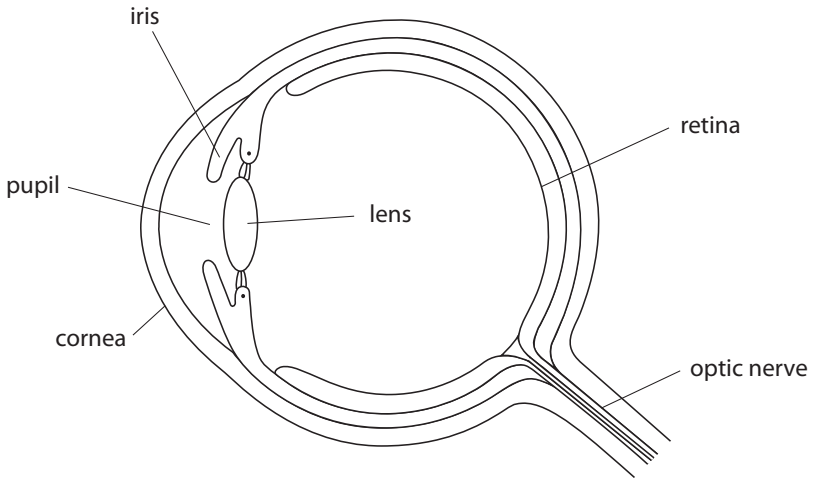
OLevels.com SCHOOL

2. Question 3 · part a · 1 mark

June 2019 Paper 1

Further practice · Attempt part a only.

3 The human eye contains structures that help to focus light on the retina.



(a) Which structures help to focus light on the retina?

(1)

- ☒ **A** cornea and lens
- ☒ **B** iris and cornea
- ☒ **C** iris and pupil
- ☒ **D** pupil and lens

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2. Question 3 · part a · 1 mark

June 2019 Paper 1

Further practice · Attempt part a only.

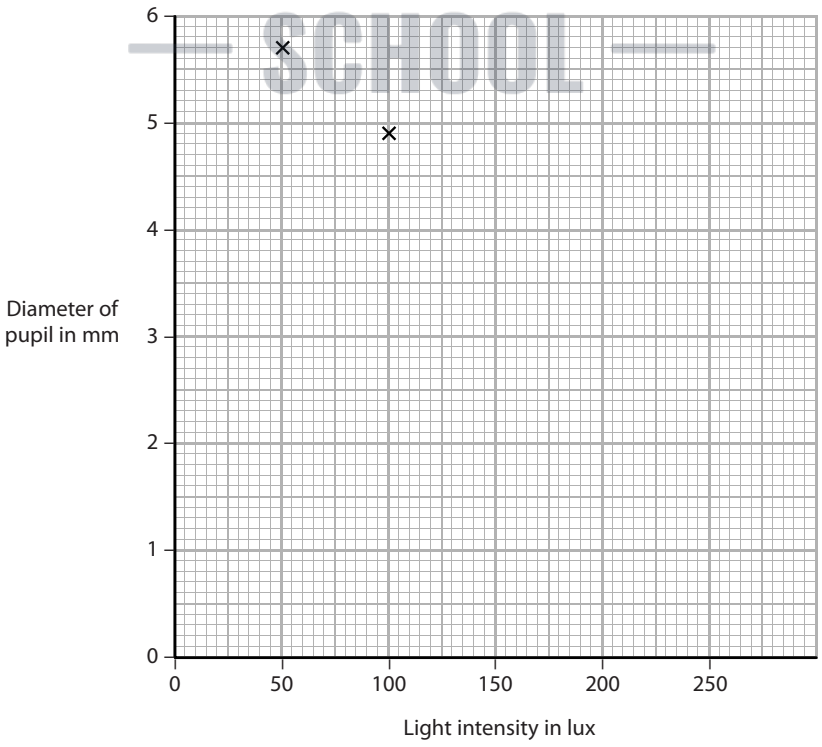
(b) The intensity of light entering the eye can be measured in a unit called lux.

The table shows the results of an investigation on how light intensity affects the diameter of the pupil in the eye.

Light intensity in lux	Diameter of pupil in mm
50	5.7
100	4.9
150	2.7
200	3.2
250	2.3

(i) Complete the graph by plotting the results for 150, 200 and 250 lux. (1)

(ii) Draw a straight line of best fit. (1)



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2. Question 3 · part a · 1 mark · Working and answer

Working and answer

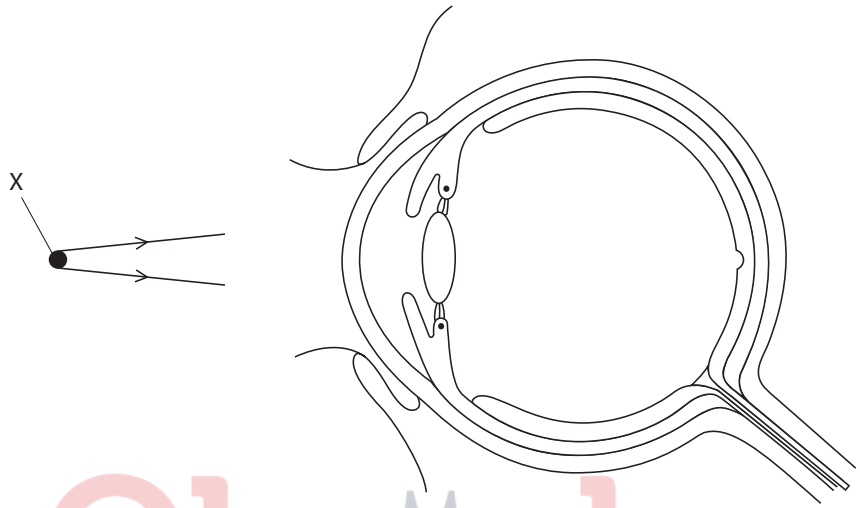
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3. Question 1 · part a · 5 marks

June 2019 Paper 2R

Further practice · Attempt part a only.

1 The diagram shows a section through an eye.



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(a) The table lists descriptions of parts of the eye.

Use the letters, A, B, C, D and E, to label the parts of the eye on the diagram.

(5)

Description of part	Letter
the layer that contains light sensitive cells	A
the muscles that change lens shape	B
the structure that controls the amount of light entering the eye	C
the structure that carries nerve impulses away from the eye	D
the structure that protects the eye from dust and grit	E

(b) A person looks at object X.

Complete the diagram by continuing the rays of light to show how the image of object X is formed on the retina.

(3)

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3. Question 1 · part a · 5 marks · Working and answer

Working and answer

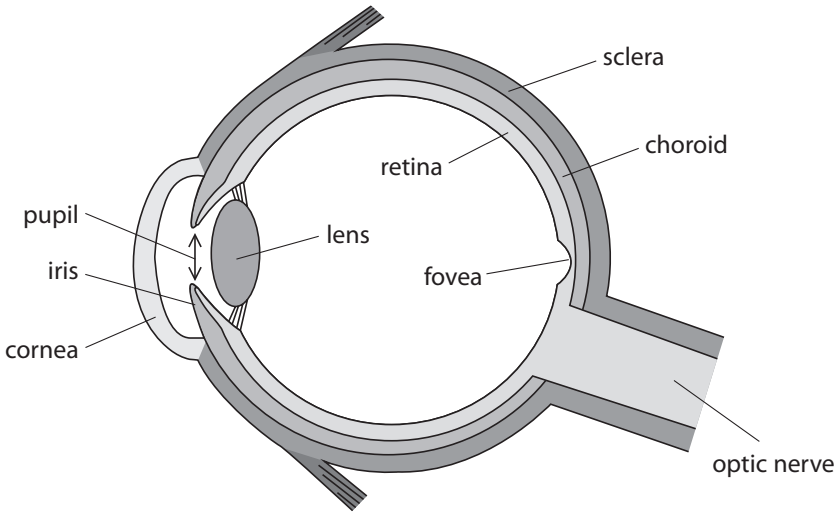
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4. Question 3 · part a.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part a.i only.

3 (a) The diagram shows the human eye, with some structures labelled.



(i) Which structure helps to focus light at the back of the eye? **(1)**

- ☒ **A** fovea
- ☒ **B** iris
- ☒ **C** lens
- ☒ **D** pupil

(ii) The box lists structures of the eye. **(3)**

sclera	retina	choroid	iris	pupil

Use words from the box to complete the passage about the eye.

The ..... contains receptor cells that are sensitive to light.

In bright light the ..... gets smaller to control the amount of light entering the eye.

The ..... contains blood vessels that deliver oxygen to receptor cells.

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4. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

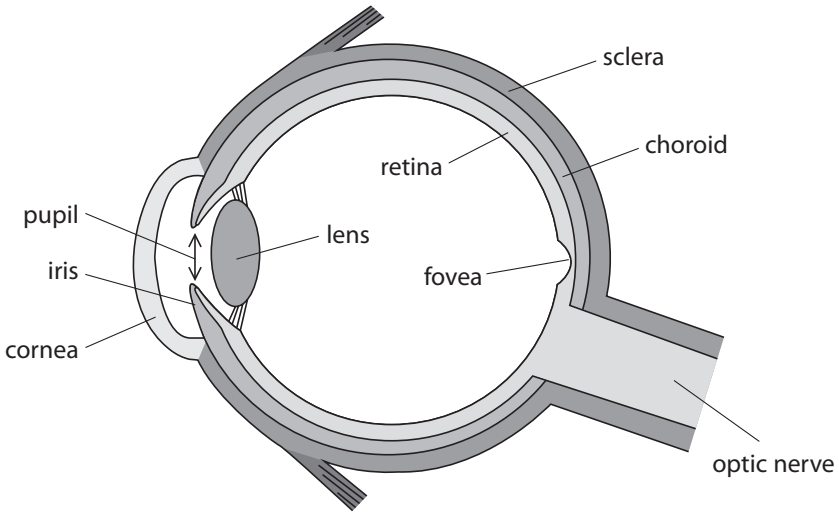
OLevels.com SCHOOL

5. Question 3 · part a.ii · 3 marks

January 2020 Paper 2

Further practice · Attempt part a.ii only.

3 (a) The diagram shows the human eye, with some structures labelled.



(i) Which structure helps to focus light at the back of the eye?

(1)

- ☒ **A** fovea
- ☒ **B** iris
- ☒ **C** lens
- ☒ **D** pupil

(ii) The box lists structures of the eye.

(3)

sclera	retina	choroid	iris	pupil

Use words from the box to complete the passage about the eye.

The ..... contains receptor cells that are sensitive to light.

In bright light the ..... gets smaller to control the amount of light entering the eye.

The ..... contains blood vessels that deliver oxygen to receptor cells.

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P 5 9 9 1 0 A 0 6 2 4

5. Question 3 · part a.ii · 3 marks

January 2020 Paper 2

Further practice · Attempt part a.ii only.

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(b) The photomicrograph gives the thickness of the two outer layers of the eye.



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Estimate the thickness of the retina.

(1)

thickness of the retina = ... µm

(c) The optic nerve contains sensory neurones.

(i) Describe the function of the sensory neurones in the optic nerve.

(2)

.....

.....

.....

.....

(ii) Glaucoma is a disease that damages the optic nerve.

Explain why damage to the optic nerve will affect vision.

(2)

.....

.....

.....

.....

(Total for Question 3 = 9 marks)

5. Question 3 · part a.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 6 · part a · 1 mark

June 2021 Paper 2

Further practice · Attempt part a only.

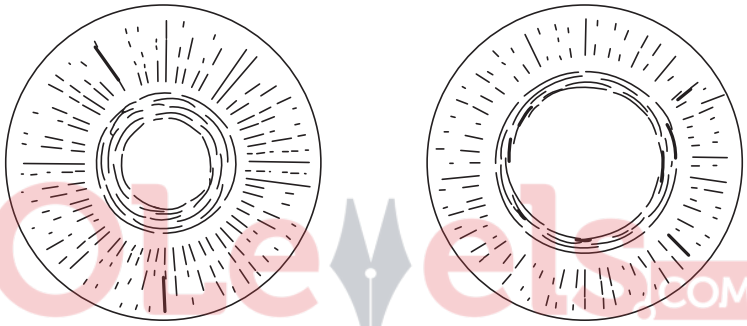
6 The human eye consists of many parts.

(a) Which part has the highest concentration of light sensitive cells?

(1)

- ☒ **A** blind spot
- ☒ **B** choroid
- ☒ **C** fovea
- ☒ **D** optic nerve

(b) The diagram shows the iris of an eye in two different light conditions, A and B.



light condition A

light condition B

(i) Describe what happens in the eye when the light condition changes from A to B.

(5)

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6. Question 6 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 8 · part a.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part a.ii only.

8 Skin and hair contain a pigment called melanin.

(a) (i) Describe the function of the pigment melanin in the skin.

(2)

.....

.....

.....

.....

(ii) State one other place, apart from hair and skin, where pigment is found in the body.

(1)

.....

(b) A condition known as white forelock is caused by a dominant allele H.
People with this condition have a white patch in their hair.

The condition causes a deficiency of the skin pigment melanin.

A woman who is heterozygous for white forelock produces children with a father who does not have the condition.

(i) Draw a genetic diagram to show the possible offspring from this cross.

The condition is not sex-linked.

(4)

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7. Question 8 · part a.ii · 1 mark · Working and answer

Working and answer

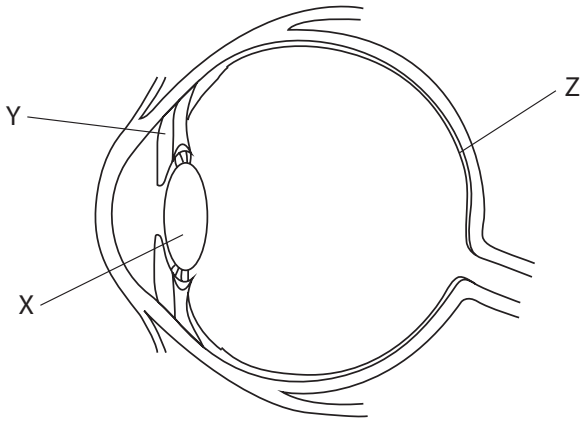
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8. Question 1 · part a.i · 3 marks

January 2023 Paper 1

Further practice · Attempt part a.i only.

1 (a) The diagram shows a human eye.



(i) Draw one line from each structure to its correct function.

(3)

Structure	Function
structure X	refracts light rays
structure Y	protects the eye from pathogens
structure Z	detects light rays
	controls the shape of the lens
	transmits electrical impulses to the brain
	controls the size of the pupil

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8. Question 1 · part a.i · 3 marks

January 2023 Paper 1

Further practice · Attempt part a.i only.

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(ii) Most humans have stereoscopic vision.

Which of these statements occurs because of stereoscopic vision?

(1)

- ☒ **A** being able to judge depth
- ☒ **B** being able to see in colour
- ☒ **C** having a wide field of view
- ☒ **D** to be able to see near and far objects

(b) The diagram shows a healthy eye and an eye with a cataract.



Healthy eye Eye with cataract

(i) Describe the cause of cataracts.

(2)

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(ii) State one treatment for cataracts.

(1)

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(iii) It is estimated that 39 million people in the world are blind.

Of these people, 43% are blind because of cataracts.

Determine the estimated number of people who are blind because of cataracts.

(2)

number of people = ...

(iv) Some people are blind because of a vitamin deficiency.

Name the vitamin that is needed to prevent this type of blindness.

(1)

.....

(Total for Question 1 = 10 marks)

8. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

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9. Question 6 · part b.ii · 1 mark

January 2023 Paper 2

Further practice · Attempt part b.ii only.

6 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

In 2012, Shinya Yamanaka, a Japanese research scientist, won a Nobel Prize for his work on stem cells. He demonstrated that it was possible to take a sample of blood or skin cells from a person and induce these cells to have the properties of stem cells.

In the past, stem cell research scientists needed to obtain human embryonic stem cells from embryos produced by IVF. This required the agreement of those people involved. Embryonic cells can continually divide and can become any human body cell. Scientists have been exploring the use of embryonic stem cells as medical treatments for many different diseases and conditions.

Using the work of Yamanaka and his team, scientists can now produce cells called induced pluripotent stem cells (iPS cells). Cells from the patient are treated with four factors which control the expression of genes. This produces iPS cells from the patient's own cells, which can then be used in stem cell therapies.

This technique is easy to replicate but there are concerns that unlicensed clinics are offering stem cell treatments. Many of these treatments are expensive and success rates are not proven. In addition, stem cells have the potential to continue to divide in the body. Many of the diseases and conditions that stem cell therapies could be used to treat have no other effective treatments. This makes stem cell therapy a desirable option.

(i) Describe the benefits of producing iPS cells from a patient's own cells.

(3)

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9. Question 6 · part b.ii · 1 mark

January 2023 Paper 2

Further practice · Attempt part b.ii only.

(ii) Give two possible risks of using stem cells to treat diseases and medical conditions.

(2)

1 .....

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2 .....

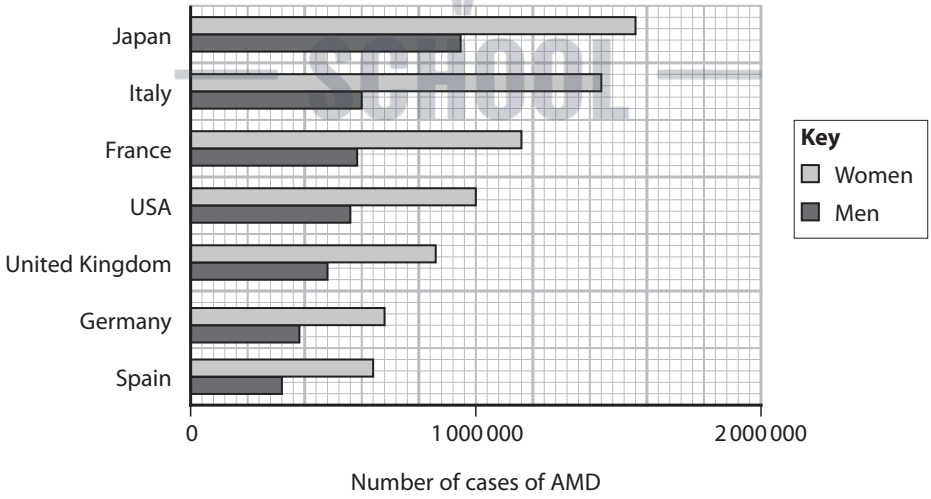
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(b) One condition that stem cells could be used to treat is age-related macular degeneration (AMD).

This is the main cause of vision loss in people over the age of 50.

The graph shows the number of people affected by AMD in some countries.



(i) Calculate the difference between the number of women and the number of men affected by AMD in the United Kingdom.

(2)

difference = ...

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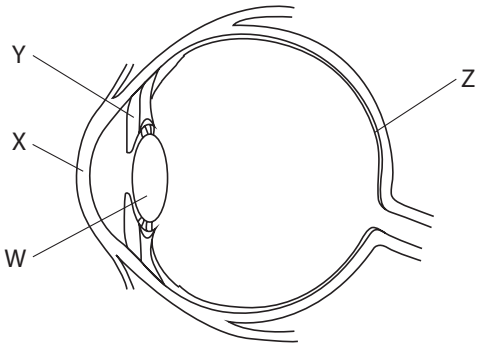
9. Question 6 · part b.ii · 1 mark

January 2023 Paper 2

Further practice · Attempt part b.ii only.

(ii) The diagram shows a human eye.

AMD affects the retina of the eye.



Which of these structures is affected by AMD?

(1)

- ☒ **A** structure W
- ☒ **B** structure X
- ☒ **C** structure Y
- ☒ **D** structure Z

(c) A deficiency of vitamin A can also cause blindness.

A genetically engineered rice product can be used to prevent deficiency of vitamin A.

Describe how genetic engineering is used to produce this rice product.

(3)

(Total for Question 6 = 11 marks)

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9. Question 6 · part b.ii · 1 mark · Working and answer

Working and answer

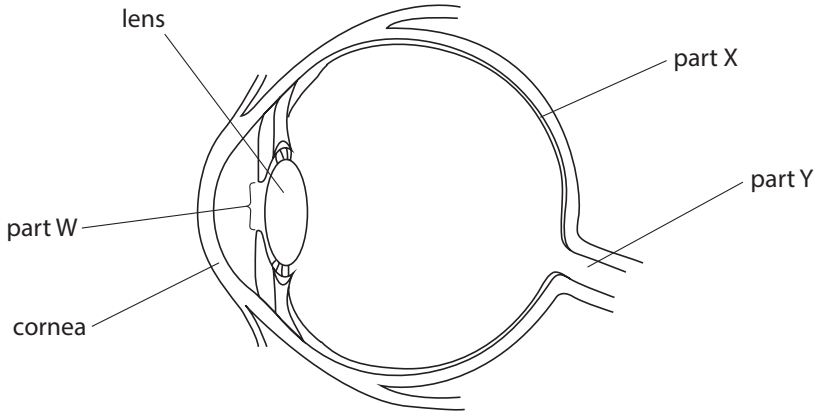
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10. Question 3 · part a.i · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.i only.

3 (a) The diagram shows the human eye.



(i) Name part W and part X.

(2)

W .....

X .....

(ii) Describe the function of part Y.

(2)

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(iii) Explain the change in the shape of the lens when the eye views a distant object.

(2)

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10. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

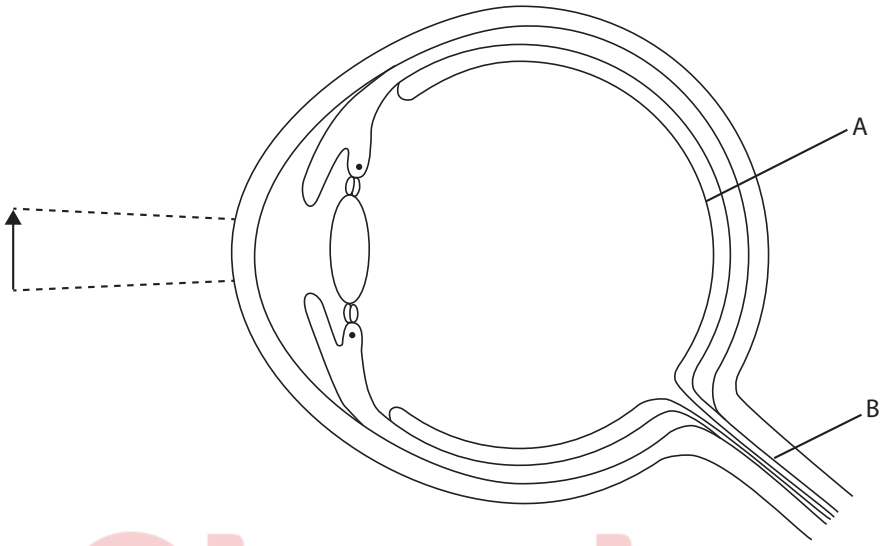
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11. Question 6 · part a.i · 4 marks

June 2025 Paper 2

Further practice · Attempt part a.i only.

6 The diagram shows a section through the eye.



(a) (i) Describe the functions of structures A and B. **(4)**

A .....

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B .....

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(ii) Complete the diagram to show how an image of the arrow is formed at the back of the eye.

(4)

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11. Question 6 · part a.i · 4 marks · Working and answer

Working and answer

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12. Question 1 · part b · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b only.

1 (a) The box lists some parts of the eye and the ear.

auditory nerve	ciliary body	cochlea	iris	lens	optic nerve
ossicles	pupil	retina	semi-circular canals	tympanum	

Complete the table using words from the box to give the missing information.

(6)

Description	Part
contains light-sensitive cells	
membrane that transfers vibrations to middle ear	
transmits impulses from the retina to the brain	
helps the body to balance	
contains muscle tissue that controls light entering the eye	
converts vibrations into nerve impulses	

(b) Explain what is meant by the blind spot in the eye.

(2)

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(Total for Question 1 = 8 marks)

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12. Question 1 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 1 · part a · 6 marks

June 2023 Paper 1R

Further practice · Attempt part a only.

1 (a) The box lists some parts of the eye and the ear.

auditory nerve	ciliary body	cochlea	iris	lens	optic nerve
ossicles	pupil	retina	semi-circular canals	tympanum	

Complete the table using words from the box to give the missing information.

(6)

Description	Part
contains light-sensitive cells	
membrane that transfers vibrations to middle ear	
transmits impulses from the retina to the brain	
helps the body to balance	
contains muscle tissue that controls light entering the eye	
converts vibrations into nerve impulses	

(b) Explain what is meant by the blind spot in the eye.

(2)

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(Total for Question 1 = 8 marks)

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13. Question 1 · part a · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 1 · part a · 7 marks

November 2023 Paper 2

Further practice · Attempt part a only.

1 (a) The box lists parts of the eye and parts of the ear.

auditory nerve	ciliary body	cochlea	cornea
eardrum	optic nerve	ossicles	
pupil	retina	semicircular canals	

The table has descriptions of parts of the eye and ear.

Complete the table by matching the descriptions, using parts from the box.

(7)

Description	Part
contains light sensitive cells	
vibrates when receives sound waves	
helps the body balance	
controls the amount of light entering the eye	
transfers vibrations across the middle ear	
turns vibrations into nerve impulses	
refracts light entering the eye	

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14. Question 1 · part a · 7 marks

November 2023 Paper 2

Further practice · Attempt part a only.

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(b) Explain why a person using both eyes finds it easier to catch a ball than a person who is blind in one eye.

(3)

(Total for Question 1 = 10 marks)

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14. Question 1 · part a · 7 marks · Working and answer

Working and answer

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Practice workbook · Match hormones to glands and effects, and contrast them with nerves

Naming the gland or system producing insulin, glucagon, ADH, adrenaline or thyroxine, stating their effects, and comparing hormonal with nervous signalling.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATCH-HORMONE-SOURCE-AND-EFFECT

1. Question 1 · part whole · 7 marks

November 2024 Paper 1

Worked-example retry · Attempt part whole only.

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1 The table gives some body processes.
Each substance or structure listed in the box is responsible for a body process.

Use information from the box to complete the table.

ADH	adrenaline	cerebellum	cerebrum	cochlea
haemoglobin	liver	semicircular canals	stomach	thyroxine

Body processes	Substance or structure responsible
regulation of water levels in the blood	
conscious thought	
transport of oxygen	
urea formation	
control of metabolic rate	
detection of body position	
reduction of blood flow to the gut	

(Total for Question 1 = 7 marks)

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1. Question 1 · part whole · 7 marks · Working and answer

Working and answer

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2. Question 6 · part a.ii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.ii only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The effects of alcohol

Humans have been drinking alcohol for thousands of years. Drinking too much alcohol over many years can cause diabetes, malnutrition and diseases of the central nervous system and the liver. One immediate side effect of drinking too much alcohol is increased urination.

Beer is about 95% water and only 5% alcohol. The liver converts that 5% of alcohol into the same volume of water and some carbon dioxide. So if a person drinks 200 cm^3 of beer (one unit), the end result is 200 cm^3 of water. However, this person will not just urinate 200 cm^3 of urine, in fact the person urinates a total of 320 cm^3 of urine. So in general each unit of alcohol makes the person urinate an extra 120 cm^3 of urine more than the normal urine output.

A person with a mass of 60 kg produces 60 cm^3 of urine an hour and a person with a mass of 80 kg produces 80 cm^3 an hour and so on. Therefore if a 60 kg person drinks 200 cm^3 of beer in an hour they will produce 60 cm^3 of urine plus 320 cm^3, making a total of 380 cm^3.

Alcohol interferes with the mechanism that regulates water levels in our body because alcohol affects a small gland called the pituitary gland. One hormone released by the pituitary gland is anti-diuretic hormone (ADH). Alcohol reduces how much ADH is released by the pituitary. Each unit of alcohol that a person drinks forces the kidneys to produce an extra 120 cm^3 of urine more than the normal output.

Logic suggests that drinking lots of water would replace the 120 cm^3 of urine produced, but it is not that simple. The body only retains half the water that a person drinks. The other half leaves the body in urine, so the person will become dehydrated even if they drink lots of water.

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2. Question 6 · part a.ii · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.ii only.

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(a) (i) Where in the body is the pituitary gland (line 16)?

(1)

- ☒ **A** brain
- ☒ **B** kidney
- ☒ **C** liver
- ☒ **D** pancreas

(ii) Which system includes the pituitary gland (line 16)?

(1)

- ☒ **A** digestive
- ☒ **B** endocrine
- ☒ **C** nervous
- ☒ **D** skeletal

(b) A person weighs 75 kg and drinks 400 cm^3 of beer in an hour (lines 11–14).

(i) Calculate the volume of urine produced by the person in this hour.

(3)

volume of urine = ... cm^3

(ii) Calculate the percentage increase in the volume of urine produced by this person in this hour.

(3)

percentage increase = ...%

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2. Question 6 · part a.ii · 1 mark · Working and answer

Working and answer

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3. Question 4 · part b.i · 1 mark

June 2021 Paper 2

Further practice · Attempt part b.i only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Obesity and diabetes have been linked to a 20% increase in the number of leg and foot amputations.

Data from Diabetes UK shows that there were 26 378 lower limb amputations related to diabetes from the period 2014 to 2017. This is an increase from the period 2010 to 2013 when there were 22 092 amputations related to diabetes.

Minor lower limb amputations are below the ankle and major lower limb amputations are below the knee. Minor lower limb amputations increased by 26.5% and major lower limb amputations increased by 4.0%.

In 2019 in the United Kingdom 3.8 million people had been diagnosed with diabetes and 90% of these people had Type 2 diabetes.

(a) (i) Describe the causes of obesity.

(3)

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(ii) Give two health risks, other than diabetes, associated with obesity.

(2)

1 .....

2 .....

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3. Question 4 · part b.i · 1 mark

June 2021 Paper 2

Further practice · Attempt part b.i only.

(b) Diabetes is caused as a result of insulin not being secreted or not working correctly.

(i) Which organ secretes insulin?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

(ii) Insulin stimulates the conversion of glucose into glycogen.

In which organ does this occur?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

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(c) (i) Calculate the percentage increase in the number of lower limb amputations between the period 2010 to 2013 and the period 2014 to 2017.

(3)

percentage increase = ...%

(ii) Calculate the number of people in 2019 who had been diagnosed with Type 2 diabetes in the United Kingdom.

(1)

number of people = ... million

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3. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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4. Question 3 · part a.iii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part a.iii only.

3 (a) In England, 3 319 266 people have diabetes.

(i) Give this value to 3 significant figures.

(1)

people with diabetes = ...

(ii) In the rest of the United Kingdom, 600 239 people have diabetes.

Give this value in standard form.

(2)

people with diabetes = ...

(iii) Diabetes can be caused by the inability of an organ to produce insulin.

Name the organ that produces insulin.

(1)

.....

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4. Question 3 · part a.iii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part a.iii only.

(b) The body mass index (BMI) is a measure that uses height and body mass to work out if the body mass is healthy.

The table shows the BMI values and the categories into which a person can be placed.

BMI	BMI category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25.0 to 29.9	overweight
30.0 to 39.9	obese
40.0 and over	morbidly obese

(i) Calculate the BMI of a person with a body mass of 120 kg and a height of 1.95 m.

Use the equation

$$\text{BMI} = \frac{\text{body mass}}{\text{height}^2}$$

(2)

BMI = ...

(ii) Determine the BMI category that this person belongs to.

(1)

.....

.....

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4. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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5. Question 3 · part f · 2 marks

June 2024 Paper 2

Further practice · Attempt part f only.

3 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

People make lifestyle choices that can sometimes have a negative effect on their health. People who drink too much alcohol are at risk of brain and liver damage, certain types of cancer and mental illness. Liver damage can affect the digestion of fat. Alcohol can also affect the eyes, increasing the risk of developing cataracts that cause the lenses in the eyes to become cloudy. Alcohol is also a depressant. It increases the activity of a chemical in parts of the brain that control balance, speech, motor skills and memory. In addition, alcohol suppresses an area of the brain called the amygdala. The amygdala is responsible for the fight or flight function which prepares the body for action, initiating the release of stress hormones. These hormones increase glucose usage by the brain and other parts of the body.

When a person stops drinking alcohol after years of drinking a lot of alcohol, the amygdala in the brain becomes fully active again, which is the cause of all forms of alcohol withdrawal. Over time, if a person regularly drinks heavily, the central nervous system gets used to the suppressing effect of the alcohol, which means the brain and body are affected if the alcohol level suddenly drops. An individual can go straight into fight or flight mode as the alcohol leaves the system and levels of stress hormones begin to return to normal. Symptoms of withdrawal can be as severe as seizures and hallucinations or less severe symptoms such as loss of appetite and insomnia (an inability to sleep).

(a) Drinking alcohol and diet are two lifestyle choices.

Describe how a lifestyle choice, other than drinking alcohol and diet, can have a harmful effect on health (Lines 1 and 2).

(2)

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5. Question 3 · part f · 2 marks

June 2024 Paper 2

Further practice · Attempt part f only.

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(b) Explain how drinking alcohol affects the digestion of fat (Lines 3 to 4).

(2)

(c) State how cataracts affect vision (Lines 4 to 5).

(1)

(d) Name a part of the brain, other than the amygdala, that is affected by alcohol (Lines 5 to 7).

(1)

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(e) Explain how alcohol acts as a depressant on the nervous system (Line 5).

(3)

5. Question 3 · part f · 2 marks

June 2024 Paper 2

Further practice · Attempt part f only.

(f) After a person stops drinking alcohol, the amygdala becomes fully active again.

The release of stress hormones, such as adrenaline, returns to normal.

Give two effects of adrenaline on the body.

(2)

1 .....

.....

2 .....

.....

(Total for Question 3 = 11 marks)

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Working and answer

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6. Question 7 · part c · 1 mark

June 2025 Paper 2R

Further practice · Attempt part c only.

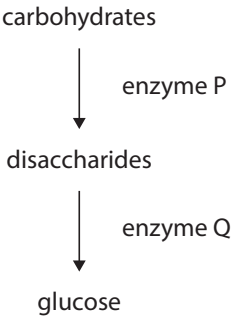
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7 (a) The diagram shows how carbohydrates are broken down to glucose by two different enzymes, enzyme P and enzyme Q.

These enzymes are found in different regions of the digestive system.



Name enzyme P.

(1)

.....

(b) People with diabetes can use glucose testing strips to test their urine.

Describe how enzymes are used in glucose testing strips.

(2)

.....

.....

.....

.....

(c) People with type 1 diabetes are unable to produce hormones to control their blood glucose levels.

Name the organ responsible for producing hormones to control levels of glucose in the blood.

(1)

.....

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6. Question 7 · part c · 1 mark

June 2025 Paper 2R

Further practice · Attempt part c only.

(d) Insulin is a hormone used to treat people with diabetes.

Until recently insulin was obtained from animals for human use.

Suggest why this type of insulin can cause health complications in people with diabetes.

(2)

(e) Describe how yeast cells can be modified to produce the human form of insulin.

(2)

(Total for Question 7 = 8 marks)

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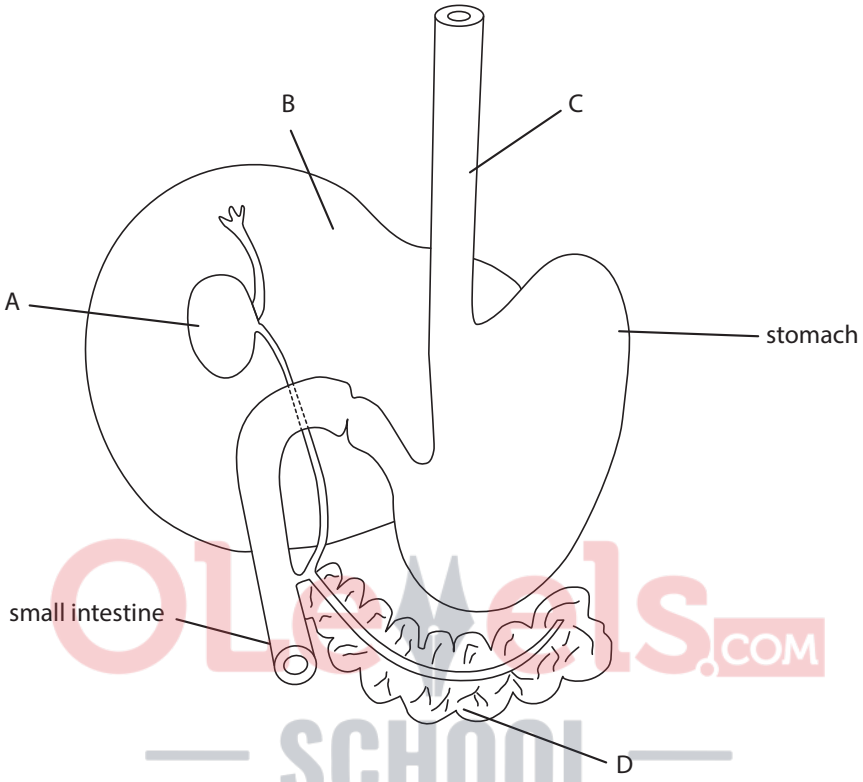
Working and answer

7. Question 2 · part b.ii · 2 marks

June 2025 Paper 2

Further practice · Attempt part b.ii only.

2 The diagram shows structures involved in digestion.



(a) (i) Identify the structures labelled A, B, C and D.

(4)

A .....

B .....

C .....

D .....

(ii) Name the fluid stored in structure A.

(1)

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7. Question 2 · part b.ii · 2 marks

June 2025 Paper 2

Further practice · Attempt part b.ii only.

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(iii) Explain the function of the fluid stored in structure A in digestion.

(3)

(b) (i) Describe the role of structure D in digestion.

(4)

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(ii) Name two substances secreted by structure D that are not involved in digestion.

(2)

1 .....

2 .....

(Total for Question 2 = 14 marks)

7. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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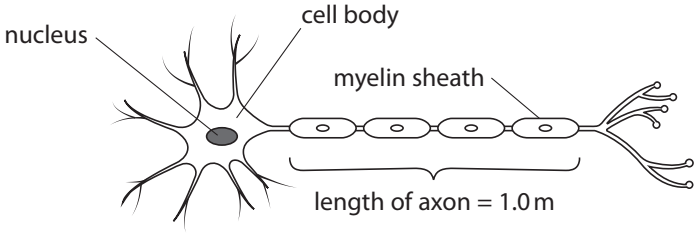
8. Question 5 · part b.i · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

5 The diagram shows a motor neurone.

not to scale



(a) (i) The actual diameter of the nucleus of the motor neurone is 12 µm.

A student observes the motor neurone using a ×50 microscope lens.

Calculate the diameter of the image seen after magnification.

Give your answer in millimetres.

(2)

OLevels.COM — SCHOOL —

diameter = ... mm

(ii) Neurones transmit electrical impulses from one part of the body to another.

Suggest one function of the myelin sheath surrounding the axon.

(1)

.....

.....

(b) Nerve impulses travel at an average speed of 1.2×10^8 µm per second.

Hormones travel at an average speed of 1.5×10^6 µm per second.

(i) Explain this difference in speed.

(2)

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8. Question 5 · part b.i · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

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(ii) Calculate the time taken for a nerve impulse to travel along the axon of the motor neurone.

(3)

time taken = ... s

(Total for Question 5 = 8 marks)

OLevels.com SCHOOL

Working and answer

9. Question 6 · part a · 3 marks

January 2022 Paper 2

Further practice · Attempt part a only.

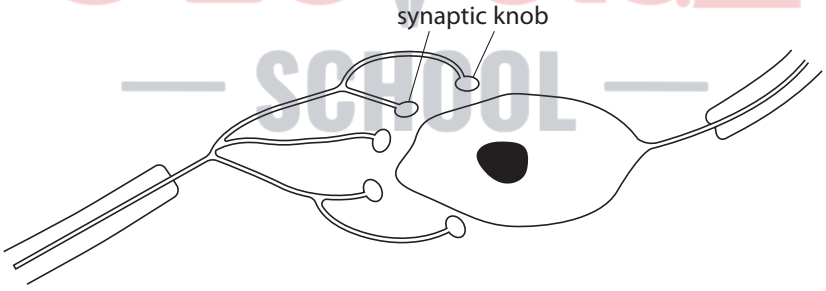
6 (a) Co-ordination of the activities of the body involves the nervous system and the hormonal system.

Complete the table to compare the two systems.

(3)

	Nervous system	Hormonal system
Route of transmission		
Type of signal		
Relative speed		

(b) The diagram shows a synapse.



(i) Draw an arrow on the diagram to show the direction an impulse travels.

(1)

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9. Question 6 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Draw, label and compare the structure of a neurone

Drawing a labelled motor neurone, naming a part such as the axon, giving the role of the myelin sheath, and spotting structural differences between two neurone drawings.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DRAW-LABEL-NEURONE-PARTS

1. Question 3 · part a.i · 4 marks

June 2025 Paper 2R

Worked-example retry · Attempt part a.i only.

3 (a) The nervous system is made up of sensory, motor and relay neurones.

(i) Draw a diagram of a motor neurone.

You should label these structures in your diagram.

- nucleus
- axon
- myelin sheath

(4)

OLevels.com SCHOOL

(ii) Describe the function of a motor neurone.

(2)

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1. Question 3 · part a.i · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 7 · part b.ii · 4 marks

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

7 (a) The nervous system sends electrical impulses to parts of the body in response to an external stimulus.

(i) Name the structure in the nervous system that detects an external stimulus.

(1)

(ii) Name the structure that transmits electrical impulses directly to the central nervous system.

(1)

(b) Neurones are specialised cells.

(i) Describe what is meant by a specialised cell.

(2)

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OLevels.com SCHOOL

2. Question 7 · part b.ii · 4 marks

June 2024 Paper 1R

Further practice · Attempt part b.ii only.

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(ii) A motor neurone is a specialised cell.

Draw a diagram of a motor neurone.

Your diagram should include these labels.

- nucleus
- cell body
- myelin sheath

(4)

OLevels.COM SCHOOL

(iii) Describe the function of a motor neurone in the nervous system.

(2)

2. Question 7 · part b.ii · 4 marks · Working and answer

Working and answer

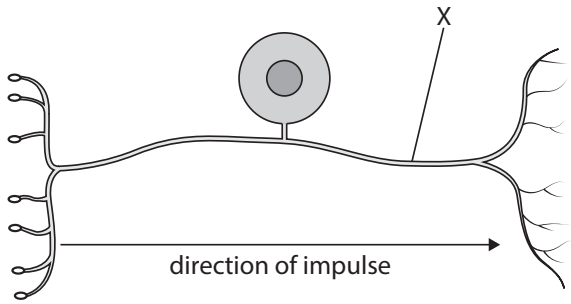
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3. Question 1 · part a.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.ii only.

1 (a) The diagram shows a cell found in the human nervous system.



(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

(1)

.....

(ii) Name the structure labelled X.

(1)

.....

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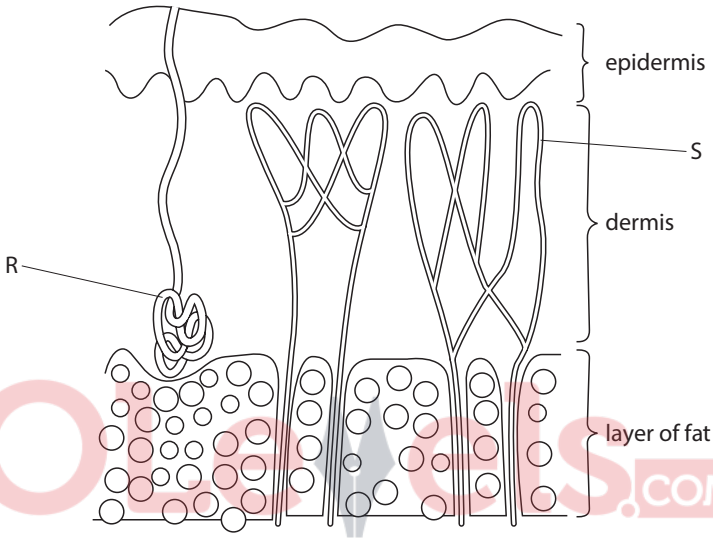
3. Question 1 · part a.ii · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.ii only.

(b) Structures in the skin respond to information from the nervous system.

The diagram shows a section of the skin.



(Source: adapted from Human Physiology and Health, Author David Wright, Publisher Heinemann page 113)

(i) Name part R and part S.

(2)

part R

.....

part S

.....

(ii) Describe how part R helps to cool the body.

(2)

.....

.....

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3. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

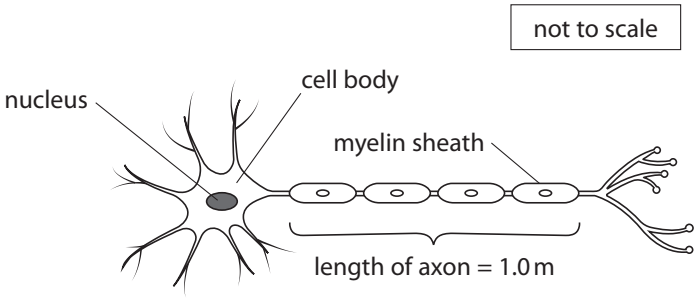
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4. Question 5 · part a.ii · 1 mark

January 2020 Paper 1

Further practice · Attempt part a.ii only.

5 The diagram shows a motor neurone.



(a) (i) The actual diameter of the nucleus of the motor neurone is 12 µm.

A student observes the motor neurone using a ×50 microscope lens.

Calculate the diameter of the image seen after magnification.

Give your answer in millimetres.

(2)

OLevels.com SCHOOL

diameter = ... mm

(ii) Neurones transmit electrical impulses from one part of the body to another.

Suggest one function of the myelin sheath surrounding the axon.

(1)

.....

.....

(b) Nerve impulses travel at an average speed of 1.2×10^8 µm per second.

Hormones travel at an average speed of 1.5×10^6 µm per second.

(i) Explain this difference in speed.

(2)

.....

.....

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4. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 5 · part c.i · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.i only.

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5 This question is about the nervous system.

(a) Describe what is meant by a reflex action.

(3)

(b) Diagram 1 shows a reflex arc.

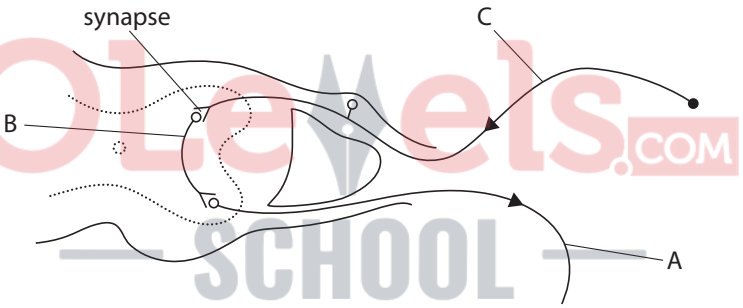


Diagram 1

Name the types of neurone labelled A, B and C.

(3)

A .....

B .....

C .....

5. Question 5 · part c.i · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.i only.

(c) Diagram 2 shows a neurone from a healthy brain and a neurone from the brain of someone with Alzheimer's disease.

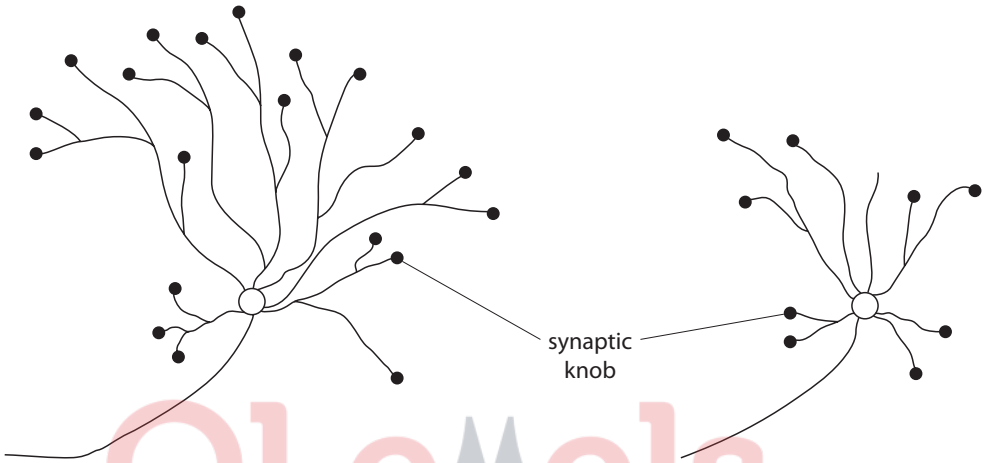


Diagram 2

(i) Give three differences between the two neurones. **(3)**

1 .....

.....

2 .....

.....

3 .....

.....

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5. Question 5 · part c.i · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.i only.

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(ii) Acetylcholine is a neurotransmitter.

Explain the importance of acetylcholine in the functioning of the brain.

(3)

(iii) One symptom of Alzheimer's disease is that less acetylcholine is produced in the brain.

Explain why there is less acetylcholine produced in people with Alzheimer's disease.

Use information from diagram 2 to help your answer.

(3)

(Total for Question 5 = 15 marks)

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5. Question 5 · part c.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Define a reflex and name the neurones on the receptor-to-effector path

What a reflex action is and why it matters, and naming or describing sensory, relay and motor neurones from position or role in a reflex arc.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-REFLEX-ARC-AND-NEURONE-TYPES

1. Question 5 · part b · 3 marks

November 2020 Paper 1

Worked-example retry · Attempt part b only.

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5 This question is about the nervous system.

(a) Describe what is meant by a reflex action.

(3)

(b) Diagram 1 shows a reflex arc.

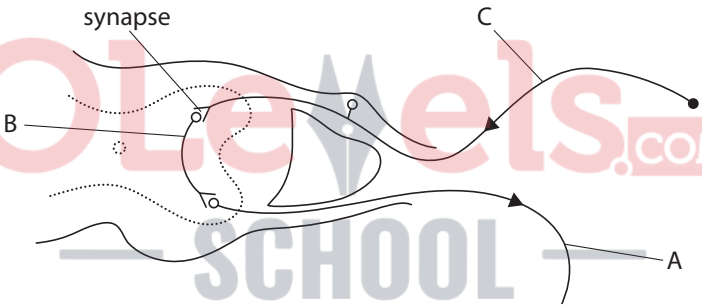


Diagram 1

Name the types of neurone labelled A, B and C.

(3)

A .....

B .....

C .....

1. Question 5 · part b · 3 marks

November 2020 Paper 1

Worked-example retry · Attempt part b only.

(c) Diagram 2 shows a neurone from a healthy brain and a neurone from the brain of someone with Alzheimer's disease.

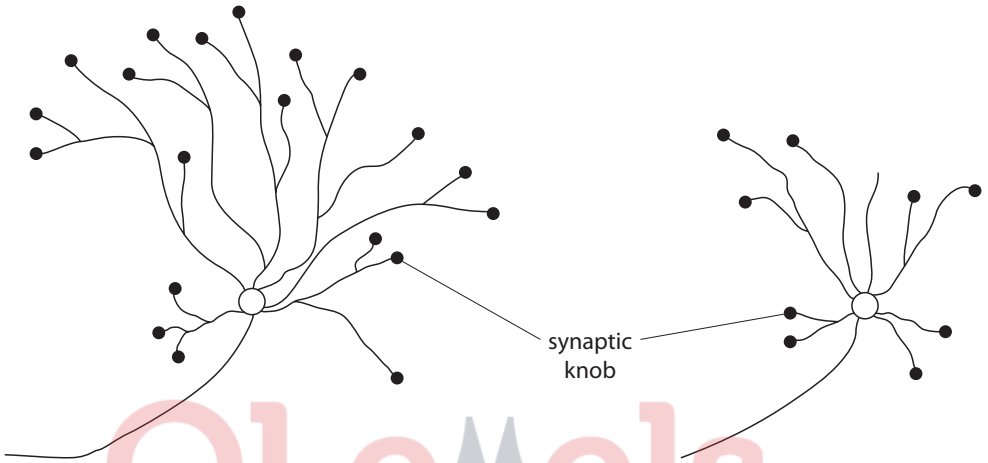


Diagram 2

(i) Give three differences between the two neurones. **(3)**

1 .....

.....

2 .....

.....

3 .....

.....

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1. Question 5 · part b · 3 marks · Working and answer

Working and answer

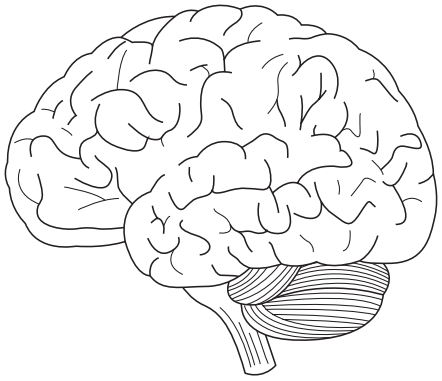
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2. Question 7 · part b.i · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.i only.

7 (a) The diagram shows a human brain.



Add lines labelled X, Y and Z to the diagram to show the areas of the brain that control these functions.

(3)

X voluntary actions

Y balance

Z breathing rate

(b) Parkinson's disease affects the cells in the brain that help to control body movement.

The affected cells are unable to communicate effectively with neurones that cause muscles to contract.

(i) Name the type of neurone that causes muscles to contract.

(1)

.....

(ii) Give one difference between neurones that cause muscles to contract and neurones that transmit nerve impulses from receptor organs.

(1)

.....

.....

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2. Question 7 · part b.i · 1 mark · Working and answer

Working and answer

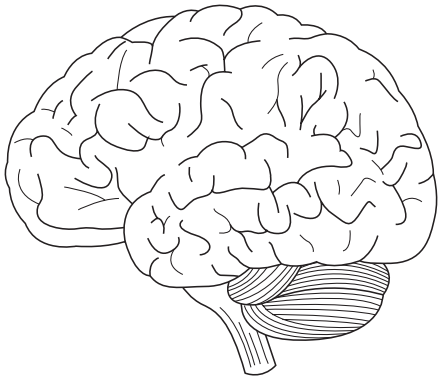
OLevels.com SCHOOL

3. Question 7 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

7 (a) The diagram shows a human brain.



Add lines labelled X, Y and Z to the diagram to show the areas of the brain that control these functions.

(3)

X voluntary actions

Y balance

Z breathing rate

(b) Parkinson's disease affects the cells in the brain that help to control body movement.

The affected cells are unable to communicate effectively with neurones that cause muscles to contract.

(i) Name the type of neurone that causes muscles to contract.

(1)

.....

(ii) Give one difference between neurones that cause muscles to contract and neurones that transmit nerve impulses from receptor organs.

(1)

.....

.....

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3. Question 7 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

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(iii) Explain how one neurone communicates with another neurone.

(3)

(c) Recent scientific research into the treatment of Parkinson's disease has involved the use of stem cells.

Explain how stem cells could be used to reduce the symptoms of Parkinson's disease.

(3)

(Total for Question 7 = 11 marks)

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Working and answer

4. Question 3 · part b · 1 mark

November 2021 Paper 2

Further practice · Attempt part b only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Sweating is normal when we are hot. It is part of our natural cooling mechanism. If you sweat visibly when you do not need to cool down, it can be a problem.

The sweating process is controlled by the brain, which sends signals down the nerves to the sweat glands.

Sweating is crucial to survival. We sweat to control body temperature and the sweat glands in our armpits can produce several litres of sweat in 24 hours. How much we sweat varies hugely from individual to individual and even from day to day.

Hyperhidrosis occurs when some areas of the body start sweating excessively. This affects about 1% of the population. There are two main types. The most common type is primary focal hyperhidrosis, which affects many parts of the body. There is no known cause, but it usually begins in childhood and often runs in families. Another type is secondary hyperhidrosis, which often doesn't begin until after the teenage years, and usually has an underlying medical cause. These causes can include diabetes, menopause or chronic infection. It can also be a result of eating certain foods. Stress can play a major role too.

(a) Which part of the brain controls sweating (line 4)?

(1)

- ☒ **A** cerebellum
- ☒ **B** cerebral hemispheres
- ☒ **C** hypothalamus
- ☒ **D** pituitary gland

(b) Which type of nerve sends signals to the sweat glands (lines 4 and 5)?

(1)

- ☒ **A** connector
- ☒ **B** motor
- ☒ **C** relay
- ☒ **D** sensory

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4. Question 3 · part b · 1 mark

November 2021 Paper 2

Further practice · Attempt part b only.

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(c) Describe how sweating controls body temperature (lines 6 and 7).

(3)

(d) Explain the evidence in the passage that indicates primary focal hyperhidrosis is a genetic condition.

(3)

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(e) Secondary hyperhidrosis can be caused by diabetes.

Give one risk factor for diabetes (line 16).

(1)

4. Question 3 · part b · 1 mark · Working and answer

Working and answer

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5. Question 7 · part a.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.ii only.

7 (a) The nervous system sends electrical impulses to parts of the body in response to an external stimulus.

(i) Name the structure in the nervous system that detects an external stimulus.

(1)

.....

(ii) Name the structure that transmits electrical impulses directly to the central nervous system.

(1)

.....

(b) Neurones are specialised cells.

(i) Describe what is meant by a specialised cell.

(2)

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5. Question 7 · part a.ii · 1 mark · Working and answer

Working and answer

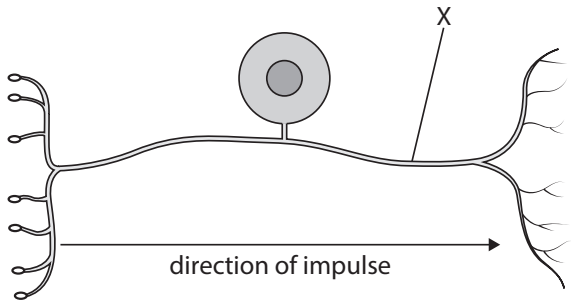
OLevels.com SCHOOL

6. Question 1 · part a.i · 1 mark

June 2024 Paper 1

Further practice · Attempt part a.i only.

1 (a) The diagram shows a cell found in the human nervous system.



(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

(1)

.....

(ii) Name the structure labelled X.

(1)

.....

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6. Question 1 · part a.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 3 · part b.i · 3 marks

June 2025 Paper 2

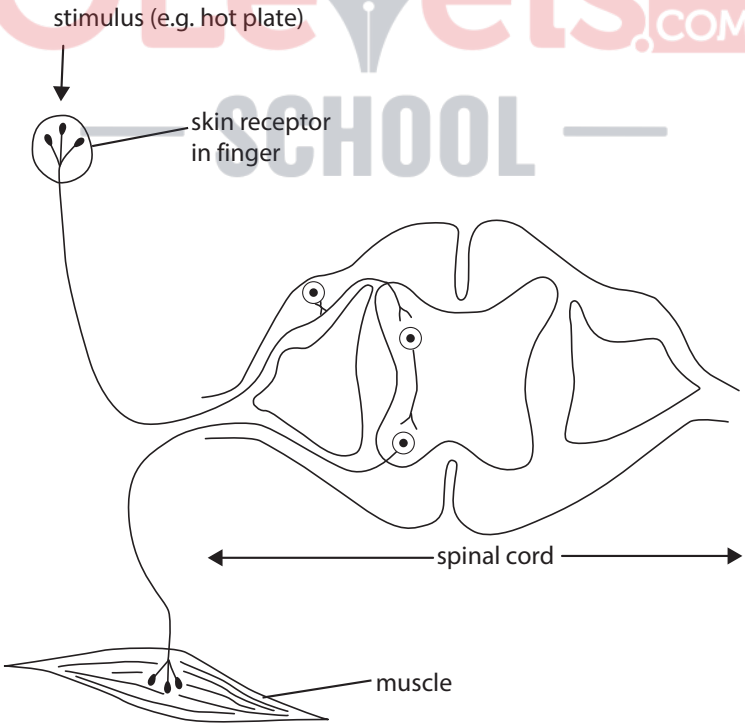
Further practice · Attempt part b.i only.

3 (a) Describe the meaning of the term **reflex action**.

(3)

(b) The diagram shows a reflex arc.

The reflex arc contains three types of neurone.



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P 7 8 7 6 9 A 0 6 2 0

7. Question 3 · part b.i · 3 marks

June 2025 Paper 2

Further practice · Attempt part b.i only.

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(i) Use labels to identify each type of neurone in the diagram.

(3)

(ii) Draw an arrow on the diagram to show the direction of an impulse as it travels through the reflex arc.

(1)

(iii) State which structure in a reflex arc prevents the impulse travelling in the wrong direction.

(1)

(Total for Question 3 = 8 marks)

Working and answer

OLevels.com SCHOOL

8. Question 3 · part a · 4 marks

June 2022 Paper 2

Further practice · Attempt part a only.

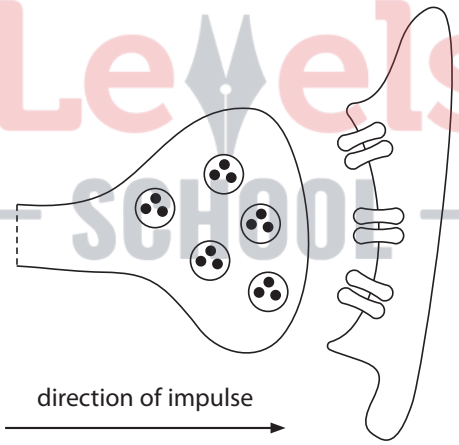
3 (a) The nervous system is adapted to transfer information rapidly from one part of the body to another.

Complete the passage about the nervous system by giving the correct words.

(4)

A change in the external environment is detected by ..... in sense organs. These structures convert stimuli into electrical impulses which then travel along ..... neurones to the central nervous system. The central nervous system coordinates the incoming information using ..... neurones. These neurones then transfer the electrical impulses to ..... neurones.

(b) The diagram shows the junction between two neurones.



(i) State the name of the junction between two neurones.

(1)

.....

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8. Question 3 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 7 · part b.iii · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b.iii only.

7 (a) The nervous system sends electrical impulses to parts of the body in response to an external stimulus.

(i) Name the structure in the nervous system that detects an external stimulus.

(1)

(ii) Name the structure that transmits electrical impulses directly to the central nervous system.

(1)

(b) Neurones are specialised cells.

(i) Describe what is meant by a specialised cell.

(2)

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9. Question 7 · part b.iii · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b.iii only.

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(ii) A motor neurone is a specialised cell.

Draw a diagram of a motor neurone.

Your diagram should include these labels.

- nucleus
- cell body
- myelin sheath

(4)

OLevels.com SCHOOL

(iii) Describe the function of a motor neurone in the nervous system.

(2)

9. Question 7 · part b.iii · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b.iii only.

(c) Neurotransmitters allow different neurones to communicate with each other.

The distance between two neurones is 100 nm.

Neurotransmitters diffuse across the gap between neurones at a speed of 4.0×10^6 nm/s.

Calculate the time taken, in milliseconds, for a neurotransmitter to cross the gap between two neurones.

Give your answer in standard form.

(4)

OLevels.COM — SCHOOL —

time taken = ... ms

(Total for Question 7 = 14 marks)

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Working and answer

10. Question 3 · part a.ii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.ii only.

3 (a) The nervous system is made up of sensory, motor and relay neurones.

(i) Draw a diagram of a motor neurone.

You should label these structures in your diagram.

- nucleus
- axon
- myelin sheath

(4)

OLevels.com SCHOOL

(ii) Describe the function of a motor neurone.

(2)

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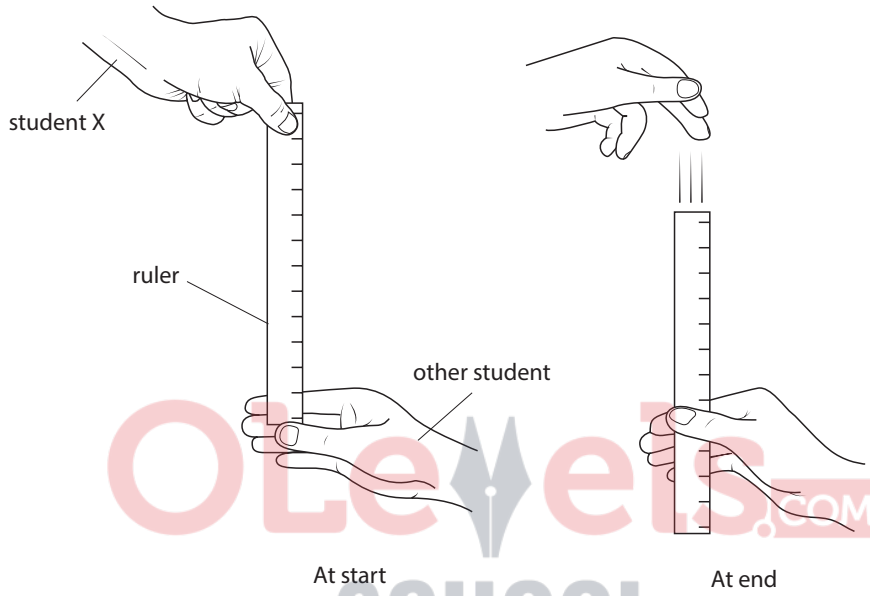
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10. Question 3 · part a.ii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.ii only.

(b) Student X uses this equipment to investigate the reaction time of another student.



Student X knows that the reaction time can be calculated using this formula.

$$\text{reaction time} = \sqrt{\frac{\text{measurement from ruler}}{5}}$$

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10. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 5 · part a · 3 marks

November 2020 Paper 1

Further practice · Attempt part a only.

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5 This question is about the nervous system.

(a) Describe what is meant by a reflex action.

(3)

(b) Diagram 1 shows a reflex arc.

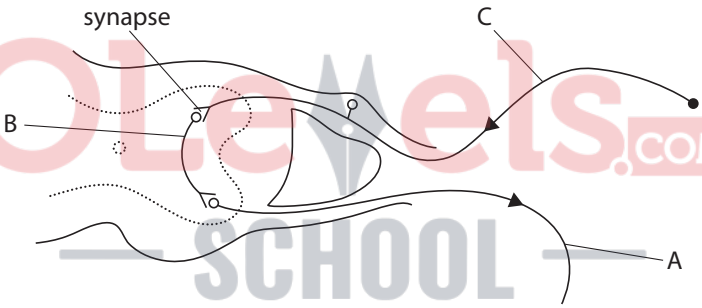


Diagram 1

Name the types of neurone labelled A, B and C.

(3)

A .....

B .....

C .....

11. Question 5 · part a · 3 marks · Working and answer

Working and answer

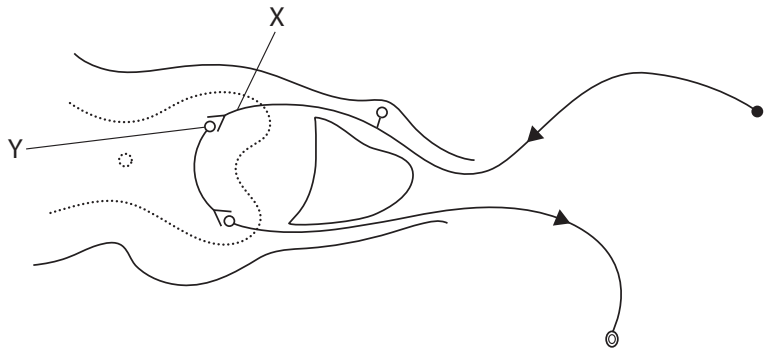
OLevels.com SCHOOL

12. Question 8 · part a · 3 marks

June 2021 Paper 1

Further practice · Attempt part a only.

8 The diagram shows a reflex arc involved in a reflex action.



(a) Describe the function of a reflex action.

(3)

(b) Using a line labelled Z, show on the diagram which part of the reflex arc would be connected to a receptor.

(1)

(c) Describe how the nerve impulse passes from position X to position Y in the reflex arc.

(6)

(Total for Question 8 = 10 marks)

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12. Question 8 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 3 · part a · 3 marks

June 2025 Paper 2

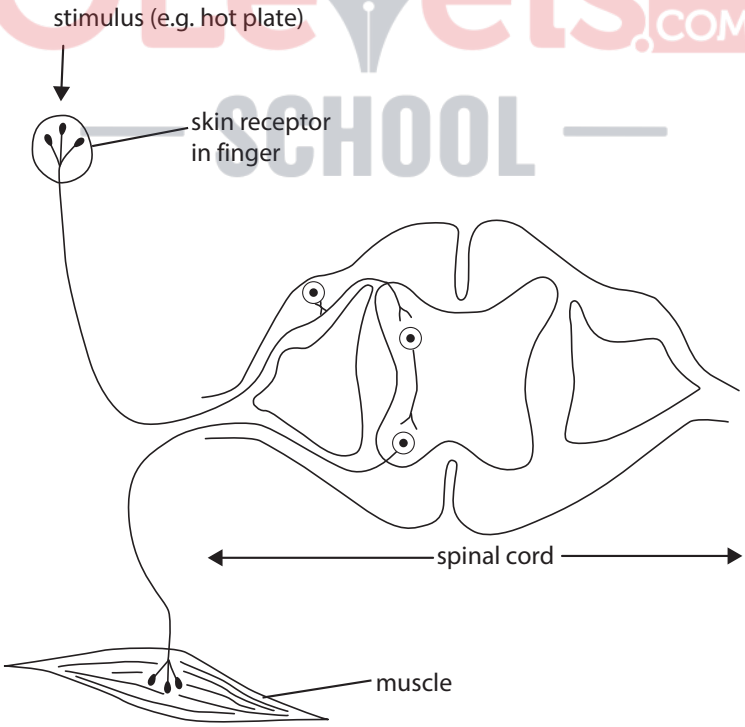
Further practice · Attempt part a only.

3 (a) Describe the meaning of the term **reflex action**.

(3)

(b) The diagram shows a reflex arc.

The reflex arc contains three types of neurone.



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P 7 8 7 6 9 A 0 6 2 0

13. Question 3 · part a · 3 marks · Working and answer

Working and answer

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Practice workbook · Describe how an impulse crosses a synapse

Naming the synapse, defining a neurotransmitter, and sequencing transmission from an arriving impulse to a new impulse in the next neurone, including through a reflex arc.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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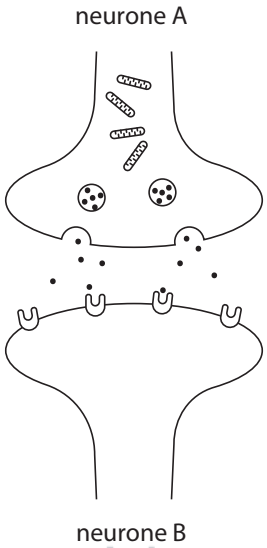
Type ID: HBI-EXPLAIN-SYNAPTIC-TRANSMISSION-SEQUENCE

1. Question 3 · part a.iii · 6 marks

November 2024 Paper 2

Worked-example retry · Attempt part a.iii only.

3 (a) The diagram shows the gap between two neurones, A and B.



(i) Name this gap. **(1)**

(ii) Draw an arrow on the diagram to show the direction of travel of an impulse. **(1)**

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P 7 5 9 5 9 A 0 8 2 4

1. Question 3 · part a.iii · 6 marks

November 2024 Paper 2

Worked-example retry · Attempt part a.iii only.

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(iii) Describe how a nerve impulse travels from one side of the gap to the other.

(6)

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(iv) Explain why an impulse can only travel in one direction across the gap.

(3)

1. Question 3 · part a.iii · 6 marks · Working and answer

Working and answer

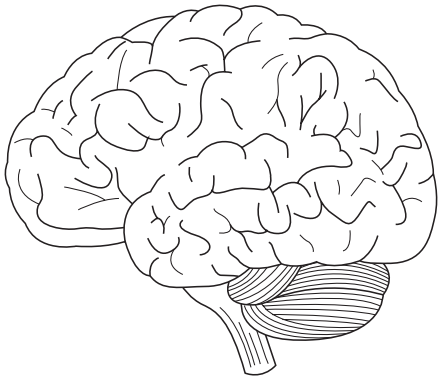
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2. Question 7 · part b.iii · 3 marks

June 2019 Paper 1

Further practice · Attempt part b.iii only.

7 (a) The diagram shows a human brain.



Add lines labelled X, Y and Z to the diagram to show the areas of the brain that control these functions.

(3)

X voluntary actions

Y balance

Z breathing rate

(b) Parkinson's disease affects the cells in the brain that help to control body movement.

The affected cells are unable to communicate effectively with neurones that cause muscles to contract.

(i) Name the type of neurone that causes muscles to contract.

(1)

(ii) Give one difference between neurones that cause muscles to contract and neurones that transmit nerve impulses from receptor organs.

(1)

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2. Question 7 · part b.iii · 3 marks

June 2019 Paper 1

Further practice · Attempt part b.iii only.

(iii) Explain how one neurone communicates with another neurone.

(3)

(c) Recent scientific research into the treatment of Parkinson's disease has involved the use of stem cells.

Explain how stem cells could be used to reduce the symptoms of Parkinson's disease.

(3)

(Total for Question 7 = 11 marks)

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2. Question 7 · part b.iii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 5 · part c.ii · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.ii only.

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5 This question is about the nervous system.

(a) Describe what is meant by a reflex action.

(3)

(b) Diagram 1 shows a reflex arc.

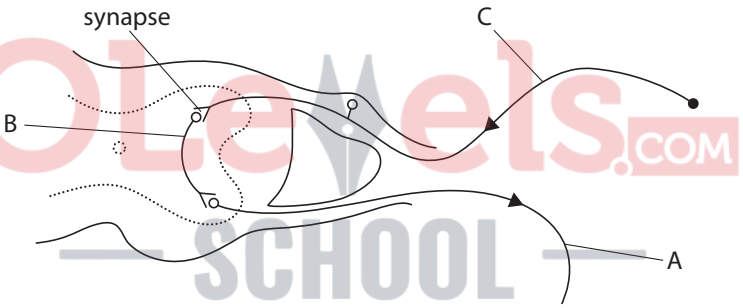


Diagram 1

Name the types of neurone labelled A, B and C.

(3)

A .....

B .....

C .....

3. Question 5 · part c.ii · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.ii only.

(c) Diagram 2 shows a neurone from a healthy brain and a neurone from the brain of someone with Alzheimer's disease.

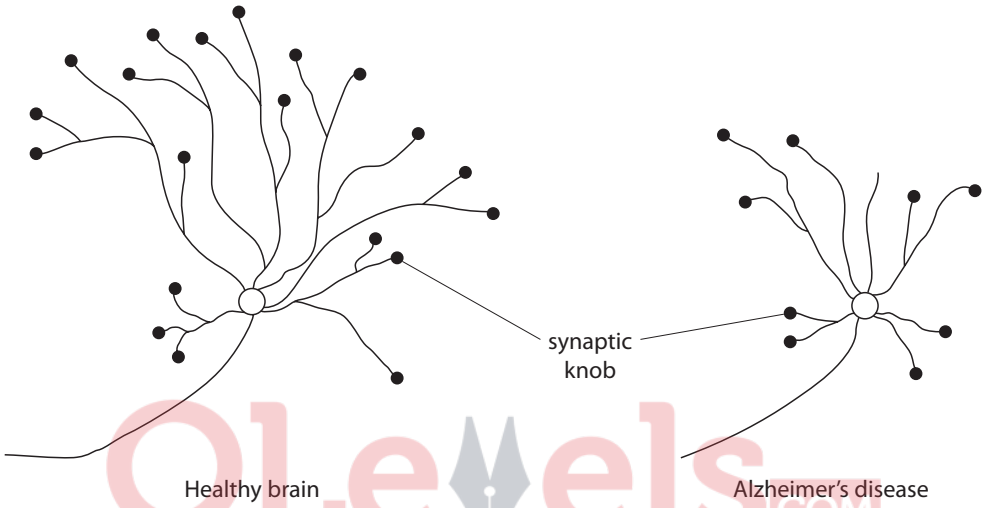


Diagram 2

(i) Give three differences between the two neurones. **(3)**

1 .....

.....

2 .....

.....

3 .....

.....

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P 6 2 0 5 1 A 0 1 4 2 4

3. Question 5 · part c.ii · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.ii only.

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(ii) Acetylcholine is a neurotransmitter.

Explain the importance of acetylcholine in the functioning of the brain.

(3)

(iii) One symptom of Alzheimer's disease is that less acetylcholine is produced in the brain.

Explain why there is less acetylcholine produced in people with Alzheimer's disease.

Use information from diagram 2 to help your answer.

(3)

(Total for Question 5 = 15 marks)

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3. Question 5 · part c.ii · 3 marks · Working and answer

Working and answer

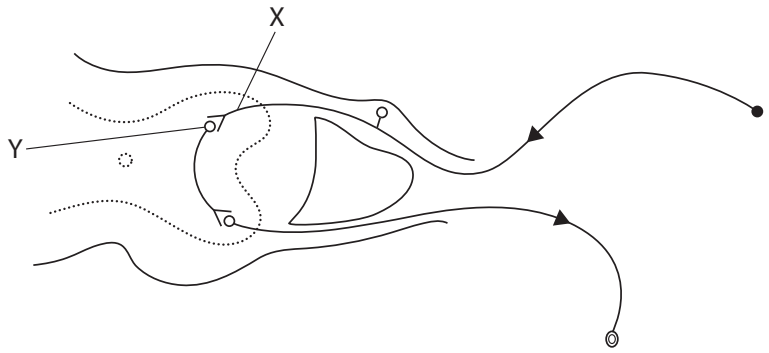
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4. Question 8 · part c · 6 marks

June 2021 Paper 1

Further practice · Attempt part c only.

8 The diagram shows a reflex arc involved in a reflex action.



(a) Describe the function of a reflex action.

(3)

(b) Using a line labelled Z, show on the diagram which part of the reflex arc would be connected to a receptor.

(1)

(c) Describe how the nerve impulse passes from position X to position Y in the reflex arc.

(6)

(Total for Question 8 = 10 marks)

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4. Question 8 · part c · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 6 · part b.ii · 4 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

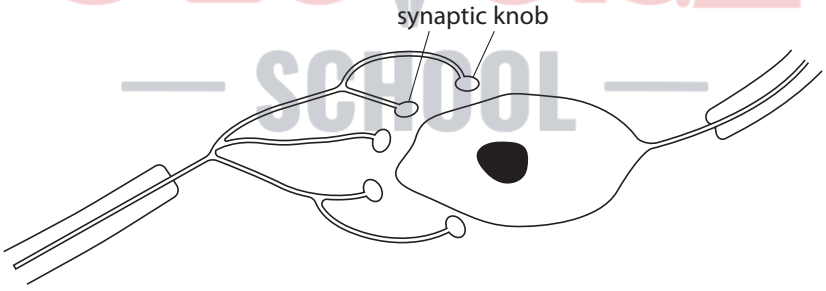
6 (a) Co-ordination of the activities of the body involves the nervous system and the hormonal system.

Complete the table to compare the two systems.

(3)

	Nervous system	Hormonal system
Route of transmission		
Type of signal		
Relative speed		

(b) The diagram shows a synapse.



(i) Draw an arrow on the diagram to show the direction an impulse travels.

(1)

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5. Question 6 · part b.ii · 4 marks

January 2022 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Explain how an impulse travels across a synapse.

(4)

(iii) Explain why an impulse can only travel in one direction across a synapse.

(4)

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(Total for Question 6 = 12 marks)

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5. Question 6 · part b.ii · 4 marks · Working and answer

Working and answer

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6. Question 3 · part b.i · 1 mark

June 2022 Paper 2

Further practice · Attempt part b.i only.

3 (a) The nervous system is adapted to transfer information rapidly from one part of the body to another.

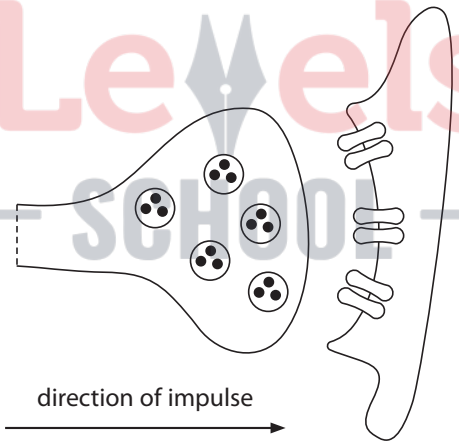
Complete the passage about the nervous system by giving the correct words.

(4)

A change in the external environment is detected by ..... in sense organs. These structures convert stimuli into electrical impulses which then travel along ..... neurones to the central nervous system.

The central nervous system coordinates the incoming information using ..... neurones. These neurones then transfer the electrical impulses to ..... neurones.

(b) The diagram shows the junction between two neurones.



(i) State the name of the junction between two neurones.

(1)

.....

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6. Question 3 · part b.i · 1 mark

June 2022 Paper 2

Further practice · Attempt part b.i only.

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(ii) Describe how an impulse reaching the end of one neurone produces an impulse in the next neurone.

(2)

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.....

(c) A nerve impulse travels at a speed of 120 metres per second.

Calculate the distance that this nerve impulse will travel in 5.0 minutes.

(2)

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distance = ... m

(Total for Question 3 = 9 marks)

Working and answer

7. Question 3 · part b.ii · 2 marks

June 2022 Paper 2

Further practice · Attempt part b.ii only.

3 (a) The nervous system is adapted to transfer information rapidly from one part of the body to another.

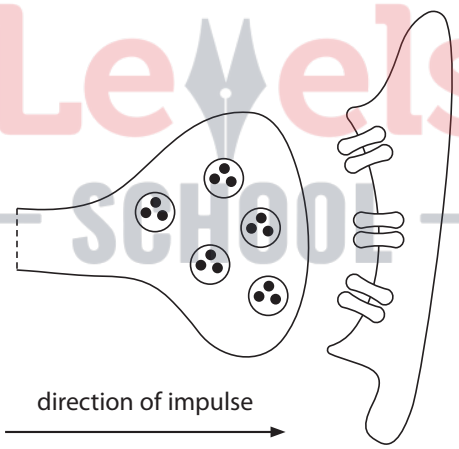
Complete the passage about the nervous system by giving the correct words.

(4)

A change in the external environment is detected by ..... in sense organs. These structures convert stimuli into electrical impulses which then travel along ..... neurones to the central nervous system.

The central nervous system coordinates the incoming information using ..... neurones. These neurones then transfer the electrical impulses to ..... neurones.

(b) The diagram shows the junction between two neurones.



(i) State the name of the junction between two neurones.

(1)

.....

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7. Question 3 · part b.ii · 2 marks

June 2022 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Describe how an impulse reaching the end of one neurone produces an impulse in the next neurone.

(2)

.....

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.....

(c) A nerve impulse travels at a speed of 120 metres per second.

Calculate the distance that this nerve impulse will travel in 5.0 minutes.

(2)

distance = ... m

(Total for Question 3 = 9 marks)

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7. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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8. Question 5 · part a.i · 2 marks

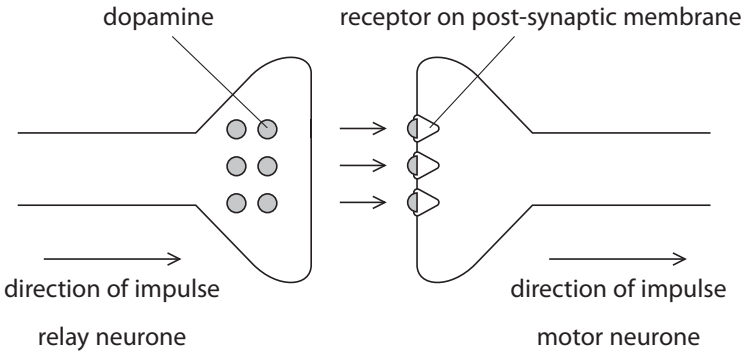
June 2023 Paper 2R

Further practice · Attempt part a.i only.

5 (a) Dopamine is a neurotransmitter produced by nerve cells in the brain.

The nerve cells that produce dopamine are involved in stimulating movement and increasing heart rate.

The diagram shows how dopamine travels from a relay neurone to a motor neurone in the brain.



(i) Describe what is meant by a neurotransmitter.

(2)

(ii) Describe how information is transmitted from the relay neurone to the motor neurone.

(2)

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P 7 2 5 8 8 A 0 1 4 2 4

8. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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9. Question 5 · part a.ii · 2 marks

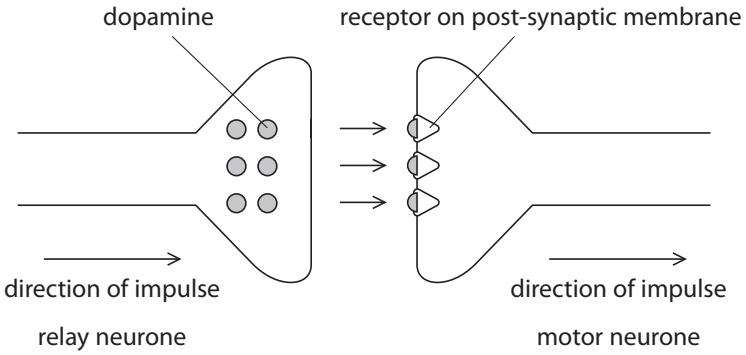
June 2023 Paper 2R

Further practice · Attempt part a.ii only.

5 (a) Dopamine is a neurotransmitter produced by nerve cells in the brain.

The nerve cells that produce dopamine are involved in stimulating movement and increasing heart rate.

The diagram shows how dopamine travels from a relay neurone to a motor neurone in the brain.



(i) Describe what is meant by a neurotransmitter.

(2)

(ii) Describe how information is transmitted from the relay neurone to the motor neurone.

(2)

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9. Question 5 · part a.ii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a.ii only.

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(iii) Suggest how an excess of dopamine will affect bodily functions.

(2)

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.....

.....

(b) In 2017 in the United Kingdom, 145 000 people were diagnosed with Parkinson's disease.

This figure is expected to have an 18% increase by 2025.

(i) Give two symptoms of Parkinson's disease.

(2)

1 .....

.....

2 .....

.....

(ii) Calculate the number of people expected to be diagnosed with Parkinson's disease in 2025.

(2)

number of people = ...

(Total for Question 5 = 10 marks)

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9. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

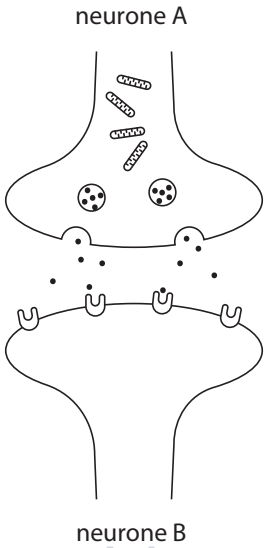
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10. Question 3 · part a.i · 1 mark

November 2024 Paper 2

Further practice · Attempt part a.i only.

3 (a) The diagram shows the gap between two neurones, A and B.



(i) Name this gap. **(1)**

.....

(ii) Draw an arrow on the diagram to show the direction of travel of an impulse. **(1)**

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P 7 5 9 5 9 A 0 8 2 4

10. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain how the eye focuses on near and distant objects

Ciliary muscle, suspensory ligament and lens changes for near and distant focus, as tables, drawings, data-based or written explanations.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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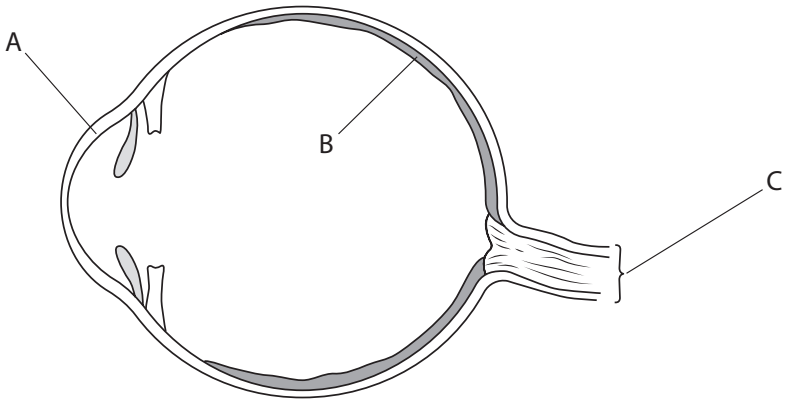
Type ID: HBI-EXPLAIN-LENS-ACCOMMODATION

1. Question 2 · part b · 4 marks

June 2025 Paper 1R

Worked-example retry · Attempt part b only.

2 The diagram shows a section through the eye with the lens missing.



(a) (i) Complete the diagram by drawing in the lens and suspensory ligaments as they would appear when the eye is looking at a near object.

(2)

(ii) Name the parts labelled A, B and C.

(3)

A .....

B .....

C .....

(b) A person is looking at a near object.

Describe the changes that occur in the eye if the person then looks at a distant object.

(4)

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.....

(Total for Question 2 = 9 marks)

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1. Question 2 · part b · 4 marks · Working and answer

Working and answer

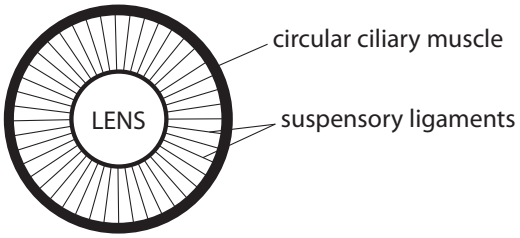
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2. Question 5 · part a · 3 marks

January 2022 Paper 2

Further practice · Attempt part a only.

5 (a) The diagram shows a front view of structures in the eye that help to form an image on the retina.



The three structures shown in the diagram change when focusing on near and distant objects.

Complete the table by putting ticks (✓) in the correct boxes to show the condition of each structure of the eye when focusing on near and distant objects.

(3)

	Focusing on a near object	Focusing on a distant object
Lens	thin ☐ thick ☐	thin ☐ thick ☐
Circular ciliary muscle	contracted ☐ relaxed ☐	contracted ☐ relaxed ☐
Suspensory ligament	stretched ☐ loose ☐	stretched ☐ loose ☐

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2. Question 5 · part a · 3 marks

January 2022 Paper 2

Further practice · Attempt part a only.

(b) Humans have stereoscopic vision.

Explain what is meant by the term **stereoscopic vision**.

(3)

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2. Question 5 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 4 · part b.ii · 1 mark

June 2023 Paper 1

Further practice · Attempt part b.ii only.

4 The eye contains receptors that can detect light.

(a) Name two other senses.

(2)

1 .....

2 .....

(b) The table shows how the focal length of the eye lens changes as the thickness of the lens changes.

Thickness of lens in mm	Focal length in mm
10	12.5
9	14.0
8	15.5
7	18.0
6	21.0
5	25.0

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(i) Describe the relationship between the thickness of the lens and its focal length.

(2)

.....

.....

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.....

(ii) Give the effect on the diameter of the lens as its thickness increases.

(1)

.....

.....

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3. Question 4 · part b.ii · 1 mark

June 2023 Paper 1

Further practice · Attempt part b.ii only.

(iii) A person is looking at a distant object and then looks at a near object.

Use the data in the table to explain the changes that occur in the eye so the person is able to form an image of the near object on the retina.

(4)

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3. Question 4 · part b.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 4 · part b.iii · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.iii only.

4 The eye contains receptors that can detect light.

(a) Name two other senses.

(2)

1 .....

2 .....

(b) The table shows how the focal length of the eye lens changes as the thickness of the lens changes.

Thickness of lens in mm	Focal length in mm
10	12.5
9	14.0
8	15.5
7	18.0
6	21.0
5	25.0

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(i) Describe the relationship between the thickness of the lens and its focal length.

(2)

.....

.....

.....

.....

(ii) Give the effect on the diameter of the lens as its thickness increases.

(1)

.....

.....

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4. Question 4 · part b.iii · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.iii only.

(iii) A person is looking at a distant object and then looks at a near object.

Use the data in the table to explain the changes that occur in the eye so the person is able to form an image of the near object on the retina.

(4)

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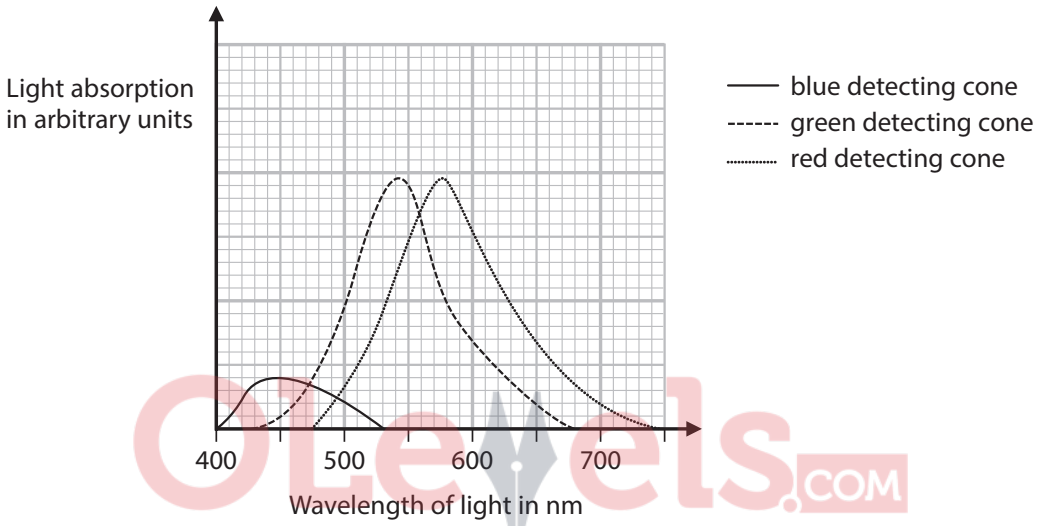
OLevels.com SCHOOL

4. Question 4 · part b.iii · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.iii only.

(c) Cone cells are found in the retina of the eye. They detect colour. The graph shows the absorption of different wavelengths of light by three types of cone cell.



(i) Determine the wavelength of light at which a red detecting cone absorbs most light.

(1)

wavelength of light = ... nm

(ii) Determine the maximum difference in the wavelengths of the light that can be absorbed by blue detecting cones.

(2)

difference in wavelengths = ... nm

(iii) Suggest why it is better to have three types of cone rather than just one type of cone.

(2)

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(Total for Question 4 = 14 marks)

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4. Question 4 · part b.iii · 4 marks · Working and answer

Working and answer

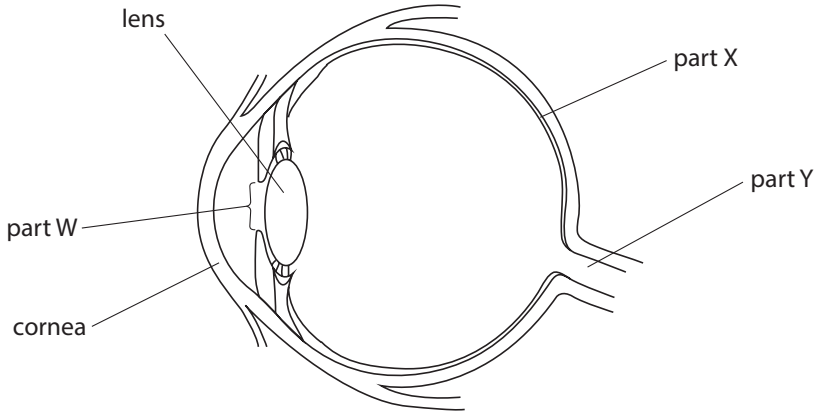
OLevels.com SCHOOL

5. Question 3 · part a.iii · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.iii only.

3 (a) The diagram shows the human eye.



(i) Name part W and part X.

(2)

W .....

X .....

(ii) Describe the function of part Y.

(2)

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(iii) Explain the change in the shape of the lens when the eye views a distant object.

(2)

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5. Question 3 · part a.iii · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.iii only.

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(iv) Describe how corneal transplants help to restore normal vision for people with corneal disease.

(2)

(b) The iris reflex controls the amount of light entering the eye.

Describe a method that could be used to investigate how the eye responds to different light intensities.

(4)

(Total for Question 3 = 12 marks)

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5. Question 3 · part a.iii · 2 marks · Working and answer

Working and answer

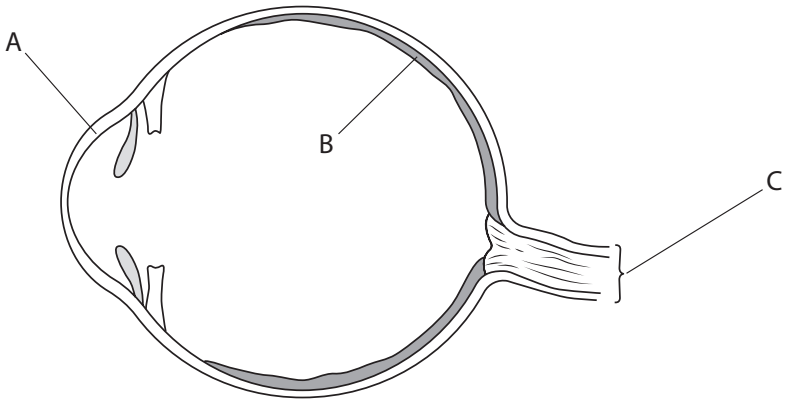
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6. Question 2 · part a.i · 2 marks

June 2025 Paper 1R

Further practice · Attempt part a.i only.

2 The diagram shows a section through the eye with the lens missing.



(a) (i) Complete the diagram by drawing in the lens and suspensory ligaments as they would appear when the eye is looking at a near object.

(2)

(ii) Name the parts labelled A, B and C.

(3)

A .....

B .....

C .....

(b) A person is looking at a near object.

Describe the changes that occur in the eye if the person then looks at a distant object.

(4)

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(Total for Question 2 = 9 marks)

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6. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

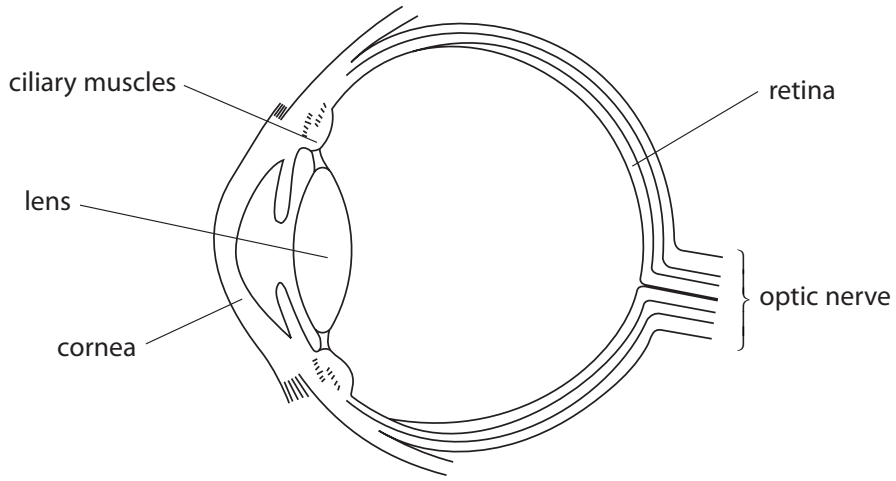
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7. Question 6 · part whole · 7 marks

June 2022 Paper 1R

Further practice · Attempt part whole only.

6 The diagram shows a section through the human eye.



Describe the role of the labelled structures in seeing a clear image of an object.

(Total for Question 6 = 7 marks)

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7. Question 6 · part whole · 7 marks · Working and answer

Working and answer

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Practice workbook · Explain eye defects and how they are corrected or treated

Short sight, long sight, astigmatism, cataracts and corneal disease: where the image falls, cause, effect on vision, lenses, surgery and transplant risks.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EYE-DEFECTS-AND-TREATMENT

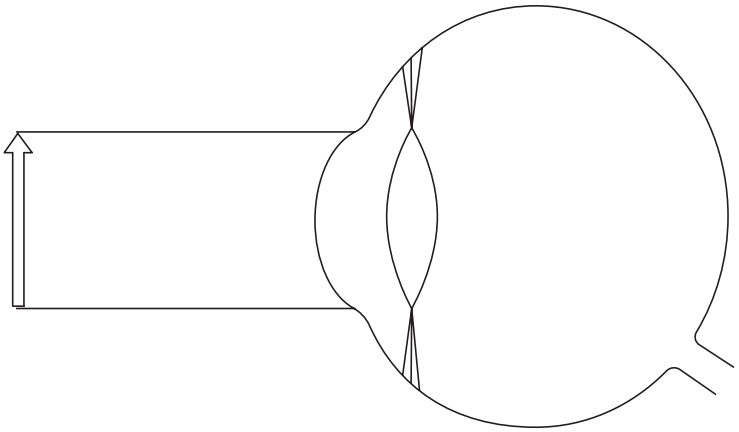
1. Question 8 · part b.iii · 2 marks

November 2021 Paper 1

Worked-example retry · Attempt part b.iii only.

8 The diagram shows a section through the eye.

The arrow is a near object.



(a) (i) Complete the diagram by continuing the rays from the arrow to show how the image of the arrow appears on the retina.

(5)

(ii) Describe the function of the optic nerve in allowing a person to see the image of the arrow.

(2)

(b) (i) State where the image of the arrow would be formed for someone with the condition of long sight.

(1)

(ii) Give a feature of the eye that can lead to long sight.

(1)

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1. Question 8 · part b.iii · 2 marks

November 2021 Paper 1

Worked-example retry · Attempt part b.iii only.

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(iii) Describe a method of treating long sight.

(2)

.....

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.....

.....

(c) Corneal transplants can be used to treat eye defects.

Suggest two risks associated with corneal transplants.

(2)

1 .....

2 .....

(Total for Question 8 = 13 marks)

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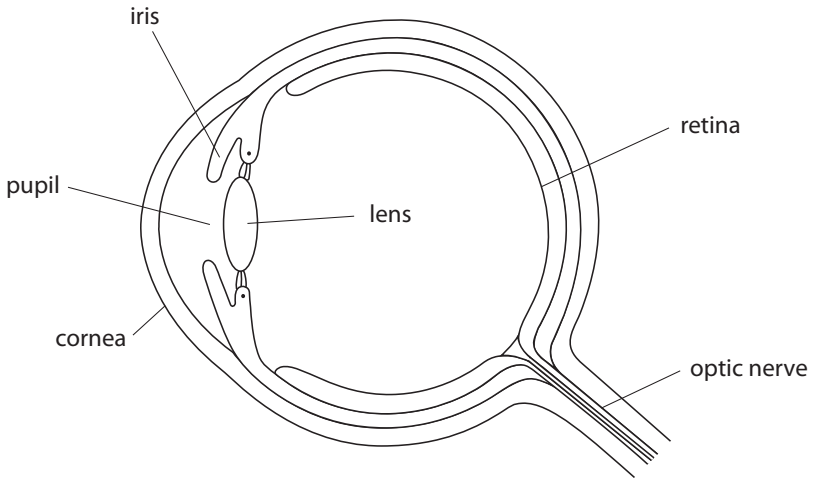
Working and answer

2. Question 3 · part c · 2 marks

June 2019 Paper 1

Further practice · Attempt part c only.

3 The human eye contains structures that help to focus light on the retina.



(a) Which structures help to focus light on the retina? **(1)**

- ☒ **A** cornea and lens
- ☒ **B** iris and cornea
- ☒ **C** iris and pupil
- ☒ **D** pupil and lens

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2. Question 3 · part c · 2 marks

June 2019 Paper 1

Further practice · Attempt part c only.

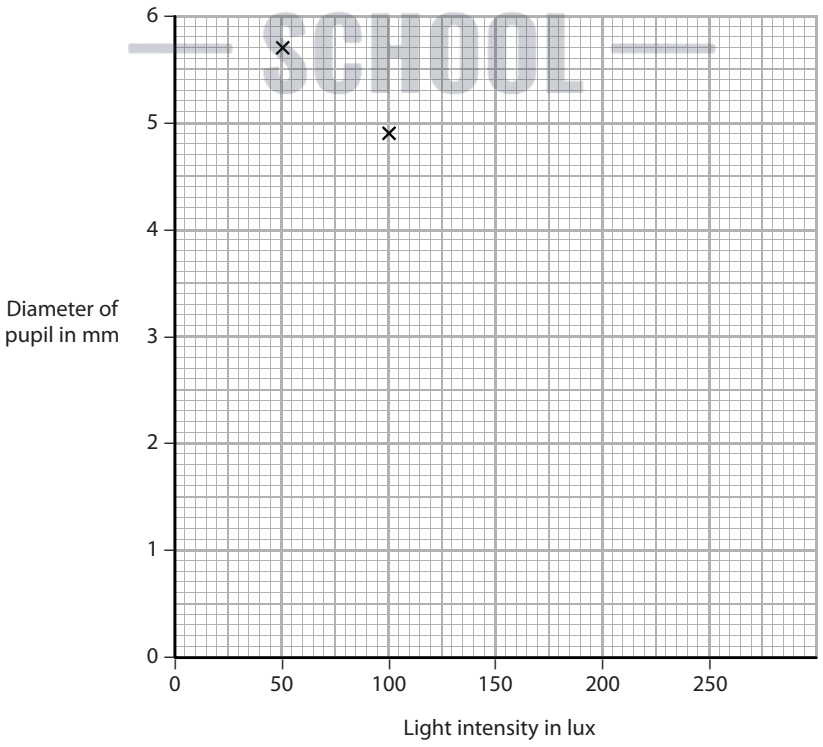
(b) The intensity of light entering the eye can be measured in a unit called lux.

The table shows the results of an investigation on how light intensity affects the diameter of the pupil in the eye.

Light intensity in lux	Diameter of pupil in mm
50	5.7
100	4.9
150	2.7
200	3.2
250	2.3

(i) Complete the graph by plotting the results for 150, 200 and 250 lux. (1)

(ii) Draw a straight line of best fit. (1)



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8

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2. Question 3 · part c · 2 marks

June 2019 Paper 1

Further practice · Attempt part c only.

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(iii) Explain which result is anomalous.

(2)

(iv) State the effect of light intensity on the diameter of the pupil.

(1)

(c) Short-sightedness is a common eye condition.

Explain how short-sightedness affects vision.

(2)

(Total for Question 3 = 8 marks)

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2. Question 3 · part c · 2 marks · Working and answer

Working and answer

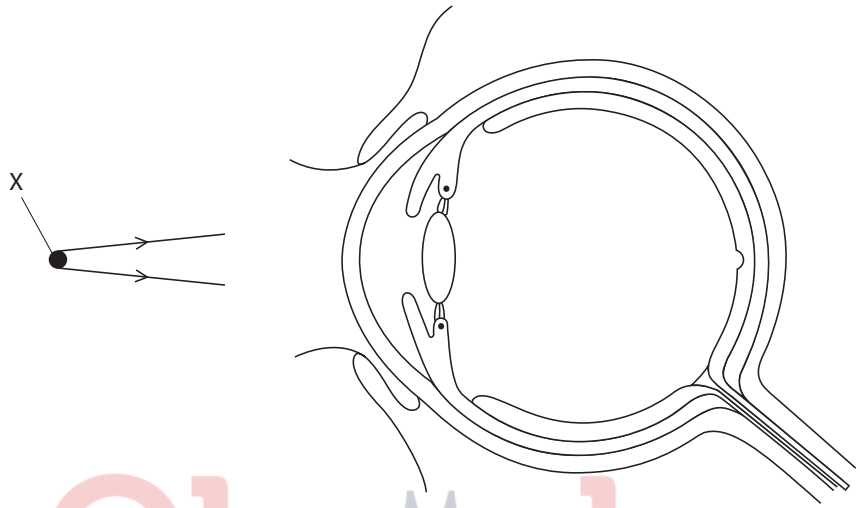
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3. Question 1 · part c · 3 marks

June 2019 Paper 2R

Further practice · Attempt part c only.

1 The diagram shows a section through an eye.



(a) The table lists descriptions of parts of the eye.

Use the letters, A, B, C, D and E, to label the parts of the eye on the diagram.

(5)

Description of part	Letter
the layer that contains light sensitive cells	A
the muscles that change lens shape	B
the structure that controls the amount of light entering the eye	C
the structure that carries nerve impulses away from the eye	D
the structure that protects the eye from dust and grit	E

(b) A person looks at object X.

Complete the diagram by continuing the rays of light to show how the image of object X is formed on the retina.

(3)

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3. Question 1 · part c · 3 marks

June 2019 Paper 2R

Further practice · Attempt part c only.

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(c) Name three defects of the eye that can be corrected by using glasses.

(3)

1 .....

2 .....

3 .....

(Total for Question 1 = 11 marks)

Working and answer

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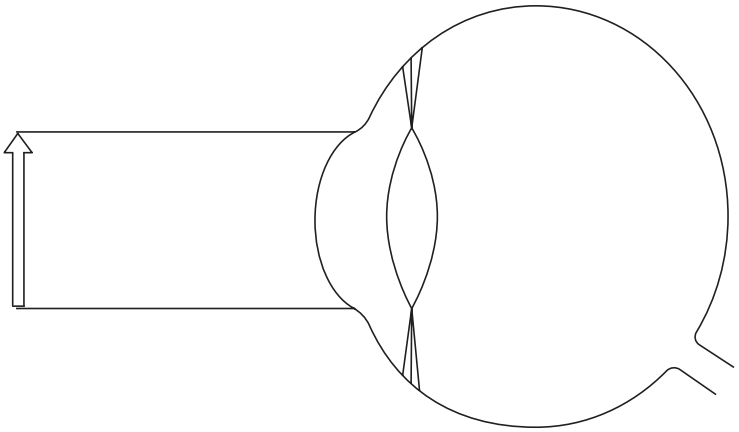
4. Question 8 · part b.i · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.i only.

8 The diagram shows a section through the eye.

The arrow is a near object.



(a) (i) Complete the diagram by continuing the rays from the arrow to show how the image of the arrow appears on the retina.

(5)

(ii) Describe the function of the optic nerve in allowing a person to see the image of the arrow.

(2)

(b) (i) State where the image of the arrow would be formed for someone with the condition of long sight.

(1)

(ii) Give a feature of the eye that can lead to long sight.

(1)

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4. Question 8 · part b.i · 1 mark · Working and answer

Working and answer

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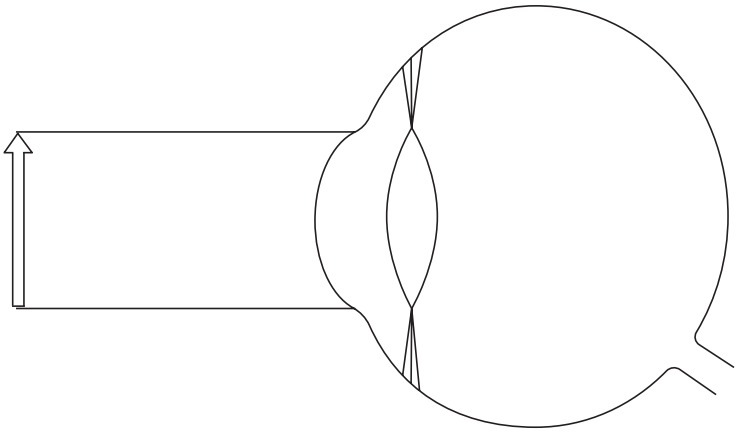
5. Question 8 · part b.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.ii only.

8 The diagram shows a section through the eye.

The arrow is a near object.



(a) (i) Complete the diagram by continuing the rays from the arrow to show how the image of the arrow appears on the retina.

(5)

(ii) Describe the function of the optic nerve in allowing a person to see the image of the arrow.

(2)

(b) (i) State where the image of the arrow would be formed for someone with the condition of long sight.

(1)

(ii) Give a feature of the eye that can lead to long sight.

(1)

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P 6 7 0 6 0 R R A 0 1 8 2 0

5. Question 8 · part b.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.ii only.

(iii) Describe a method of treating long sight.

(2)

.....

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(c) Corneal transplants can be used to treat eye defects.

Suggest two risks associated with corneal transplants.

(2)

1 .....

2 .....

(Total for Question 8 = 13 marks)

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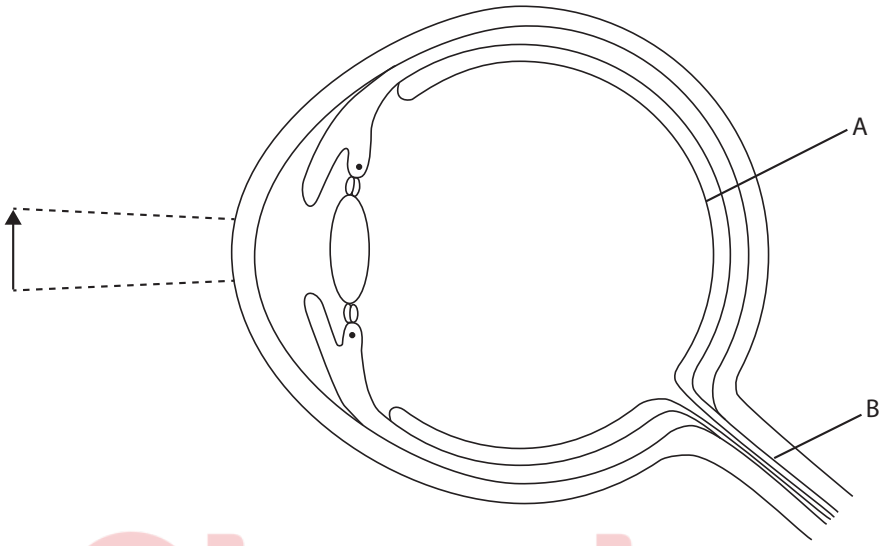
Working and answer

6. Question 6 · part c · 2 marks

June 2025 Paper 2

Further practice · Attempt part c only.

6 The diagram shows a section through the eye.



(a) (i) Describe the functions of structures A and B. **(4)**

A .....

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B .....

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(ii) Complete the diagram to show how an image of the arrow is formed at the back of the eye.

(4)

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P 7 8 7 6 9 A 0 1 4 2 0

6. Question 6 · part c · 2 marks

June 2025 Paper 2

Further practice · Attempt part c only.

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(b) Describe a simple experiment to show that stereoscopic vision would help a person to catch a ball thrown towards them.

(4)

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(c) Describe what is meant by the term **astigmatism**.

(2)

(Total for Question 6 = 14 marks)

6. Question 6 · part c · 2 marks · Working and answer

Working and answer

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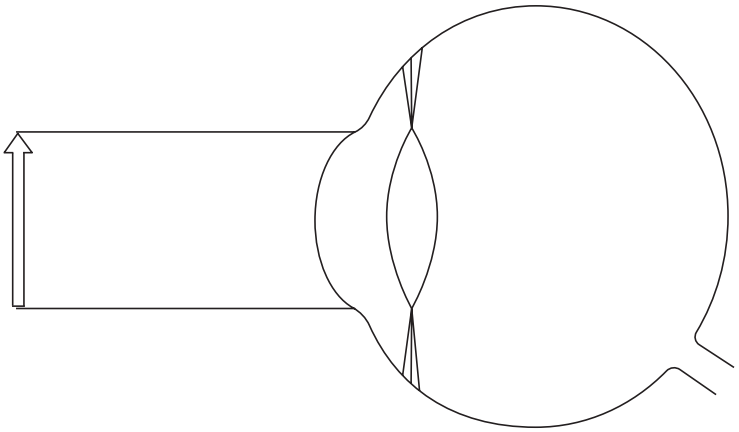
7. Question 8 · part c · 2 marks

November 2021 Paper 1

Further practice · Attempt part c only.

8 The diagram shows a section through the eye.

The arrow is a near object.



(a) (i) Complete the diagram by continuing the rays from the arrow to show how the image of the arrow appears on the retina.

(5)

(ii) Describe the function of the optic nerve in allowing a person to see the image of the arrow.

(2)

(b) (i) State where the image of the arrow would be formed for someone with the condition of long sight.

(1)

(ii) Give a feature of the eye that can lead to long sight.

(1)

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7. Question 8 · part c · 2 marks

November 2021 Paper 1

Further practice · Attempt part c only.

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(iii) Describe a method of treating long sight.

(2)

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(c) Corneal transplants can be used to treat eye defects.

Suggest two risks associated with corneal transplants.

(2)

1 .....

2 .....

(Total for Question 8 = 13 marks)

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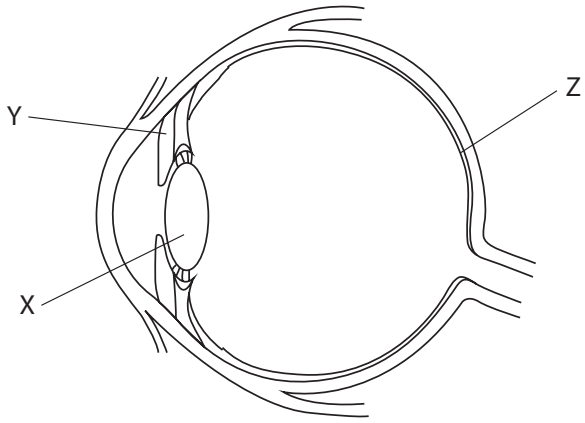
Working and answer

8. Question 1 · part b.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.i only.

1 (a) The diagram shows a human eye.



(i) Draw one line from each structure to its correct function.

(3)

Structure	Function
structure X	refracts light rays
structure Y	protects the eye from pathogens
structure Z	detects light rays
	controls the shape of the lens
	transmits electrical impulses to the brain
	controls the size of the pupil

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8. Question 1 · part b.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.i only.

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(ii) Most humans have stereoscopic vision.

Which of these statements occurs because of stereoscopic vision?

(1)

- ☒ **A** being able to judge depth
- ☒ **B** being able to see in colour
- ☒ **C** having a wide field of view
- ☒ **D** to be able to see near and far objects

(b) The diagram shows a healthy eye and an eye with a cataract.



Healthy eye Eye with cataract

(i) Describe the cause of cataracts.

(2)

.....

.....

.....

(ii) State one treatment for cataracts.

(1)

.....

(iii) It is estimated that 39 million people in the world are blind.

Of these people, 43% are blind because of cataracts.

Determine the estimated number of people who are blind because of cataracts.

(2)

number of people = ...

(iv) Some people are blind because of a vitamin deficiency.

Name the vitamin that is needed to prevent this type of blindness.

(1)

.....

(Total for Question 1 = 10 marks)

8. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

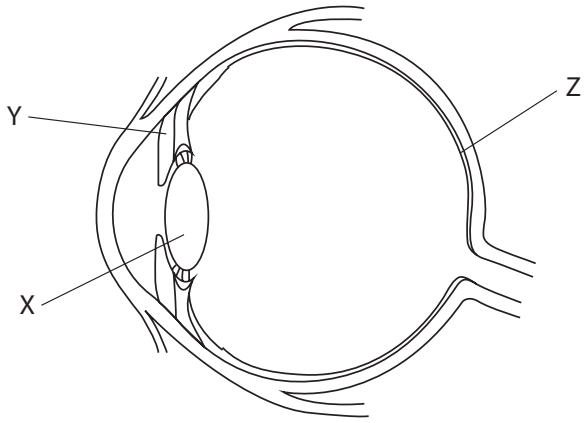
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9. Question 1 · part b.ii · 1 mark

January 2023 Paper 1

Further practice · Attempt part b.ii only.

1 (a) The diagram shows a human eye.



(i) Draw one line from each structure to its correct function.

(3)

Structure	Function
structure X	refracts light rays
structure Y	protects the eye from pathogens
structure Z	detects light rays
	controls the shape of the lens
	transmits electrical impulses to the brain
	controls the size of the pupil

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9. Question 1 · part b.ii · 1 mark

January 2023 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Most humans have stereoscopic vision.

Which of these statements occurs because of stereoscopic vision?

(1)

- ☒ **A** being able to judge depth
- ☒ **B** being able to see in colour
- ☒ **C** having a wide field of view
- ☒ **D** to be able to see near and far objects

(b) The diagram shows a healthy eye and an eye with a cataract.



Healthy eye Eye with cataract

(i) Describe the cause of cataracts.

(2)

.....

.....

.....

(ii) State one treatment for cataracts.

(1)

.....

(iii) It is estimated that 39 million people in the world are blind.

Of these people, 43% are blind because of cataracts.

Determine the estimated number of people who are blind because of cataracts.

(2)

number of people = ...

(iv) Some people are blind because of a vitamin deficiency.

Name the vitamin that is needed to prevent this type of blindness.

(1)

.....

(Total for Question 1 = 10 marks)

9. Question 1 · part b.ii · 1 mark · Working and answer

Working and answer

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10. Question 3 · part c · 1 mark

June 2024 Paper 2

Further practice · Attempt part c only.

3 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

People make lifestyle choices that can sometimes have a negative effect on their health. People who drink too much alcohol are at risk of brain and liver damage, certain types of cancer and mental illness. Liver damage can affect the digestion of fat. Alcohol can also affect the eyes, increasing the risk of developing cataracts that cause the lenses in the eyes to become cloudy. Alcohol is also a depressant. It increases the activity of a chemical in parts of the brain that control balance, speech, motor skills and memory. In addition, alcohol suppresses an area of the brain called the amygdala. The amygdala is responsible for the fight or flight function which prepares the body for action, initiating the release of stress hormones. These hormones increase glucose usage by the brain and other parts of the body.

When a person stops drinking alcohol after years of drinking a lot of alcohol, the amygdala in the brain becomes fully active again, which is the cause of all forms of alcohol withdrawal. Over time, if a person regularly drinks heavily, the central nervous system gets used to the suppressing effect of the alcohol, which means the brain and body are affected if the alcohol level suddenly drops. An individual can go straight into fight or flight mode as the alcohol leaves the system and levels of stress hormones begin to return to normal. Symptoms of withdrawal can be as severe as seizures and hallucinations or less severe symptoms such as loss of appetite and insomnia (an inability to sleep).

(a) Drinking alcohol and diet are two lifestyle choices.

Describe how a lifestyle choice, other than drinking alcohol and diet, can have a harmful effect on health (Lines 1 and 2).

(2)

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10. Question 3 · part c · 1 mark

June 2024 Paper 2

Further practice · Attempt part c only.

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(b) Explain how drinking alcohol affects the digestion of fat (Lines 3 to 4).

(2)

(c) State how cataracts affect vision (Lines 4 to 5).

(1)

(d) Name a part of the brain, other than the amygdala, that is affected by alcohol (Lines 5 to 7).

(1)

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(e) Explain how alcohol acts as a depressant on the nervous system (Line 5).

(3)

10. Question 3 · part c · 1 mark · Working and answer

Working and answer

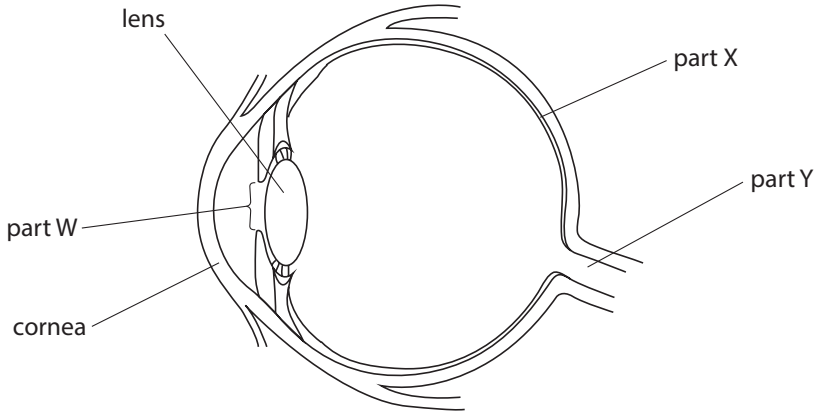
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11. Question 3 · part a.iv · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.iv only.

3 (a) The diagram shows the human eye.



(i) Name part W and part X.

(2)

W .....

X .....

(ii) Describe the function of part Y.

(2)

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(iii) Explain the change in the shape of the lens when the eye views a distant object.

(2)

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11. Question 3 · part a.iv · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.iv only.

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(iv) Describe how corneal transplants help to restore normal vision for people with corneal disease.

(2)

(b) The iris reflex controls the amount of light entering the eye.

Describe a method that could be used to investigate how the eye responds to different light intensities.

(4)

(Total for Question 3 = 12 marks)

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11. Question 3 · part a.iv · 2 marks · Working and answer

Working and answer

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Practice workbook · Explain the effects and dangers of alcohol and other drugs

Alcohol slowing reactions and driving, its absorption and liver damage; dangers of cocaine and other drugs; caffeine and painkiller data on speed of effect.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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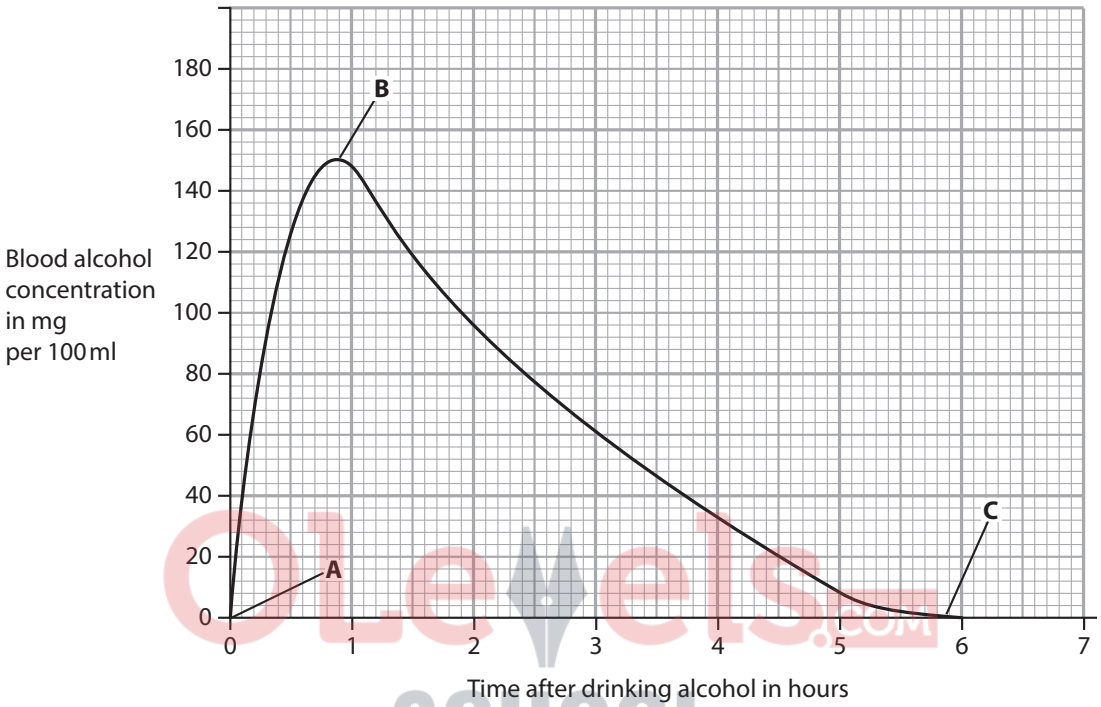
Type ID: HBI-EXPLAIN-DRUG-AND-ALCOHOL-EFFECTS

1. Question 6 · part b.ii · 4 marks

June 2025 Paper 1

Worked-example retry · Attempt part b.ii only.

6 The graph shows the changes in blood alcohol concentration of a person who drinks alcohol.



(a) (i) Explain the shape of the graph between points A and B.

(2)

(ii) Explain the shape of the graph between points B and C.

(3)

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1. Question 6 · part b.ii · 4 marks

June 2025 Paper 1

Worked-example retry · Attempt part b.ii only.

(b) In the United Kingdom the legal limit to drive is less than 80 mg per 100 ml of alcohol in the blood.

(i) The person shown in the graph drinks alcohol and the alcohol level in the blood goes above the legal limit to drive.

Determine how many hours after drinking alcohol it would be legal for the person shown in the graph to drive.

(1)

(ii) Explain why it is against the law to drive with high levels of alcohol in the blood.

(4)

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1. Question 6 · part b.ii · 4 marks

June 2025 Paper 1

Worked-example retry · Attempt part b.ii only.

(c) Alcohol reduces the production of ADH.

Explain the effect on the urine produced by a person drinking alcohol.

(5)

(Total for Question 6 = 15 marks)

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1. Question 6 · part b.ii · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 7 · part b.ii · 2 marks

June 2019 Paper 2

Further practice · Attempt part b.ii only.

7 The volume of alcohol in alcoholic drinks can be measured in units.

One unit is equivalent to 10 cm^3 of pure alcohol.

The diagram shows a bottle of wine with 14% alcohol content and a glass containing 250 cm^3 of this wine.



250 cm^3 glass of Ruby Red wine

(a) Calculate the number of units of alcohol in the glass of wine.

(2)

number of units = ...

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2. Question 7 · part b.ii · 2 marks

June 2019 Paper 2

Further practice · Attempt part b.ii only.

(b) It is illegal for a person to drive a motor vehicle if they have more than the legal limit of alcohol in their bloodstream.

It takes one hour for the body to process one unit of alcohol so that there is no alcohol left in the bloodstream.

(i) A person drinks three units of alcohol. This puts her at the legal limit.

Calculate the time it would take before a person can legally drive after drinking four glasses of Ruby Red wine.

(2)

time = ... hours

(ii) Suggest why it is illegal for people to drive a motor vehicle with more than the legal limit of alcohol in their bloodstream.

(2)

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O Levels .COM SCHOOL

2. Question 7 · part b.ii · 2 marks

June 2019 Paper 2

Further practice · Attempt part b.ii only.

(c) Drinking a lot of alcohol over a long period of time can cause a condition called cirrhosis.

This condition prevents the cells of the liver from functioning normally.

Explain the effect of cirrhosis on the digestion of food molecules.

(5)

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(Total for Question 7 = 11 marks)

2. Question 7 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 9 · part a · 3 marks

June 2022 Paper 2

Further practice · Attempt part a only.

9 (a) Hepatitis is a disease of the liver.

This disease can be caused by drinking too much alcohol.

Explain the damaging effects of alcohol on the liver.

(3)

(b) Hepatitis B is one type of virus that can cause hepatitis.

Describe the reproduction of a hepatitis B virus.

(6)

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3. Question 9 · part a · 3 marks · Working and answer

Working and answer

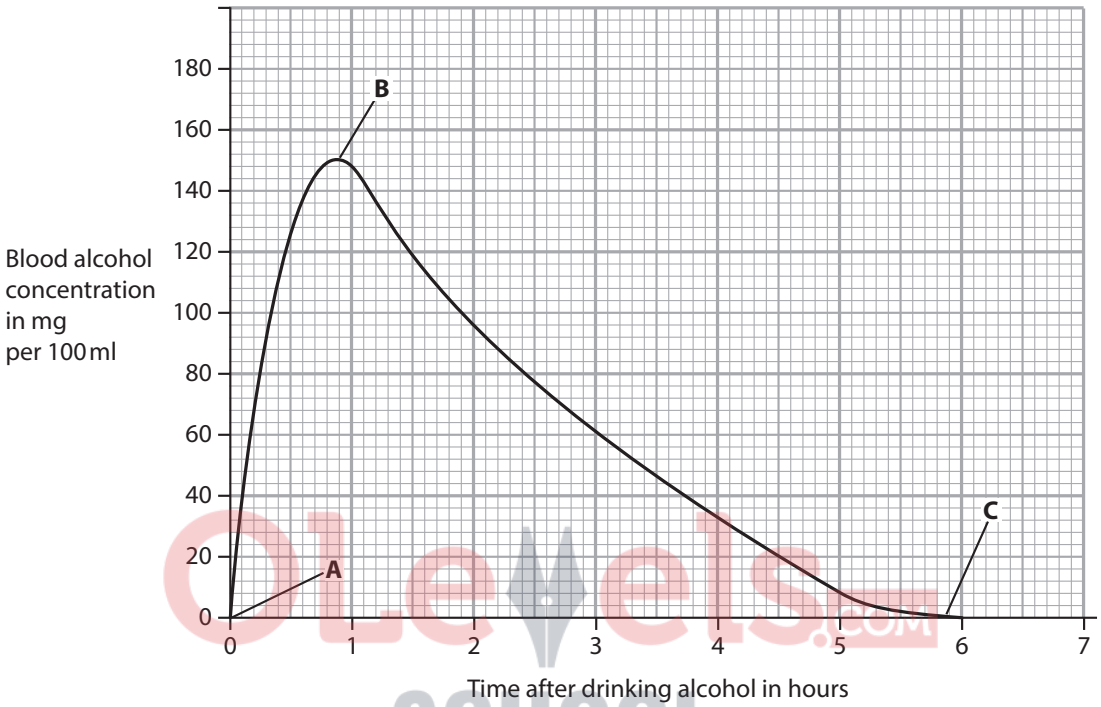
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4. Question 6 · part a.i · 2 marks

June 2025 Paper 1

Further practice · Attempt part a.i only.

6 The graph shows the changes in blood alcohol concentration of a person who drinks alcohol.



(a) (i) Explain the shape of the graph between points A and B.

(2)

(ii) Explain the shape of the graph between points B and C.

(3)

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4. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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5. Question 4 · part b · 3 marks

November 2020 Paper 2

Further practice · Attempt part b only.

4 Aspirin is the active ingredient used in some tablets to relieve pain.

Aspirin becomes acidic when it dissolves in the body. The faster aspirin dissolves, the faster it can relieve pain.

A student investigates if it is better to mix sucrose or starch with the active ingredient to make an aspirin tablet.

This is the student's method.

- add distilled water to three beakers, A, B and C
- add 1 aspirin tablet to each beaker
- add 5 g of sucrose to beaker B
- add 5 g of starch to beaker C
- measure the pH of the contents of each beaker at the start of the investigation
- measure the pH of the contents of each beaker every minute for 5 minutes

The table shows the student's results.

Beaker	pH of contents of beaker					
	at start	1 minute	2 minutes	3 minutes	4 minutes	5 minutes
A – aspirin	7.6	7.1	6.6	6.2	5.8	5.4
B – aspirin and sucrose	7.2	6.7	6.3	5.7	5.2	4.7
C – aspirin and starch	7.1	7.2	7.3	7.3	7.4	7.4

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5. Question 4 · part b · 3 marks

November 2020 Paper 2

Further practice · Attempt part b only.

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(a) (i) Plot a graph of the student's results.

Join the points with straight lines.

(5)



(ii) State two variables that should be controlled in this investigation.

(2)

1 .....

2 .....

(iii) Name the dependent variable in this investigation.

(1)

.....

5. Question 4 · part b · 3 marks

November 2020 Paper 2

Further practice · Attempt part b only.

(b) Explain how the results of this investigation could help a manufacturer improve the effectiveness of its aspirin tablets.

(3)

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(Total for Question 4 = 11 marks)

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Working and answer

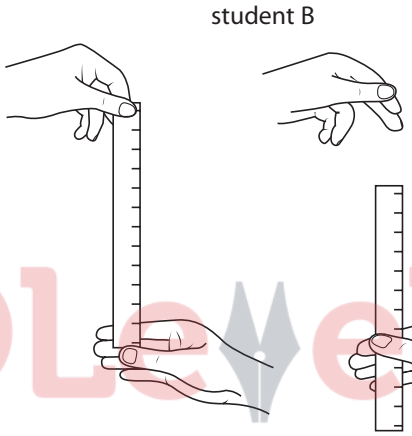
6. Question 5 · part b · 1 mark

November 2021 Paper 2

Further practice · Attempt part b only.

5 Two students use this method to investigate if caffeine affects the speed of a reflex action.

- student A sits in a chair holding out a hand with their thumb and forefinger open
- student B holds a metre rule with the 0 mm mark just above the open thumb and forefinger of student A
- student B releases the ruler
- student A catches the ruler as quickly as possible by closing their thumb and forefinger



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The method is repeated 10 times.

The student catching the ruler now drinks a cup of coffee and waits for 30 minutes.

The investigation is then repeated.

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6. Question 5 · part b · 1 mark

November 2021 Paper 2

Further practice · Attempt part b only.

The table shows the student's results.

Conditions \ Trial	Distance travelled by ruler in mm										
	1	2	3	4	5	6	7	8	9	10	mean
without coffee	210	190	200	205	207	195	198	195	199	201	200
with coffee	165	160	163	167	165	170	175	170	180	175	

(a) Complete the table by calculating the missing mean value.

(2)

(b) Coffee contains caffeine.

State the effect of caffeine on the reflex action.

(1)

(c) Explain why the student waited 30 minutes after drinking the coffee before continuing with the investigation.

(3)

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6. Question 5 · part b · 1 mark · Working and answer

Working and answer

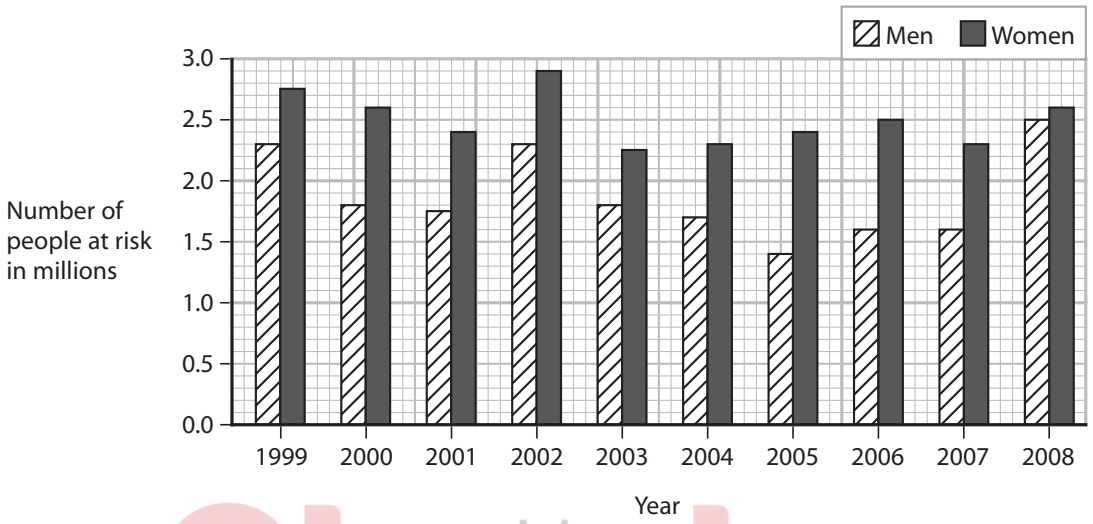
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7. Question 2 · part c · 1 mark

June 2022 Paper 2

Further practice · Attempt part c only.

2 The graph shows the number of working-age people at risk of developing a mental illness.



(a) (i) Determine the year when the greatest number of women were at risk of developing a mental illness.

(1)

.....

(ii) Determine the year when there was the greatest difference between the number of males and the number of females at risk of developing a mental illness.

(1)

.....

(iii) Determine the ratio of females at risk of developing a mental illness in 1999 compared with 2008.

Give your answer in the form n : 1

(2)

ratio = ...

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P 7 0 9 7 4 A 0 4 2 0

7. Question 2 · part c · 1 mark

June 2022 Paper 2

Further practice · Attempt part c only.

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(iv) Suggest one possible reason why the number of females at risk of developing a mental illness increased from 2007 to 2008.

(1)

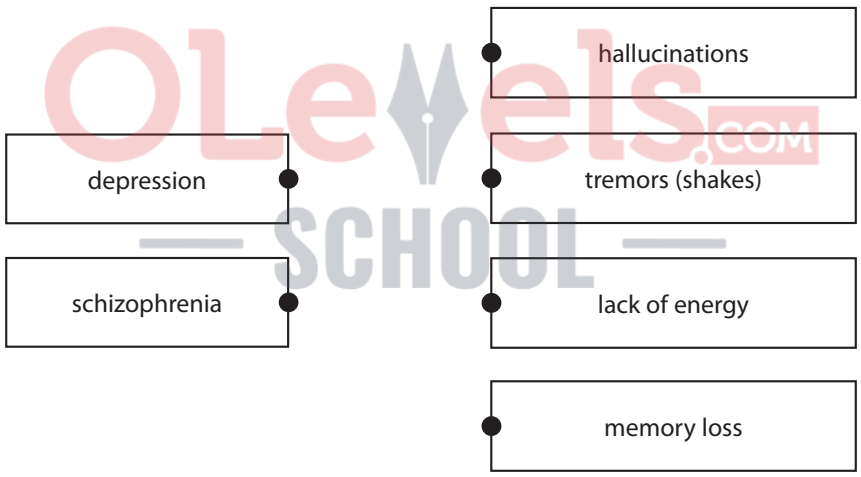
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(b) The boxes list two mental illnesses and some symptoms.

Draw one straight line from each mental illness to a symptom of the illness.

(2)



(c) Which combination of drugs is most likely to affect mental health?

(1)

- ☒ **A** alcohol and heroin
- ☒ **B** antibiotics and alcohol
- ☒ **C** cannabis and paracetamol
- ☒ **D** paracetamol and antibiotics

(d) People who use the drug cocaine often have a raised body temperature.

Name two body structures that help to control body temperature.

(2)

1 .....

2 .....

(Total for Question 2 = 10 marks)

7. Question 2 · part c · 1 mark · Working and answer

Working and answer

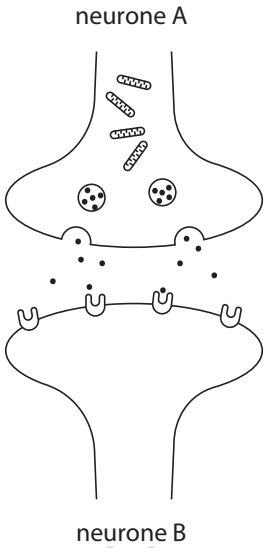
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8. Question 3 · part b · 2 marks

November 2024 Paper 2

Further practice · Attempt part b only.

3 (a) The diagram shows the gap between two neurones, A and B.



(i) Name this gap. **(1)**

(ii) Draw an arrow on the diagram to show the direction of travel of an impulse. **(1)**

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P 7 5 9 5 9 A 0 8 2 4

8. Question 3 · part b · 2 marks

November 2024 Paper 2

Further practice · Attempt part b only.

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(iii) Describe how a nerve impulse travels from one side of the gap to the other.

(6)

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(iv) Explain why an impulse can only travel in one direction across the gap.

(3)

P 7 5 9 5 9 A 0 9 2 4

8. Question 3 · part b · 2 marks

November 2024 Paper 2

Further practice · Attempt part b only.

(b) Some people take the illegal drug known as cocaine.

State two dangers of taking cocaine.

(2)

1 .....

.....

2 .....

.....

(Total for Question 3 = 13 marks)

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Working and answer

OLevels.com SCHOOL

Practice workbook · Show how an image forms on the retina and reaches the brain

Continuing rays through cornea and lens to an inverted image on the retina, and the optic nerve carrying impulses to the brain, including effects of damage.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EYE-IMAGE-FORMATION-AND-OPTIC-NERVE

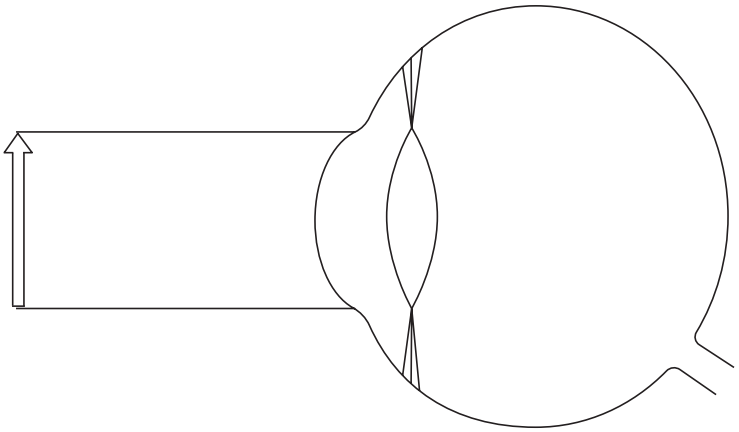
1. Question 8 · part a.i · 5 marks

November 2021 Paper 1

Worked-example retry · Attempt part a.i only.

8 The diagram shows a section through the eye.

The arrow is a near object.



(a) (i) Complete the diagram by continuing the rays from the arrow to show how the image of the arrow appears on the retina.

(5)

(ii) Describe the function of the optic nerve in allowing a person to see the image of the arrow.

(2)

(b) (i) State where the image of the arrow would be formed for someone with the condition of long sight.

(1)

(ii) Give a feature of the eye that can lead to long sight.

(1)

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1. Question 8 · part a.i · 5 marks · Working and answer

Working and answer

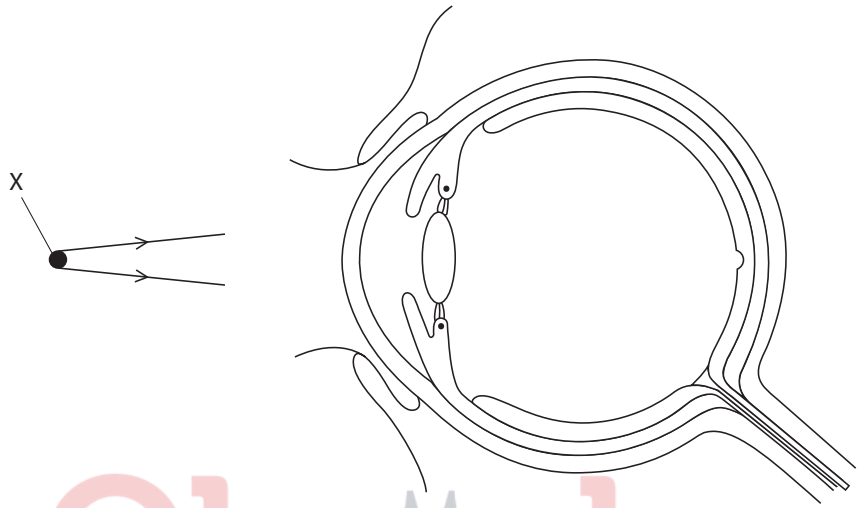
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2. Question 1 · part b · 3 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

1 The diagram shows a section through an eye.



(a) The table lists descriptions of parts of the eye.

Use the letters, A, B, C, D and E, to label the parts of the eye on the diagram.

(5)

Description of part	Letter
the layer that contains light sensitive cells	A
the muscles that change lens shape	B
the structure that controls the amount of light entering the eye	C
the structure that carries nerve impulses away from the eye	D
the structure that protects the eye from dust and grit	E

(b) A person looks at object X.

Complete the diagram by continuing the rays of light to show how the image of object X is formed on the retina.

(3)

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2. Question 1 · part b · 3 marks

June 2019 Paper 2R

Further practice · Attempt part b only.

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(c) Name three defects of the eye that can be corrected by using glasses.

(3)

1 .....

2 .....

3 .....

(Total for Question 1 = 11 marks)

Working and answer

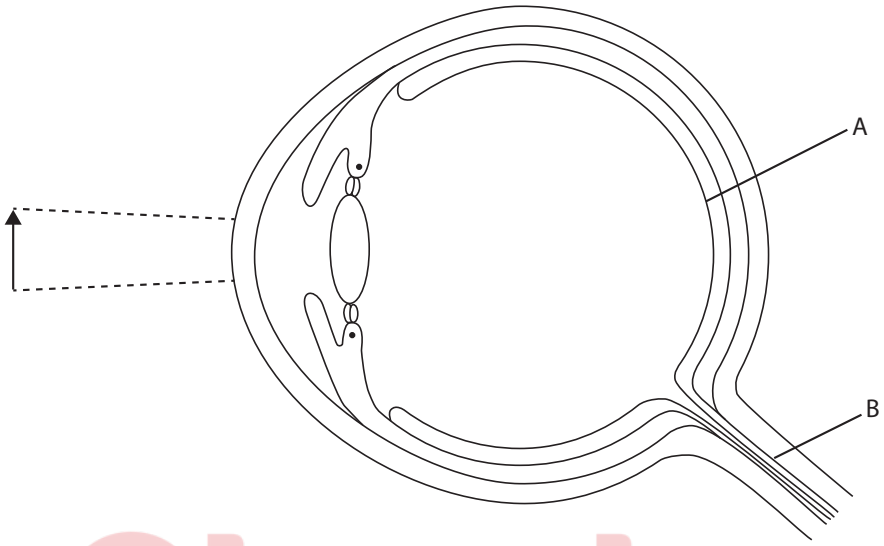
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3. Question 6 · part a.ii · 4 marks

June 2025 Paper 2

Further practice · Attempt part a.ii only.

6 The diagram shows a section through the eye.



(a) (i) Describe the functions of structures A and B. **(4)**

A .....

.....

.....

.....

B .....

.....

.....

.....

(ii) Complete the diagram to show how an image of the arrow is formed at the back of the eye.

(4)

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P 7 8 7 6 9 A 0 1 4 2 0

3. Question 6 · part a.ii · 4 marks

June 2025 Paper 2

Further practice · Attempt part a.ii only.

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(b) Describe a simple experiment to show that stereoscopic vision would help a person to catch a ball thrown towards them.

(4)

(c) Describe what is meant by the term **astigmatism**.

(2)

(Total for Question 6 = 14 marks)

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3. Question 6 · part a.ii · 4 marks · Working and answer

Working and answer

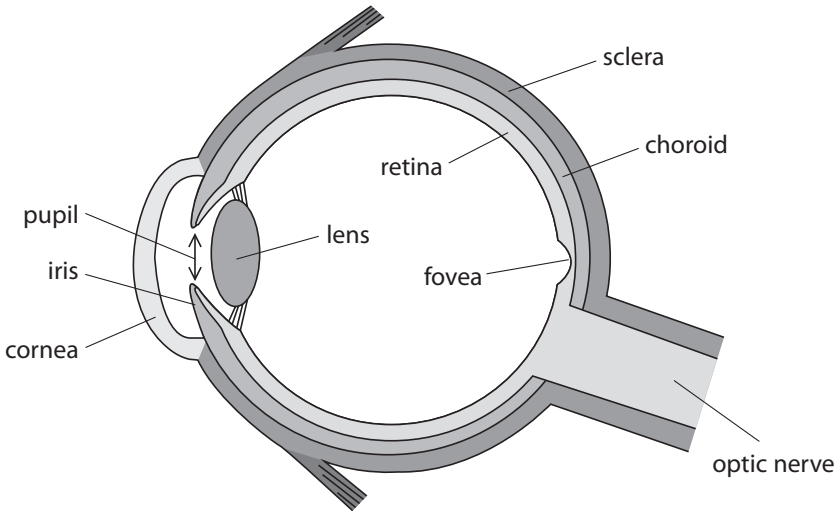
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4. Question 3 · part c.i · 2 marks

January 2020 Paper 2

Further practice · Attempt part c.i only.

3 (a) The diagram shows the human eye, with some structures labelled.



(i) Which structure helps to focus light at the back of the eye? **(1)**

- ☒ **A** fovea
- ☒ **B** iris
- ☒ **C** lens
- ☒ **D** pupil

(ii) The box lists structures of the eye. **(3)**

sclera	retina	choroid	iris	pupil

Use words from the box to complete the passage about the eye.

The ..... contains receptor cells that are sensitive to light.

In bright light the ..... gets smaller to control the amount of light entering the eye.

The ..... contains blood vessels that deliver oxygen to receptor cells.

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4. Question 3 · part c.i · 2 marks

January 2020 Paper 2

Further practice · Attempt part c.i only.

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(b) The photomicrograph gives the thickness of the two outer layers of the eye.



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Estimate the thickness of the retina.

(1)

thickness of the retina = ... µm

(c) The optic nerve contains sensory neurones.

(i) Describe the function of the sensory neurones in the optic nerve.

(2)

.....

.....

.....

.....

(ii) Glaucoma is a disease that damages the optic nerve.

Explain why damage to the optic nerve will affect vision.

(2)

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(Total for Question 3 = 9 marks)

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4. Question 3 · part c.i · 2 marks · Working and answer

Working and answer

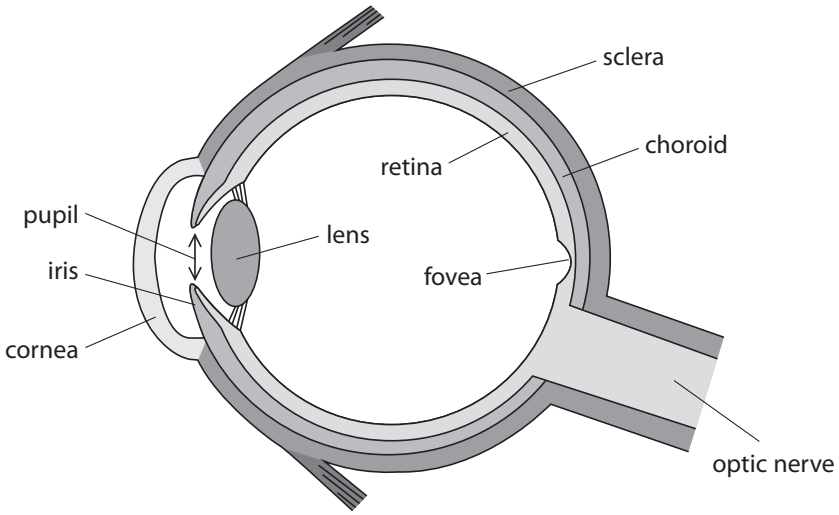
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5. Question 3 · part c.ii · 2 marks

January 2020 Paper 2

Further practice · Attempt part c.ii only.

3 (a) The diagram shows the human eye, with some structures labelled.



(i) Which structure helps to focus light at the back of the eye?

(1)

- ☒ **A** fovea
- ☒ **B** iris
- ☒ **C** lens
- ☒ **D** pupil

(ii) The box lists structures of the eye.

(3)

sclera	retina	choroid	iris	pupil

Use words from the box to complete the passage about the eye.

The ..... contains receptor cells that are sensitive to light.

In bright light the ..... gets smaller to control the amount of light entering the eye.

The ..... contains blood vessels that deliver oxygen to receptor cells.

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5. Question 3 · part c.ii · 2 marks

January 2020 Paper 2

Further practice · Attempt part c.ii only.

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(b) The photomicrograph gives the thickness of the two outer layers of the eye.



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Estimate the thickness of the retina.

(1)

thickness of the retina = ... µm

(c) The optic nerve contains sensory neurones.

(i) Describe the function of the sensory neurones in the optic nerve.

(2)

.....

.....

.....

.....

(ii) Glaucoma is a disease that damages the optic nerve.

Explain why damage to the optic nerve will affect vision.

(2)

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(Total for Question 3 = 9 marks)

5. Question 3 · part c.ii · 2 marks · Working and answer

Working and answer

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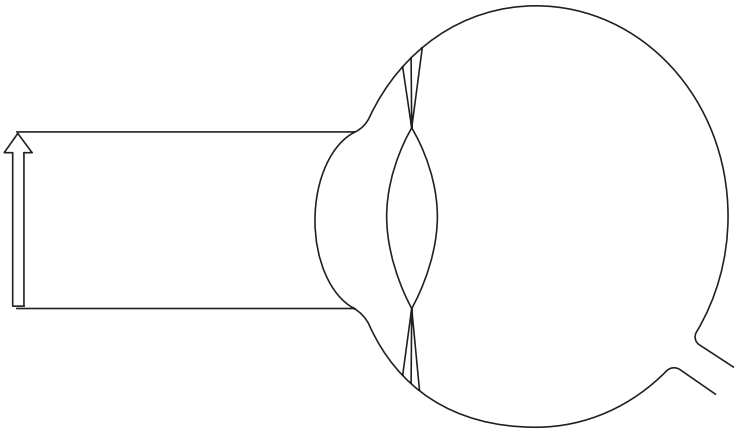
6. Question 8 · part a.ii · 2 marks

November 2021 Paper 1

Further practice · Attempt part a.ii only.

8 The diagram shows a section through the eye.

The arrow is a near object.



(a) (i) Complete the diagram by continuing the rays from the arrow to show how the image of the arrow appears on the retina.

(5)

(ii) Describe the function of the optic nerve in allowing a person to see the image of the arrow.

(2)

.....

.....

.....

.....

(b) (i) State where the image of the arrow would be formed for someone with the condition of long sight.

(1)

.....

.....

(ii) Give a feature of the eye that can lead to long sight.

(1)

.....

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6. Question 8 · part a.ii · 2 marks · Working and answer

Working and answer

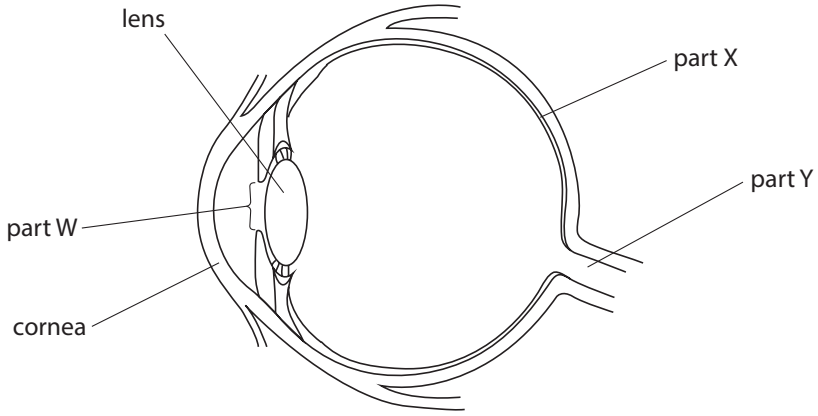
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7. Question 3 · part a.ii · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.ii only.

3 (a) The diagram shows the human eye.



(i) Name part W and part X.

(2)

W .....

X .....

(ii) Describe the function of part Y.

(2)

.....

.....

.....

.....

(iii) Explain the change in the shape of the lens when the eye views a distant object.

(2)

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7. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Plan an investigation measuring a response or treatment in people

Pulse, reaction time, pupil size, catching, hearing, sweating or urine volume in volunteers, and treated versus untreated group trials of drugs or diets.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-PLAN-INVESTIGATION-ON-PEOPLE

1. Question 5 · part b.ii · 6 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

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5 (a) A red blood cell travelling from the liver to the kidney passes through six different arteries and veins.

Complete the flow chart by naming these blood vessels in order. The first one has been done for you.

(5)

LIVER

↓

hepatic vein

↓

.....

↓

.....

↓

.....

↓

.....

↓

.....

↓

KIDNEY

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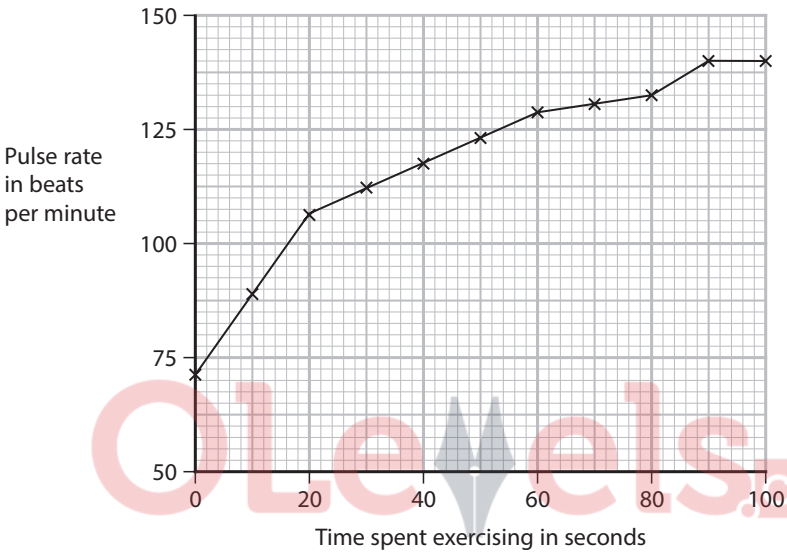
1. Question 5 · part b.ii · 6 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

(b) A student investigates the effect of time spent exercising on pulse rate.

The graph shows the student's results.



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1. Question 5 · part b.ii · 6 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

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(i) Explain the change in pulse rate recorded by the student.

(4)

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1. Question 5 · part b.ii · 6 marks

June 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

(ii) Describe how the student could carry out the investigation.

(6)

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(Total for Question 5 = 15 marks)

1. Question 5 · part b.ii · 6 marks · Working and answer

Working and answer

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2. Question 2 · part b · 4 marks

January 2020 Paper 2

Further practice · Attempt part b only.

2 A student records her pulse rate in beats per minute (bpm) after different activities.

These are the student's results.

- at rest 78 bpm
- walking 95 bpm
- jogging 138 bpm
- running 146 bpm

(a) Draw a table to show the student's results.

(3)

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2. Question 2 · part b · 4 marks

January 2020 Paper 2

Further practice · Attempt part b only.

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(b) Describe a method that the student could use to obtain these results.

(4)

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(c) State one factor that the student should control during the investigation.

(1)

(d) Name a body process, other than pulse rate, that increases during exercise.

(1)

(Total for Question 2 = 9 marks)

2. Question 2 · part b · 4 marks · Working and answer

Working and answer

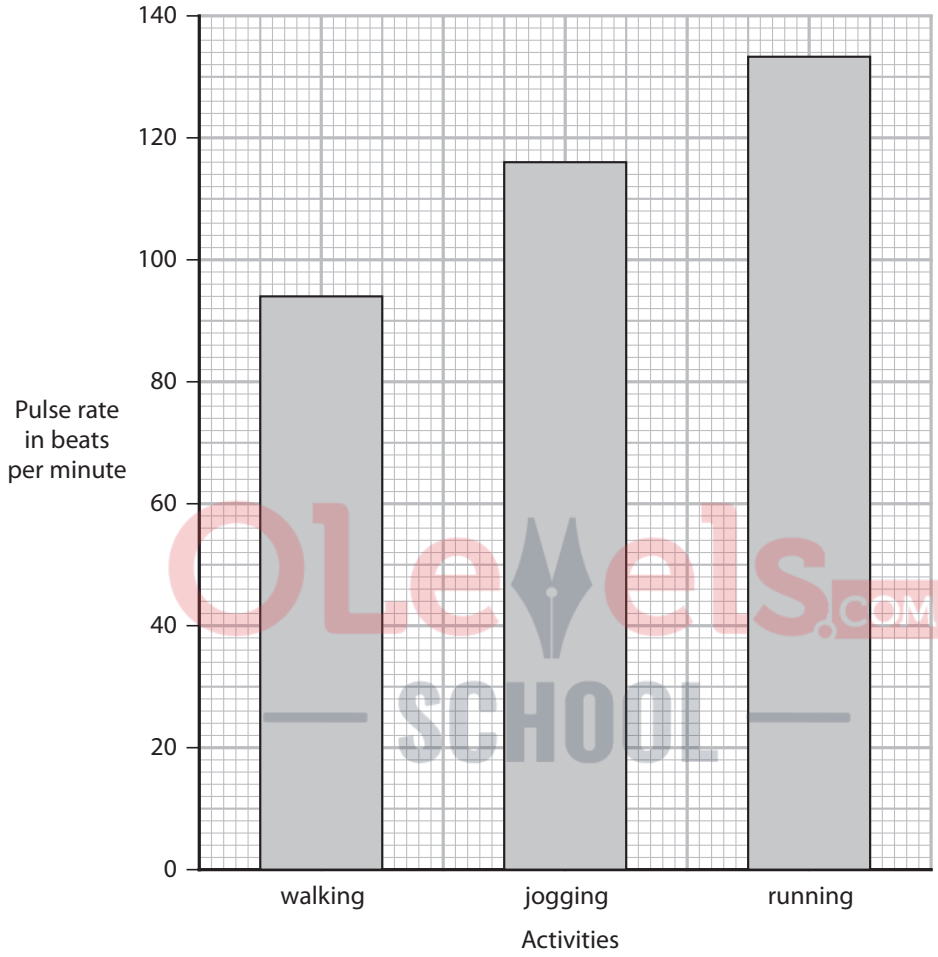
OLevels.com SCHOOL

3. Question 4 · part a.iii · 4 marks

June 2023 Paper 2R

Further practice · Attempt part a.iii only.

4 (a) The bar chart shows the pulse rate of a student at the end of three different activities.



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3. Question 4 · part a.iii · 4 marks

June 2023 Paper 2R

Further practice · Attempt part a.iii only.

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(i) Draw a table that the student could use to record these results.

(3)

(ii) Complete the table by adding the student's results.

(2)

(iii) Describe a method that the student could use to obtain the results shown in the bar chart.

(4)

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3. Question 4 · part a.iii · 4 marks

June 2023 Paper 2R

Further practice · Attempt part a.iii only.

(b) The pulse rates at the end of walking and running are different.

Explain this difference.

(4)

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(Total for Question 4 = 13 marks)

3. Question 4 · part a.iii · 4 marks · Working and answer

Working and answer

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4. Question 3 · part b.i · 6 marks

June 2025 Paper 2R

Further practice · Attempt part b.i only.

3 (a) The nervous system is made up of sensory, motor and relay neurones.

(i) Draw a diagram of a motor neurone.

You should label these structures in your diagram.

- nucleus
- axon
- myelin sheath

(4)

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(ii) Describe the function of a motor neurone.

(2)

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4. Question 3 · part b.i · 6 marks

June 2025 Paper 2R

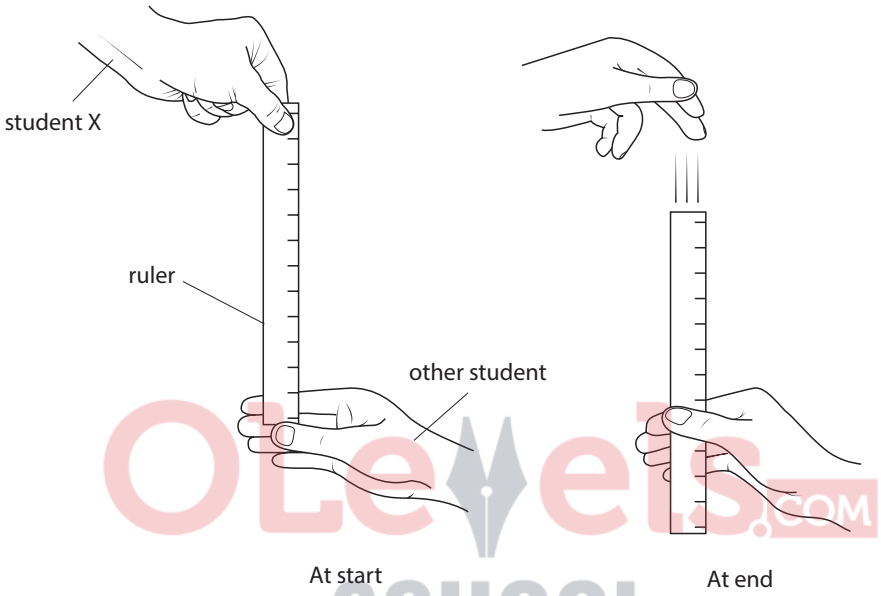
Further practice · Attempt part b.i only.

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(b) Student X uses this equipment to investigate the reaction time of another student.



Student X knows that the reaction time can be calculated using this formula.

$$\text{reaction time} = \sqrt{\frac{\text{measurement from ruler}}{5}}$$

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4. Question 3 · part b.i · 6 marks

June 2025 Paper 2R

Further practice · Attempt part b.i only.

(i) Describe a method the student could use to compare the reaction times of 10 students.

(6)

(ii) Name the dependent variable in this investigation.

(1)

(iii) Give a reason why the student should display the results using a bar chart instead of a line graph.

(1)

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4. Question 3 · part b.i · 6 marks · Working and answer

Working and answer

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5. Question 3 · part c.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part c.ii only.

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3 The skin is important in the control of body temperature.

(a) State two other functions of the skin.

(2)

1 .....

.....

2 .....

.....

(b) As a result of vigorous exercise, the colour of the skin changes.

Explain why there is this change in skin colour.

(4)

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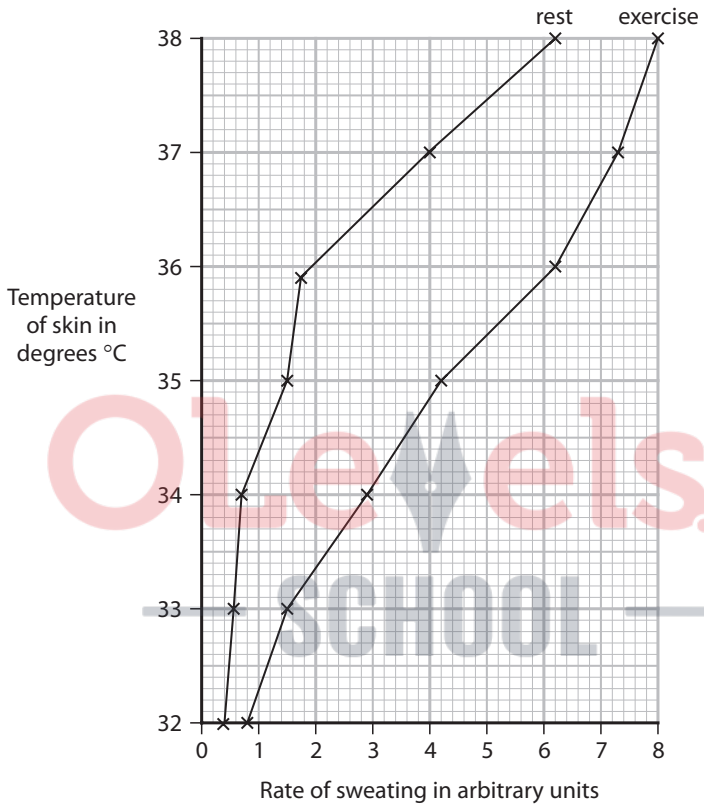
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5. Question 3 · part c.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part c.ii only.

(c) The graph shows how the temperature of the skin varies as the rate of sweating changes in a person at rest and during exercise.



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5. Question 3 · part c.ii · 2 marks

January 2022 Paper 2

Further practice · Attempt part c.ii only.

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(i) Describe how the skin temperature varies as the rate of sweating changes for the person at rest.

(2)

(ii) Suggest a method that could be used to measure the rate of sweating.

(2)

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(d) Explain the importance of maintaining a constant body temperature.

(3)

(Total for Question 3 = 13 marks)

5. Question 3 · part c.ii · 2 marks · Working and answer

Working and answer

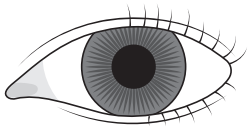
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6. Question 6 · part a · 3 marks

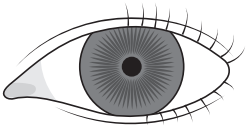
June 2024 Paper 2

Further practice · Attempt part a only.

6 The diagram shows how the size of the pupil changes in different light intensities.



Dim light



Bright light

(a) Describe a method that could be used to investigate how the size of the pupil changes in different light intensities.

(3)

(b) A student calculated the area of the pupil in bright light to be 7.1 mm^2.

Using information from the diagram, calculate the area of the pupil in dim light.

[for a circle, area = πr^2 $\pi = 3.14$]

(3)

area of pupil in dim light = ... mm^2

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6. Question 6 · part a · 3 marks · Working and answer

Working and answer

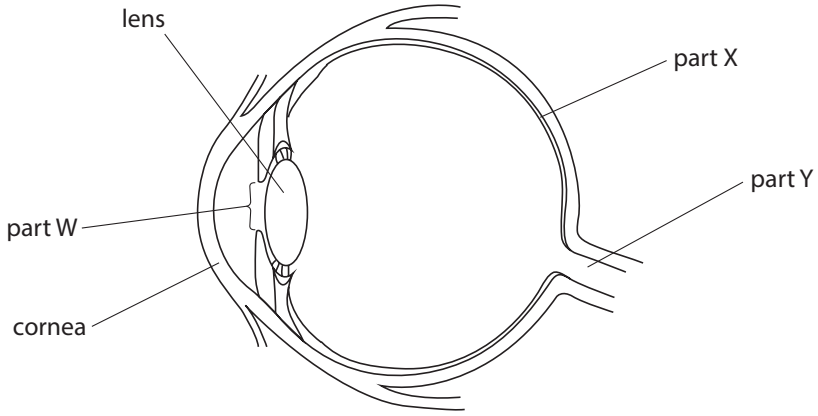
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7. Question 3 · part b · 4 marks

November 2024 Paper 1

Further practice · Attempt part b only.

3 (a) The diagram shows the human eye.



(i) Name part W and part X.

(2)

W .....

X .....

(ii) Describe the function of part Y.

(2)

.....

.....

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(iii) Explain the change in the shape of the lens when the eye views a distant object.

(2)

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7. Question 3 · part b · 4 marks

November 2024 Paper 1

Further practice · Attempt part b only.

(iv) Describe how corneal transplants help to restore normal vision for people with corneal disease.

(2)

(b) The iris reflex controls the amount of light entering the eye.

Describe a method that could be used to investigate how the eye responds to different light intensities.

(4)

(Total for Question 3 = 12 marks)

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7. Question 3 · part b · 4 marks · Working and answer

Working and answer

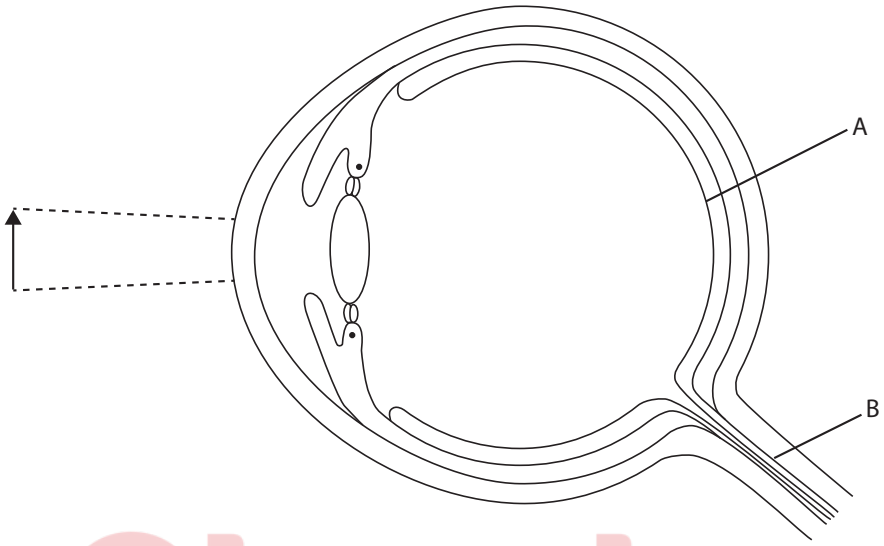
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8. Question 6 · part b · 4 marks

June 2025 Paper 2

Further practice · Attempt part b only.

6 The diagram shows a section through the eye.



(a) (i) Describe the functions of structures A and B. **(4)**

A .....

.....

.....

.....

B .....

.....

.....

.....

(ii) Complete the diagram to show how an image of the arrow is formed at the back of the eye.

(4)

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8. Question 6 · part b · 4 marks

June 2025 Paper 2

Further practice · Attempt part b only.

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(b) Describe a simple experiment to show that stereoscopic vision would help a person to catch a ball thrown towards them.

(4)

(c) Describe what is meant by the term **astigmatism**.

(2)

(Total for Question 6 = 14 marks)

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8. Question 6 · part b · 4 marks · Working and answer

Working and answer

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9. Question 5 · part a.i · 3 marks

January 2023 Paper 1

Further practice · Attempt part a.i only.

5 (a) A student reads that as people get older their ability to hear high frequency sounds decreases.

(i) Describe an investigation a student could do to show if this is correct.

(3)

(ii) Give a reason why hearing loss occurs as people get older.

(1)

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9. Question 5 · part a.i · 3 marks · Working and answer

Working and answer

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10. Question 5 · part a · 4 marks

June 2022 Paper 1

Further practice · Attempt part a only.

5 (a) There is a difference in the volume of urine produced on a hot day and the volume of urine produced on a cold day.

Design an investigation to determine the difference in the volume of urine produced on a hot day and the volume of urine produced on a cold day.

Include experimental details in your answer and write in full sentences.

(4)

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10. Question 5 · part a · 4 marks

June 2022 Paper 1

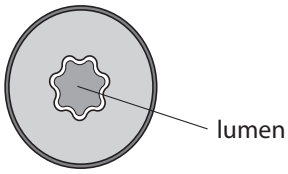
Further practice · Attempt part a only.

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(b) The diagram shows a cross-section of a blood vessel in the skin on a cold day.



Explain the changes in this blood vessel on a hot day.

(3)

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(c) Describe how ADH regulates the volume of water in the body on a hot day.

(4)

(Total for Question 5 = 11 marks)

10. Question 5 · part a · 4 marks · Working and answer

Working and answer

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11. Question 4 · part d · 3 marks

June 2019 Paper 2

Further practice · Attempt part d only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

In 1916, an American doctor, Joseph Goldberger, noticed symptoms of a mystery disease that were common among his patients. These symptoms included headaches, swollen tongues, skin rashes, upset stomachs and mental illness. In one American state, North Carolina, 40% of the 30 000 people living in this area died from this mystery disease.

Goldberger investigated whether the symptoms of the mystery disease were caused by pathogens. His investigation involved injecting himself with the blood from one patient, eating the skin rash of another and swallowing fluid taken from the intestines of a third. Goldberger did not get the disease, although he did suffer from nausea and diarrhoea.

Goldberger concluded that the symptoms shown by the patients were not caused by an infectious disease. He decided that they were more likely to be caused by the poor diet of the patients.

(a) Calculate the number of people in North Carolina who died from this mystery disease (lines 4 to 5).

(2)

number of deaths = ...

(b) State the meaning of the term pathogens (line 7).

(1)

.....

.....

(c) The people were suffering from a deficiency disease called pellagra, which is caused by a lack of vitamin B in the diet.

State why deficiency diseases such as pellagra are not infectious diseases.

(1)

.....

.....

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11. Question 4 · part d · 3 marks

June 2019 Paper 2

Further practice · Attempt part d only.

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(d) Describe a method that Goldberger could use to show that the symptoms of his patients were caused by a lack of vitamin B in their diet.

(3)

(e) Scurvy is another deficiency disease.

Scurvy affects the production of a protein called collagen.

Collagen is a component of many body tissues including cartilage.

Explain why a person with scurvy often experiences pain when walking.

(2)

(Total for Question 4 = 9 marks)

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11. Question 4 · part d · 3 marks · Working and answer

Working and answer

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12. Question 9 · part a · 3 marks

June 2022 Paper 1R

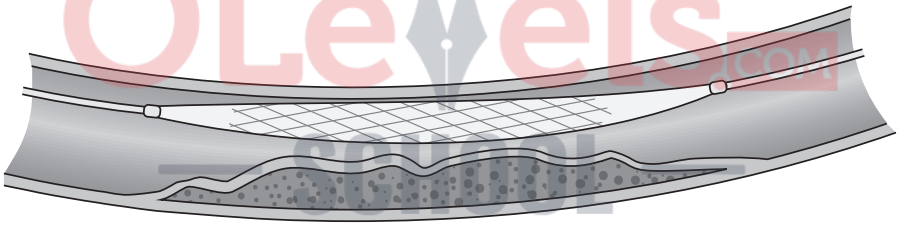
Further practice · Attempt part a only.

9 (a) Statins can be used to treat people with circulatory disorders.

Design a method that can be used to test the effectiveness of statins in treating circulatory disorders.

(3)

(b) The diagram shows a stent being used in the treatment of heart disease.



(Source: © DesignPrax/Shutterstock)

Explain how a stent can be used to treat heart disease.

(4)

(Total for Question 9 = 7 marks)

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12. Question 9 · part a · 3 marks · Working and answer

Working and answer

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13. Question 6 · part b.iii · 5 marks

June 2024 Paper 2R

Further practice · Attempt part b.iii only.

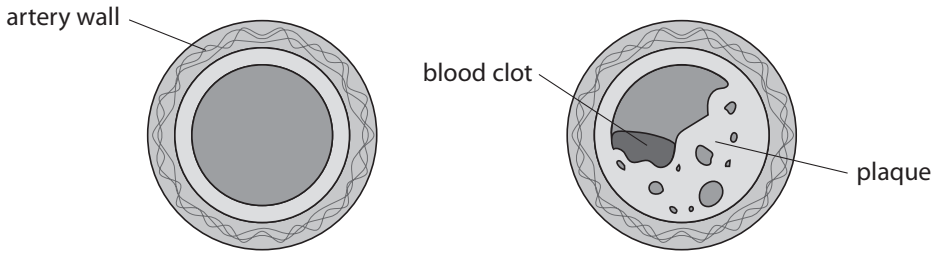
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6 (a) Cholesterol is a fatty substance that can build up inside arteries forming waxy deposits called plaques.

The diagram shows a healthy artery and an artery with a plaque.



healthy artery artery with a plaque

(Source: adapted from https://www.dreamstime.com/artery-plaque-blood-clot-medical-illustration-diseased-compared-to-healthy-cross-section-diagram-shows-image120895536)

Explain how plaques might affect the heart of an individual.

(4)

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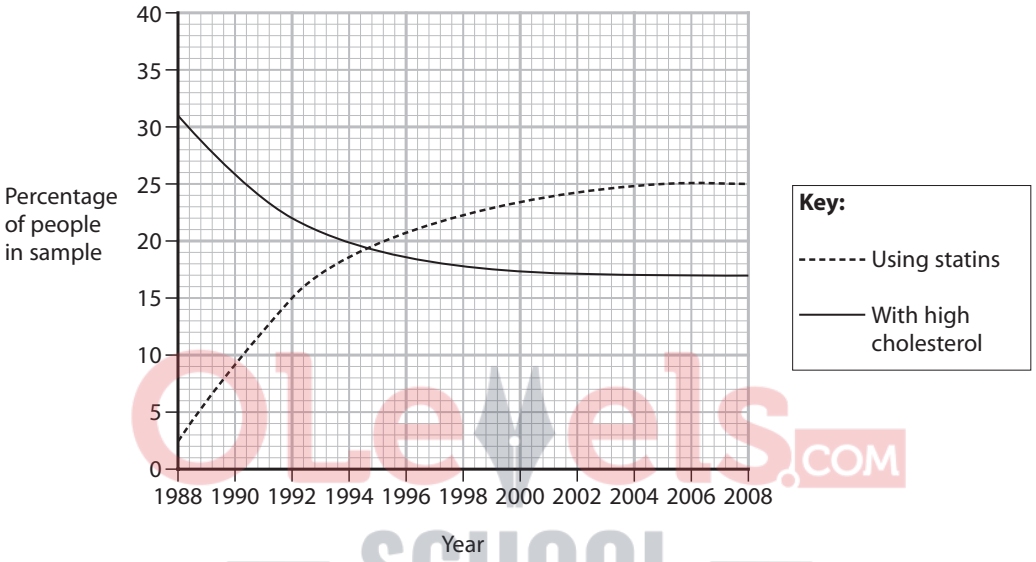
13. Question 6 · part b.iii · 5 marks

June 2024 Paper 2R

Further practice · Attempt part b.iii only.

(b) Scientists investigated the effect of statins on cholesterol levels in a sample of people.

The graph shows the results of their investigation.



(Source: https://www.health.harvard.edu/heart-health/trends-in-high-cholesterol-and-statin-use)

(i) Give the relationship between the use of statins and the percentage of people with high cholesterol levels.

(1)

.....

.....

(ii) Calculate the ratio of the percentage of people with high cholesterol levels in 1988 compared with the percentage of people with high cholesterol levels in 2005.

Give your answer in the form of 1 : n

(3)

ratio = ...

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13. Question 6 · part b.iii · 5 marks

June 2024 Paper 2R

Further practice · Attempt part b.iii only.

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(iii) Describe a method that scientists could use to obtain the data shown in the graph.

(5)

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(iv) Give two factors the scientists need to consider to make sure that their investigation is valid.

(2)

1 .....

.....

2 .....

.....

(Total for Question 6 = 15 marks)

13. Question 6 · part b.iii · 5 marks · Working and answer

Working and answer

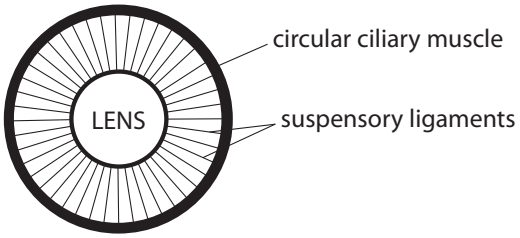
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14. Question 5 · part c · 6 marks

January 2022 Paper 2

Further practice · Attempt part c only.

5 (a) The diagram shows a front view of structures in the eye that help to form an image on the retina.



The three structures shown in the diagram change when focusing on near and distant objects.

Complete the table by putting ticks (✓) in the correct boxes to show the condition of each structure of the eye when focusing on near and distant objects.

(3)

	Focusing on a near object	Focusing on a distant object
Lens	thin ☐ thick ☐	thin ☐ thick ☐
Circular ciliary muscle	contracted ☐ relaxed ☐	contracted ☐ relaxed ☐
Suspensory ligament	stretched ☐ loose ☐	stretched ☐ loose ☐

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14. Question 5 · part c · 6 marks

January 2022 Paper 2

Further practice · Attempt part c only.

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(b) Humans have stereoscopic vision.

Explain what is meant by the term **stereoscopic vision**.

(3)

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14. Question 5 · part c · 6 marks

January 2022 Paper 2

Further practice · Attempt part c only.

(c) Design an investigation to show that it is easier for a person to catch a ball if they have stereoscopic vision.

Your answer should include experimental details and be written in complete sentences.

(6)

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(Total for Question 5 = 12 marks)

14. Question 5 · part c · 6 marks · Working and answer

Working and answer

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Practice workbook · Explain validity problems and fixes in investigations on people

Investigations using volunteers or patients where the student explains why results are not valid, why a rest or fasting step improves validity, or what to change.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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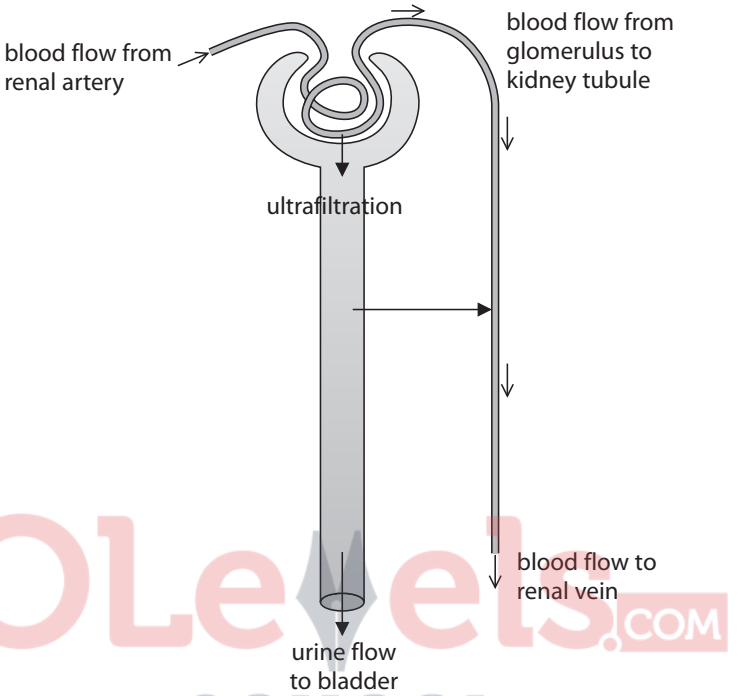
Type ID: HBI-PRAC-CONTROL-HUMAN-SUBJECT-VALIDITY

1. Question 8 · part b.ii · 2 marks

June 2019 Paper 2

Worked-example retry · Attempt part b.ii only.

8 (a) The diagram shows part of a nephron in the human excretory system.



(Source: adapted from © Alila Medical Images/Alamy)

(i) Explain how ultrafiltration occurs in the Bowman's capsule.

(2)

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1. Question 8 · part b.ii · 2 marks

June 2019 Paper 2

Worked-example retry · Attempt part b.ii only.

(ii) Explain the differences in the composition of the blood flowing through the renal artery and through the renal vein.

(4)

(b) The quantity of salt in the diet affects the concentration of urine produced.

An investigation is set up to test this statement.

This is the method used.

- ten people are each given a different food
- each food contains a different quantity of salt
- the volume of urine produced by each person is recorded each hour for six hours after eating the food

(i) State the independent variable in this investigation.

(1)

(ii) Explain why this investigation is unlikely to produce valid results.

(2)

(iii) State one change to the method that would make the results more valid.

(1)

(Total for Question 8 = 10 marks)

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1. Question 8 · part b.ii · 2 marks · Working and answer

Working and answer

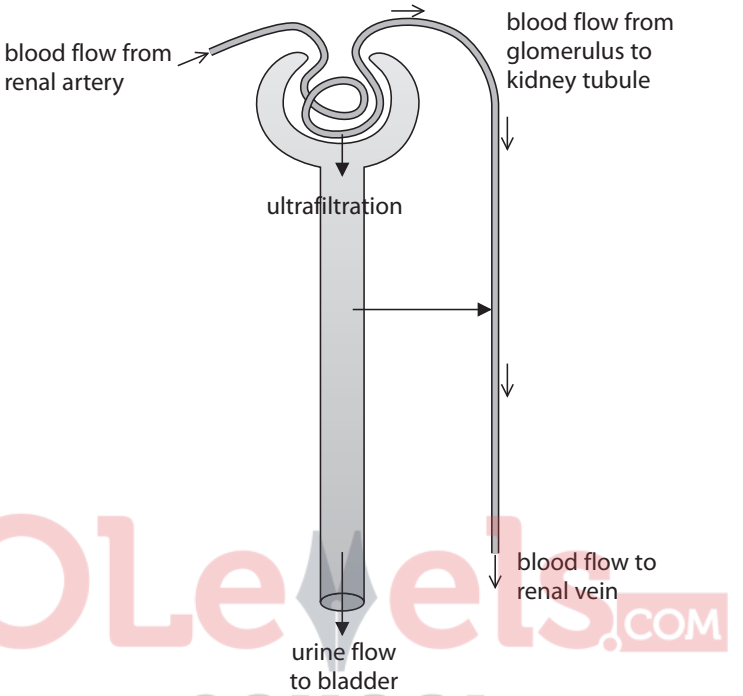
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2. Question 8 · part b.iii · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.iii only.

8 (a) The diagram shows part of a nephron in the human excretory system.



(Source: adapted from © Alila Medical Images/Alamy)

(i) Explain how ultrafiltration occurs in the Bowman's capsule.

(2)

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2. Question 8 · part b.iii · 1 mark

June 2019 Paper 2

Further practice · Attempt part b.iii only.

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(4)

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(1)

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(2)

(iii) State one change to the method that would make the results more valid.

(1)

(Total for Question 8 = 10 marks)

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2. Question 8 · part b.iii · 1 mark · Working and answer

Working and answer

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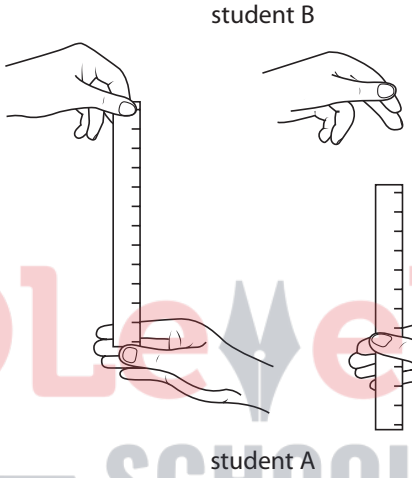
3. Question 5 · part d · 2 marks

November 2021 Paper 2

Further practice · Attempt part d only.

5 Two students use this method to investigate if caffeine affects the speed of a reflex action.

- student A sits in a chair holding out a hand with their thumb and forefinger open
- student B holds a metre rule with the 0 mm mark just above the open thumb and forefinger of student A
- student B releases the ruler
- student A catches the ruler as quickly as possible by closing their thumb and forefinger



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The method is repeated 10 times.

The student catching the ruler now drinks a cup of coffee and waits for 30 minutes.

The investigation is then repeated.

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3. Question 5 · part d · 2 marks

November 2021 Paper 2

Further practice · Attempt part d only.

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The table shows the student's results.

Conditions \ Trial	Distance travelled by ruler in mm										
	1	2	3	4	5	6	7	8	9	10	mean
without coffee	210	190	200	205	207	195	198	195	199	201	200
with coffee	165	160	163	167	165	170	175	170	180	175	

(a) Complete the table by calculating the missing mean value.

(2)

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(b) Coffee contains caffeine.

State the effect of caffeine on the reflex action.

(1)

(c) Explain why the student waited 30 minutes after drinking the coffee before continuing with the investigation.

(3)

3. Question 5 · part d · 2 marks

November 2021 Paper 2

Further practice · Attempt part d only.

(d) The student who catches the ruler does not eat or drink anything for several hours before the investigation.

Suggest why this improves the validity of the investigation.

(2)

.....

.....

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.....

(e) A cup of coffee contains 100 mg of caffeine. A can of cola contains 50 mg of caffeine.

If the student catching the ruler had drunk a can of cola instead of coffee, estimate the mean distance travelled by the ruler.

(1)

mean distance = ... mm

(Total for Question 5 = 9 marks)

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Working and answer

4. Question 5 · part c.i · 2 marks

June 2022 Paper 2

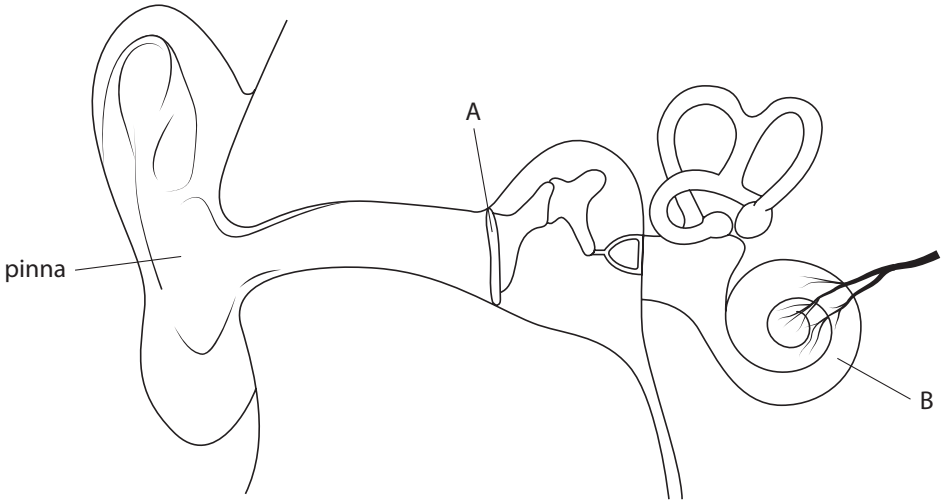
Further practice · Attempt part c.i only.

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5 The diagram shows a section through the human ear.



(a) (i) Name parts A and B. **(2)**

A .....

B .....

(ii) Describe the function of the pinna. **(2)**

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4. Question 5 · part c.i · 2 marks

June 2022 Paper 2

Further practice · Attempt part c.i only.

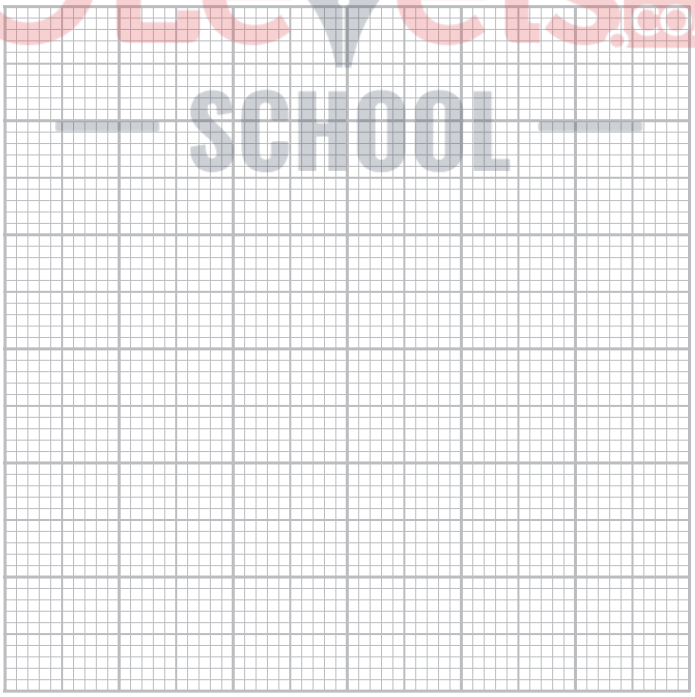
(b) The range of frequencies audible to the human ear can vary with age.

The table shows the maximum frequency detected by people of different ages.

Age in years	Maximum frequency detected in kHz
20	20
30	18
40	16
50	14

(i) Draw a bar chart to show how the maximum frequency detected varies with age.

(4)



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P 7 0 9 7 4 A 0 1 0 2 0

4. Question 5 · part c.i · 2 marks

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Further practice · Attempt part c.i only.

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(ii) Calculate the percentage change in the maximum frequency detected by people aged 20 and people aged 50.

(3)

percentage change = ... %

(c) A student suggests this method that could be used to produce the data shown in the table.

- select a person of each age to be tested
- use a signal generator to produce sounds at different frequencies
- record the maximum frequency detected by each person

(i) Give two reasons why this is unlikely to produce valid results.

(2)

1 .....

.....

2 .....

.....

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(ii) Describe how the reliability of the results could be improved.

(2)

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(Total for Question 5 = 15 marks)

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4. Question 5 · part c.i · 2 marks · Working and answer

Working and answer

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5. Question 6 · part b.iv · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.iv only.

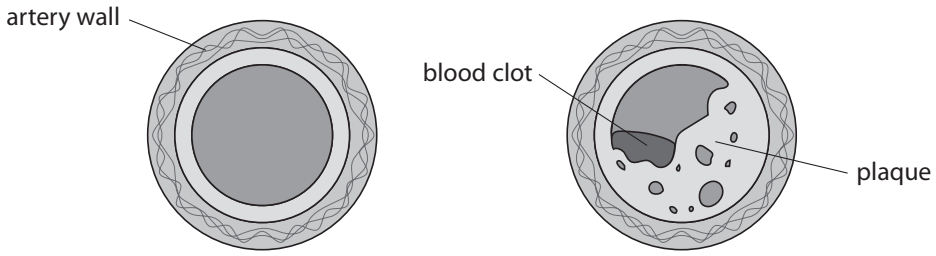
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6 (a) Cholesterol is a fatty substance that can build up inside arteries forming waxy deposits called plaques.

The diagram shows a healthy artery and an artery with a plaque.



healthy artery

artery with a plaque

(Source: adapted from https://www.dreamstime.com/artery-plaque-blood-clot-medical-illustration-diseased-compared-to-healthy-cross-section-diagram-shows-image120895536)

Explain how plaques might affect the heart of an individual.

(4)

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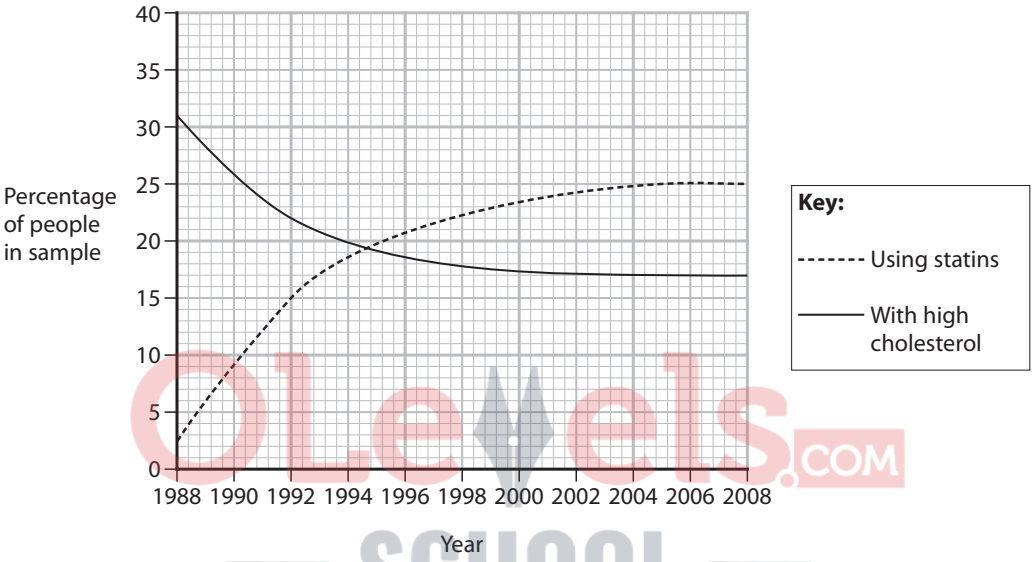
5. Question 6 · part b.iv · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.iv only.

(b) Scientists investigated the effect of statins on cholesterol levels in a sample of people.

The graph shows the results of their investigation.



(Source: https://www.health.harvard.edu/heart-health/trends-in-high-cholesterol-and-statin-use)

(i) Give the relationship between the use of statins and the percentage of people with high cholesterol levels.

(1)

.....

.....

(ii) Calculate the ratio of the percentage of people with high cholesterol levels in 1988 compared with the percentage of people with high cholesterol levels in 2005.

Give your answer in the form of 1 : n

(3)

ratio = ...

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5. Question 6 · part b.iv · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.iv only.

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(iii) Describe a method that scientists could use to obtain the data shown in the graph.

(5)

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(iv) Give two factors the scientists need to consider to make sure that their investigation is valid.

(2)

1 .....

.....

2 .....

.....

(Total for Question 6 = 15 marks)

5. Question 6 · part b.iv · 2 marks · Working and answer

Working and answer

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6. Question 5 · part a.i · 2 marks

June 2023 Paper 1R

Further practice · Attempt part a.i only.

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5 A student uses this method to investigate the effect of exercise on the production of sweat.

- place a person in a room which has a temperature of 20 °C
- ask the person to remain inactive for 5 minutes
- measure the amount of sweat produced by a 25 cm^2 area of skin
- ask the person to exercise and measure the amount of sweat produced
- ask the person to rest for 10 minutes

Repeat the method at room temperatures of 22 °C, 24 °C, 26 °C, 28 °C and 30 °C

(a) (i) Explain why the person rests for 10 minutes before the method is repeated.

(2)

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(ii) State three variables that should be controlled.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

6. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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Practice workbook · Predict how a drug, toxin or disease changes synaptic transmission

Using the synapse model to explain effects of plaques, enzyme inhibitors, curare, alcohol, or too little or too much dopamine or acetylcholine.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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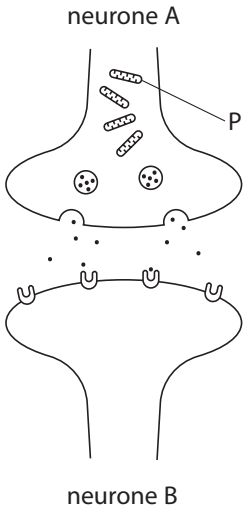
Type ID: HBI-APPLY-SYNAPSE-TO-DRUG-OR-DISEASE

1. Question 8 · part b.ii · 2 marks

November 2020 Paper 2R

Worked-example retry · Attempt part b.ii only.

8 The diagram shows the gap between two neurones, A and B.



(a) (i) Draw an arrow on the diagram to show the direction of travel of an impulse. **(1)**

(ii) Explain the function of structure P in the transmission of the impulse across the gap. **(3)**

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1. Question 8 · part b.ii · 2 marks

November 2020 Paper 2R

Worked-example retry · Attempt part b.ii only.

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(b) Curare is a poison used by South American Indians when hunting. Curare blocks receptors on the post synaptic membranes of neurones supplying skeletal muscles and prevents breathing.

(i) Explain how curare can stop the process of breathing.

(2)

.....

.....

.....

.....

(ii) Acetylcholinesterase is an enzyme found in the gap between two neurones.

Suggest why acetylcholinesterase inhibitors can be used to reverse the effects of curare poisoning.

(2)

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(Total for Question 8 = 8 marks)

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1. Question 8 · part b.ii · 2 marks · Working and answer

Working and answer

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2. Question 6 · part d · 3 marks

June 2019 Paper 2R

Further practice · Attempt part d only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Alzheimer's disease is a degenerative disease that usually starts slowly and gets worse over time. Scientists want to understand the causes of the disease and discover treatments for the memory loss that is a major feature of Alzheimer's. Scientists use the brains of people who have died to try to find out what happens in the brains of people who have Alzheimer's.

The most common early symptom is short-term memory loss, which is difficulty in remembering recent events. People with this condition are no longer able to transfer information about their daily life from their short-term to long-term memory. For example, they are unable to remember if they went shopping.

The condition causes plaques of an insoluble protein called B-amyloid to build up in the gap between neurones. Over time, these plaques appear in the neurones themselves in the regions of the brain associated with memory.

The B-amyloid is formed when an essential protein called amyloid precursor protein (APP) is not broken down correctly.

(a) Describe two differences between short-term memory and long-term memory (lines 8 to 9). **(2)**

1 .....

.....

2 .....

.....

(b) State the area of the brain responsible for memory. **(1)**

.....

(c) State two symptoms of Alzheimer's disease, other than memory loss. **(2)**

1 .....

.....

2 .....

.....

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16

2. Question 6 · part d · 3 marks

June 2019 Paper 2R

Further practice · Attempt part d only.

(d) Explain why the build-up of B-amyloid disrupts brain function (lines 10 to 11).

(3)

(e) One method of treating the early stages of Alzheimer's is the use of an enzyme inhibitor.

Explain how this treatment could help people with this condition.

(2)

(f) Describe the evidence that shows the amyloid precursor protein (APP) is involved in the development of Alzheimer's disease (lines 10 to 14).

(2)

(Total for Question 6 = 12 marks)

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2. Question 6 · part d · 3 marks · Working and answer

Working and answer

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3. Question 6 · part e · 2 marks

June 2019 Paper 2R

Further practice · Attempt part e only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Alzheimer's disease is a degenerative disease that usually starts slowly and gets worse over time. Scientists want to understand the causes of the disease and discover treatments for the memory loss that is a major feature of Alzheimer's. Scientists use the brains of people who have died to try to find out what happens in the brains of people who have Alzheimer's.

The most common early symptom is short-term memory loss, which is difficulty in remembering recent events. People with this condition are no longer able to transfer information about their daily life from their short-term to long-term memory. For example, they are unable to remember if they went shopping.

The condition causes plaques of an insoluble protein called B-amyloid to build up in the gap between neurones. Over time, these plaques appear in the neurones themselves in the regions of the brain associated with memory.

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(a) Describe two differences between short-term memory and long-term memory (lines 8 to 9). **(2)**

1 .....

.....

2 .....

.....

(b) State the area of the brain responsible for memory. **(1)**

.....

(c) State two symptoms of Alzheimer's disease, other than memory loss. **(2)**

1 .....

.....

2 .....

.....

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16

3. Question 6 · part e · 2 marks

June 2019 Paper 2R

Further practice · Attempt part e only.

(d) Explain why the build-up of B-amyloid disrupts brain function (lines 10 to 11).

(3)

(e) One method of treating the early stages of Alzheimer's is the use of an enzyme inhibitor.

Explain how this treatment could help people with this condition.

(2)

(f) Describe the evidence that shows the amyloid precursor protein (APP) is involved in the development of Alzheimer's disease (lines 10 to 14).

(2)

(Total for Question 6 = 12 marks)

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3. Question 6 · part e · 2 marks · Working and answer

Working and answer

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4. Question 5 · part c.iii · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.iii only.

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5 This question is about the nervous system.

(a) Describe what is meant by a reflex action.

(3)

(b) Diagram 1 shows a reflex arc.

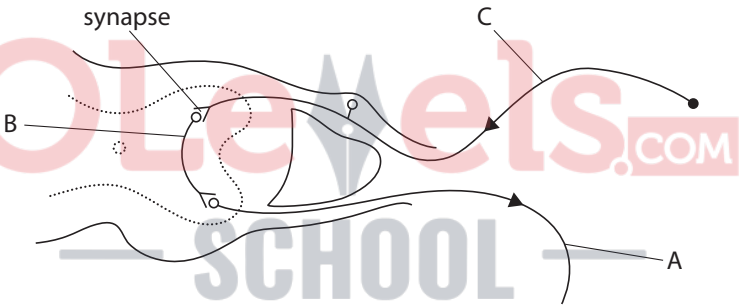


Diagram 1

Name the types of neurone labelled A, B and C.

(3)

A .....

B .....

C .....

4. Question 5 · part c.iii · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.iii only.

(c) Diagram 2 shows a neurone from a healthy brain and a neurone from the brain of someone with Alzheimer's disease.

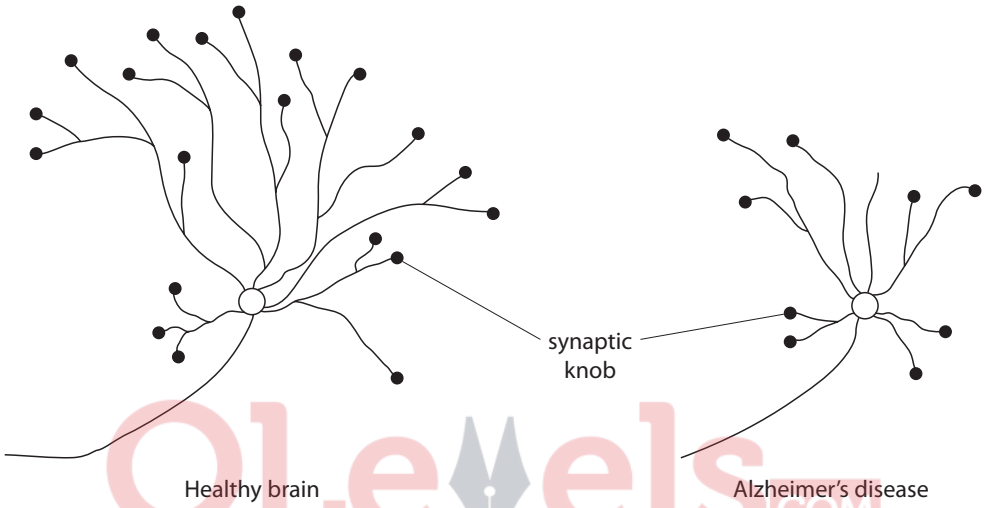


Diagram 2

(i) Give three differences between the two neurones. **(3)**

1 .....

.....

2 .....

.....

3 .....

.....

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4. Question 5 · part c.iii · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.iii only.

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(ii) Acetylcholine is a neurotransmitter.

Explain the importance of acetylcholine in the functioning of the brain.

(3)

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(iii) One symptom of Alzheimer's disease is that less acetylcholine is produced in the brain.

Explain why there is less acetylcholine produced in people with Alzheimer's disease.

Use information from diagram 2 to help your answer.

(3)

(Total for Question 5 = 15 marks)

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4. Question 5 · part c.iii · 3 marks · Working and answer

Working and answer

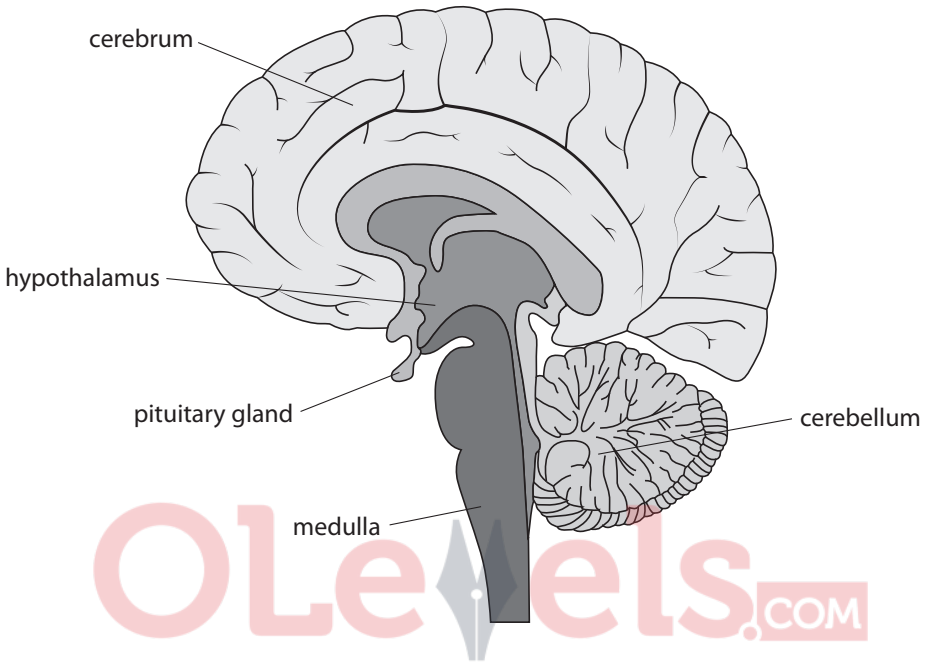
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5. Question 7 · part b · 3 marks

January 2020 Paper 2

Further practice · Attempt part b only.

7 The diagram shows parts of the human brain.



(a) (i) The table lists parts of the brain and some functions of the brain.

Complete the table by putting a tick (✓) to show if the function is correct for each part.

(3)

Part	Function	Tick
medulla	controls heart rate and breathing	
pituitary gland	involved in language	
hypothalamus	responsible for osmoregulation	
cerebellum	controls voluntary movements	

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P 5 9 9 1 0 A 0 1 8 2 4

5. Question 7 · part b · 3 marks

January 2020 Paper 2

Further practice · Attempt part b only.

(ii) The cerebrum is divided into the left and right cerebral hemispheres.

Each hemisphere contains different areas. One of these is the motor area.

State the function of the motor area.

(1)

(b) Nerve cells deep inside the brain produce the neurotransmitter dopamine.

Parkinson's disease is a brain disorder that destroys these nerve cells.

Explain why the destruction of these nerve cells causes the symptoms associated with Parkinson's disease.

(3)

(Total for Question 7 = 7 marks)

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5. Question 7 · part b · 3 marks · Working and answer

Working and answer

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6. Question 5 · part a.iii · 2 marks

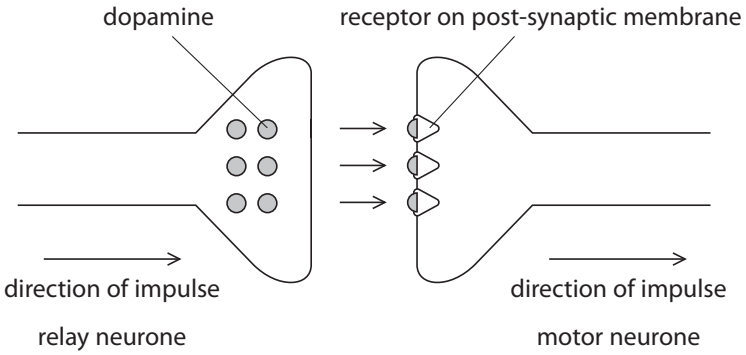
June 2023 Paper 2R

Further practice · Attempt part a.iii only.

5 (a) Dopamine is a neurotransmitter produced by nerve cells in the brain.

The nerve cells that produce dopamine are involved in stimulating movement and increasing heart rate.

The diagram shows how dopamine travels from a relay neurone to a motor neurone in the brain.



(i) Describe what is meant by a neurotransmitter.

(2)

(ii) Describe how information is transmitted from the relay neurone to the motor neurone.

(2)

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P 7 2 5 8 8 A 0 1 4 2 4

6. Question 5 · part a.iii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a.iii only.

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(iii) Suggest how an excess of dopamine will affect bodily functions.

(2)

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(b) In 2017 in the United Kingdom, 145 000 people were diagnosed with Parkinson's disease.

This figure is expected to have an 18% increase by 2025.

(i) Give two symptoms of Parkinson's disease.

(2)

1 .....

.....

2 .....

.....

(ii) Calculate the number of people expected to be diagnosed with Parkinson's disease in 2025.

(2)

number of people = ...

(Total for Question 5 = 10 marks)

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6. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

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7. Question 3 · part e · 3 marks

June 2024 Paper 2

Further practice · Attempt part e only.

3 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

People make lifestyle choices that can sometimes have a negative effect on their health. People who drink too much alcohol are at risk of brain and liver damage, certain types of cancer and mental illness. Liver damage can affect the digestion of fat. Alcohol can also affect the eyes, increasing the risk of developing cataracts that cause the lenses in the eyes to become cloudy. Alcohol is also a depressant. It increases the activity of a chemical in parts of the brain that control balance, speech, motor skills and memory. In addition, alcohol suppresses an area of the brain called the amygdala. The amygdala is responsible for the fight or flight function which prepares the body for action, initiating the release of stress hormones. These hormones increase glucose usage by the brain and other parts of the body.

When a person stops drinking alcohol after years of drinking a lot of alcohol, the amygdala in the brain becomes fully active again, which is the cause of all forms of alcohol withdrawal. Over time, if a person regularly drinks heavily, the central nervous system gets used to the suppressing effect of the alcohol, which means the brain and body are affected if the alcohol level suddenly drops. An individual can go straight into fight or flight mode as the alcohol leaves the system and levels of stress hormones begin to return to normal. Symptoms of withdrawal can be as severe as seizures and hallucinations or less severe symptoms such as loss of appetite and insomnia (an inability to sleep).

(a) Drinking alcohol and diet are two lifestyle choices.

Describe how a lifestyle choice, other than drinking alcohol and diet, can have a harmful effect on health (Lines 1 and 2).

(2)

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7. Question 3 · part e · 3 marks

June 2024 Paper 2

Further practice · Attempt part e only.

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(b) Explain how drinking alcohol affects the digestion of fat (Lines 3 to 4).

(2)

(c) State how cataracts affect vision (Lines 4 to 5).

(1)

(d) Name a part of the brain, other than the amygdala, that is affected by alcohol (Lines 5 to 7).

(1)

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(e) Explain how alcohol acts as a depressant on the nervous system (Line 5).

(3)

7. Question 3 · part e · 3 marks

June 2024 Paper 2

Further practice · Attempt part e only.

(f) After a person stops drinking alcohol, the amygdala becomes fully active again.

The release of stress hormones, such as adrenaline, returns to normal.

Give two effects of adrenaline on the body.

(2)

1 .....

.....

2 .....

.....

(Total for Question 3 = 11 marks)

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Working and answer